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Compress then Serve: Serving Thousands of LoRA Adapters with Little Overhead

Rickard Brüel Gabrielsson [1] **Jiacheng Zhu** [1] **Onkar Bhardwaj** [2] **Leshem Choshen** [1 2] **Kristjan Greenewald** [2] **Mikhail Yurochkin** [2] **Justin Solomon** [1]

Abstract

Fine-tuning large language models (LLMs) with low-rank adaptations (LoRAs) has become common practice, often yielding numerous copies of the same LLM differing only in their LoRA updates. This paradigm presents challenges for systems that serve real-time responses to queries that each involve a different LoRA. Prior works optimize the design of such systems but still require continuous loading and offloading of LoRAs, as it is infeasible to store thousands of LoRAs in GPU memory. To mitigate this issue, we investigate the efficacy of compression when serving LoRAs. We propose a method for the joint compression of LoRAs into a shared basis paired with LoRA-specific scaling matrices. We extend our algorithm to learn clusters of LoRAs that are amenable to joint compression, allowing it to scale gracefully to large LoRA collections. Our experiments with up to 1000 LoRAs demonstrate that compressed LoRAs preserve performance while offering major throughput gains in realistic serving scenarios with over a thousand LoRAs, maintaining 80% of the throughput of serving a *single* LoRA.

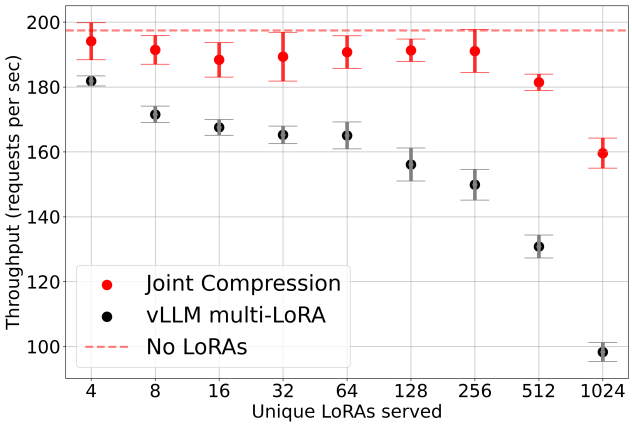

Figure 1: Throughput gains when serving 1000s of compressed LoRAs with vLLM.

1. Introduction

The myriad uses for foundation models (FMs) have led to a proliferation of specialized models, each fine-tuned to perform a downstream task. To avoid fine-tuning foundation models with billions of parameters, parameter-efficient fine-tuning (PEFT) algorithms were proposed. An especially successful PEFT method is low-rank adaptation (LoRA) (Hu et al., 2021), which learns low-rank additive changes to neural network matrices. Because of the low-rank parameterization, these matrices (called adapter weights) contain orders of magnitude fewer parameters than the base model. Still, LoRA can achieve performance on par with full fine-tuning (Hu et al., 2021).

LoRA's popularity has triggered a growing need to serve large collections of LoRA adapters at scale. Proprietary and open-source LLM providers offer fine-tuning services (OpenAI, 2024; TogetherAI, 2024; Predibase, 2024) with user bases likely in the thousands or even hundreds of thousands. As each user wants to use their own fine-tuned version of the LLM, serving a dedicated fine-tuned LLM per user becomes infeasible. To this end, S-LoRA (Sheng et al., 2023) proposes a system where only the base LLM is placed on an inference server and individual LoRA adapters are switched as needed at inference time. S-LoRA optimizes the system's inner workings via custom CUDA kernels and memory management to increase throughput when serving multiple LoRAs. Multi-LoRA system design has also been adopted in vLLM (Kwon et al., 2023), a state-of-the-art LLM serving engine. Despite optimized system designs, serving LoRAs still has a fundamental limitation: when the number of adapters is large, they need to be constantly loaded and offloaded from GPU memory to accommodate incoming requests, degrading throughput.

The problem of accommodating multiple LoRA adapters is also apparent when placing LLMs on edge devices,

[*]Equal contribution [1]MIT CSAIL [2]MIT-IBM Watson AI Lab. Correspondence to: Rickard Brüel Gabrielsson <brg@mit.edu>.

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where smaller LLMs are fine-tuned for various tasks, and the adapters are swapped depending on the task at hand (Gunter et al., 2024). In this setting, the number of adapters is smaller, e.g., a few dozen (Gunter et al., 2024), but the memory constraints are also more stringent.

In this work, we consider the problem of compressing a collection of LoRAs. We have two key objectives: (1) preserving the performance of the original LoRAs and (2) improving the throughput of serving many LoRAs. We formulate LoRA compression as a reconstruction problem, where the goal is to approximate the original adapters via collections of matrices of a smaller size. We compress LoRAs *jointly* by finding a shared basis and LoRA-specific scaling matrices and propose a joint diagonalization-based algorithm (JD). To improve reconstruction error for large numbers of LoRAs while keeping the number of parameters in check, we propose a clustering approach where each cluster is compressed independently using the joint diagonalization algorithm. Our clustering algorithm is based on alternating between optimizing the cluster assignments and the per-cluster reconstruction error.

Figure 1 showcases the benefits of joint compression. When serving up to 64 unique LoRAs, we use JD without clustering and for 128 or more, we pick the number of clusters to match the performance of compressed and original LoRAs. In each case, the GPU memory footprint of the compressed and original LoRAs is matched for a fair comparison to vLLM's multi-LoRA inference engine. When serving over 1000 LoRAs, compression increases throughput 1.6× and maintains 80% of the throughput of serving the base LLM (or a single LoRA merged into the LLM). §6 presents detailed results.

We summarize our main contributions below:

- We formulate the problem of compressing a collection of LoRAs and propose a joint compression scheme based on joint diagonalization.
- For large numbers of LoRAs, we scale joint compression by proposing a clustering algorithm where each cluster is jointly compressed to minimize reconstruction error.
- We establish theoretical guarantees for the reconstruction error of our compression formulation and relate reconstruction loss to performance empirically.
- We train a collection of more than 1000 high-quality LoRAs for `Mistral-7B-Instruct-v0.2` (Jiang et al., 2023a) on 1000 natural instruction tasks (Wang et al., 2022) and demonstrate that our compression techniques preserve the performance of the original LoRAs. We will release over a 1000 LoRAs to facilitate future work as well as the code for our method.
- We incorporate LoRA compression into a state-of-the-art LLM serving system and demonstrate that it is possible to serve over 1000 LoRAs across thousands of asynchronous requests with throughput comparable to serving a *single LoRA*.

2. Related Work

Parameter-efficient fine-tuning (PEFT) has become prevalent for updating foundation models thanks to the need for efficiency in training and communication (Lialin et al., 2023). Many PEFT methods have been proposed, e.g. (Houlsby et al., 2019; Liu et al., 2022b) and LoRA (Hu et al., 2021) became the standard, partially due to the ease of switching between LoRAs in inference time.

Several works improve LoRA (Liu et al., 2024; Wang et al., 2024), sometimes with algebraic methods like SVD (Meng et al., 2024; Zhang et al., 2023; Jiang et al., 2023b) or by leveraging its statistical properties (Zhu et al., 2024; Zeng & Lee, 2024). Relatively few, however, accelerate inference times. S-LoRA (Sheng et al., 2023) provides an efficient means of switching between LoRAs. Wen & Chaudhuri (2024) adapt training to reduce batch multiplications, accelerating inference. Our method achieves a similar outcome (see Appendix D) without changing the LoRA formulation or requiring that LoRAs be trained in a dedicated way; future improvements to LoRA will also benefit from this aspect of our work (e.g., Meng et al. (2024)).

Punica (Chen et al., 2023) introduces Segmented Gather Matrix-Vector Multiplication (SGMV) to optimize multi-LoRA serving by parallelizing feature-weight multiplications in batches and grouping requests that use the same LoRA. Our approach, by contrast, reduces parameters as a means to serve multiple LoRAs efficiently, providing an orthogonal strategy that can be seamlessly integrated with Punica's methods to enhance performance. In our vLLM experiments, we leveraged the Punica kernel for multi-LoRA implementation, demonstrating the application of our method in conjunction with Punica's optimizations.

Other research proposes alternative PEFT methods that can be more parameter-efficient than LoRA. For example, VeRA (Kopiczko et al., 2024) fine-tunes LLMs by sharing global static parameters while learning local scaling variables; $(IA)^3$ (Liu et al., 2022a) also reduces adapter parameter counts. However, none of these approaches has been as extensively tested or widely-adopted as LoRA. As a result, work that builds on LoRA enjoys a practical advantage due to its broad acceptance in practice.

There are many efforts to compress models (Cheng et al., 2017; Gholami et al., 2022; Sharma et al., 2024; Li et al., 2018). Predominantly, pruning and sparsification methods delete weights (Yadav et al., 2023a), and quantization methods reduce the weights' precision (Dettmers et al., 2024). Some works compress weights to reduce model size but typically require decompression and hence do not save

GPU memory (Hershcovitch et al., 2024). Similarly to our work, a few note increased performance and generalization after compression (Yadav et al., 2023a; Nadjahi et al., 2023; Hershcovitch et al., 2024; Sharma et al., 2024).

Our work also relates to model merging (Choshen et al., 2022; Wortsman et al., 2022; Matena & Raffel, 2021) and mixtures of experts (Muqeeth et al., 2024; Yadav et al., 2024). These methods reuse models trained by others (Choshen et al., 2023; Raffel, 2023), serving them together as one compressed model. Despite this similarity, these methods create a single general model that acts on any input, while ours yields more performant per-task solutions.

3. Rank-Based LoRA Compression

LoRA updates are parameterized by pairs of matrices A, B, whose product BA updates the fixed weight matrices $W_0 \in \mathbb{R}^{d_B \times d_A}$ of a neural network foundation model. Given an input x to a layer, the output of the LoRA-updated model at this layer is $(W_0 + BA)x$.

In formulating our compression algorithms, we consider a collection of given LoRA adapters $\{(A_i, B_i)\}_{i=1}^n$ that we would like to serve. We let r_i refer to the rank of the LoRA adapter-pair (A_i, B_i), i.e., $B_i \in \mathbb{R}^{d_B \times r_i}$, $A_i \in \mathbb{R}^{r_i \times d_A}$.

While our compression technique has access only to a collection of $\{(A_i, B_i)\}_{i=1}^n$ pairs, in our experiments we will assess the efficacy of compression by comparing how the compressed matrices perform relative to uncompressed LoRAs on typical data. For this reason, although in this section we optimize a Frobenius norm reconstruction error relative to the product B_iA_i, this is a proxy for the nonlinear and complex way that compression errors in the adapters impact transformer performance. Our experiments will thus focus on the performance of the compressed LoRAs against the uncompressed versions on real data in §6.

Our compression methods significantly reduce the overall number of parameters. Reducing parameters theoretically accelerates storage and serving of a collection of LoRAs. This reduction, however, alters the computational dynamics during inference, so parameter reduction alone does not immediately imply faster throughput. In light of the complexities of GPU optimization, we experimentally assess throughput under realistic conditions in §6.4.

3.1. Joint Diagonalization

To scale to many LoRAs, the compressed number of parameters should not scale linearly with n. Hence compressing each LoRA individually (e.g., via SVD as in our experimental baselines) is inherently limited. To address this, we suggest a Joint Diagonalization (JD) method, which optimizes a shared basis onto which we can project the set of n LoRAs. This allows structure to be shared, implicitly grouping and/or merging the collection of LoRAs.

In this model, each LoRA product B_iA_i is factorized into the form $U\Sigma_iV$, where U and V are shared across all LoRAs and Σ_i is specific to each LoRA. In this formulation, every Σ_i shares the same rank r. This allows U and V to be pre-loaded onto the GPU, with Σ_i loaded when necessary for each batch. The matrices Σ_i can be either diagonal or small square matrices, thus significantly reducing the number of LoRA-specific parameters and accelerating multi-LoRA serving.

Objective function. Motivated by the relationship of singular value decomposition to minimizing the Frobenius norm of the reconstruction error, we also propose to minimize the Frobenius norm of the adapter matrix approximation error. Specifically, we use the following objective:

$$\min_{\{\Sigma_i\}_{i=1}^n, U, V} \sum_{i=1}^n \|B_iA_i - U\Sigma_iV^\top\|_{\text{Fro}}^2. \tag{1}$$

Note this problem is *not* solved by a single matrix SVD, since U and V are shared among all terms but the Σ_i's are not. Using the Frobenius norm has the added benefit of making the objective convex in each argument separately, suggesting the possibility of efficient optimization. This objective function is underdetermined, however, so we consider two constrained regimes below.

Full Σ_i approximation. The first method we call JD-Full. Without loss of generality, U and V can be constrained to be orthogonal, so long as Σ_i remains an unconstrained full matrix. JD-Full adopts this restriction to make the optimization better posed, but note it does not restrict the expressiveness of the objective equation 1. This setting yields the following optimization problem:

$$\text{JD-Full}_r(\{B_iA_i\}_{i=1}^n) = \underset{\substack{\{\Sigma_i\}_{i=1}^n \\ U^\top U = V^\top V = I_r}}{\operatorname{argmin}} \sum_{i=1}^n \|B_iA_i - U\Sigma_iV^\top\|_{\text{Fro}}^2 \quad \text{(JD-Full)} \tag{2}$$

An efficient alternating algorithm to optimize this objective function can be found in Appendix A.

Diagonal Σ_i approximation. As an alternative, we can leave U, V unconstrained (other than to have r columns) and instead constrain the matrices Σ_i to be diagonal (but not necessarily positive). This formulation yields the following optimization problem:

$$\text{JD-Diag}_r(\{B_iA_i\}_{i=1}^n) = \underset{\{\Sigma_i\}_{i=1}^n, U, V}{\operatorname{argmin}} \sum_{i=1}^n \|B_iA_i - U\operatorname{diag}(\Sigma_i)V^\top\|_{\text{Fro}}^2 \quad \text{(JD-Diag)} \quad (3)$$

Appendix A provides an efficient alternating least squares algorithm for this objective. This diagonal version has per-LoRA parameter savings when compared to JD-Full, since the diagonal Σ_i only needs r parameters instead of r^2.

3.2. Clustering

As the number of LoRAs n grows and becomes more diverse, the rank r needed for Joint Diagonalization to achieve good performance will tend to increase. This increases the size of each Σ_i that needs to be stored, especially for JD-Full which will require $O(nr^2)$ storage for these matrices. If the necessary r grows proportionally to n, then this storage will eventually become the bottleneck.

To resolve this limitation with very large n, we propose to group the n LoRAs into k clusters C_j. Each cluster is given its own rank r for JD compression, and the clusters are chosen such that the overall reconstruction error is minimized. Specifically, the overall objective is

$$\min_{\{\{C_j\}, U_j, V_j\}, \{\Sigma_i\}} \sum_j \sum_{i \in C_j} ||B_iA_i - U_j\Sigma_iV_j||_F^2,$$

optimized by alternating between cluster assignments and the JD of each cluster; Appendix A.3 provides details. Typically, the goal with large n is to have k grow with n as r becomes fixed. Comparing k rank-r JD-Full clusters to a rank-kr JD-Full single cluster compression, the clustered approach requires $O(dkr + nr^2)$ parameters, while the single-cluster approach requires $O(dkr + nk^2r^2)$ parameters due to the increased sizes of the Σ_is. While these two approaches have the same rank, they may have different reconstruction abilities. Empirically, we find that multiple clusters significantly aid performance for $n \geq 100$.

4. Theoretical Analysis

In this section, we seek to better understand the role of the joint diagonalization method in §3.1 and how it motivates the clustering approach. We focus on the full-Σ_i case with orthogonal U, V matrices. Note that, for the same r, the r-JD-Diag has at least as large reconstruction error as r-JD-Full since it imposes an additional constraint on the Σ_i.

Firstly, note that perfect reconstruction can be achieved if and only if r is large enough, since there exist U, V such that all the B_i, A_i are in the spans of U, V resp. if and only if $r \geq \tilde{r}$:

Proposition 1. *Suppose* $\operatorname{rank}(B_iA_i) = r_i$ *for all* i*, and let*

$$\tilde{r} = \max\left\{\operatorname{rank}([A_1, \ldots, A_n]), \operatorname{rank}([B_1^\top \ldots, B_n^\top])\right\}.$$

Note $\max_i r_i \leq \tilde{r} \leq \sum_{i=1}^n r_i$*. Then JD-Full (equation 2) with* $r = \tilde{r}$ *compresses losslessly (perfect reconstruction), while* $r < \tilde{r}$ *will give nonzero reconstruction error.*

Due to training noise, $\tilde{r}$ will equal $\sum_{i=1}^n r_i$ almost always. This implies that in most realistic settings, the joint diagonalization approach is a lossy reconstruction.

This reconstruction loss can be significant, as the following theorem shows (proved in Appendix B):

Theorem 1. *Consider* n *LoRAs* $\{A_i, B_i\}_{i=1}^n$ *with* $r, n \leq d^2$*, and form the matrix* $L = \begin{bmatrix} \operatorname{vec}(B_1A_1) & \cdots & \operatorname{vec}(B_nA_n) \end{bmatrix}$*. Let* σ_j *be the singular values of* L*, sorted from largest to smallest, and let* $\bar{\sigma}_j$ *be the singular values of* $\sum_{i=1}^n B_iA_i$*. Then, using JD-Full (equation 2),*

$$\sum_{j=1}^r \bar{\sigma}_j^2 \leq \sum_{i=1}^n \|\Sigma_i\|_{\text{Fro}}^2 = \sum_{i=1}^n \|U\Sigma_iV^\top\|_{\text{Fro}}^2 \leq \sum_{j=1}^{\min(r^2,n)} \sigma_j^2,$$

implying the sum of squared Frobenius norms of the reconstructed LoRAs satisfies

$$\frac{\sum_{i=1}^n \|U\Sigma_iV^\top\|_{\text{Fro}}^2}{\sum_{i=1}^n \|B_iA_i\|_{\text{Fro}}^2} \leq \frac{\sum_{j=1}^{\min(r^2,n)} \sigma_j^2}{\sum_{j=1}^n \sigma_j^2} \leq 1, \text{ and}$$

$$\frac{\sum_{i=1}^n \|U\Sigma_iV^\top - B_iA_i\|_{\text{Fro}}^2}{\sum_{i=1}^n \|B_iA_i\|_{\text{Fro}}^2} \geq 1 - \frac{\sum_{j=1}^{\min(r^2,n)} \sigma_j^2}{\sum_{j=1}^n \sigma_j^2}.$$

In other words, reconstruction error is unavoidable if L's singular values are not concentrated in the top r^2 entries.

Remark 1 (Lower bound and merging)**.** *The lower bound* $\sum_{j=1}^r \bar{\sigma}_j^2$ *could be achieved by setting all the* Σ_i *equal, i.e., using a fully merged model instead of only merging the subspaces* U*,* V *and allowing* Σ_i *to vary with* i*.*

Remark 2 (Upper bound and grouping)**.** *The upper bound is smallest when the LoRAs are relatively* clustered*, i.e., when groups of vectors* $\operatorname{vec}(B_iA_i)$ *are similar. This situation raises the magnitude of the largest singular values of* L*, raising the upper bound in the proposition. As the LoRAs are* $d \times d$ *matrices that can be thought of as points in* $\mathbb{R}^{d^2}$*, for typical values of* d *well into the hundreds, it is likely that unrelated LoRAs will be* unclustered*, i.e., they will have relatively low inner products with each other.*

For orthogonal LoRAs, the singular values of L are the norms of the LoRAs, suggesting the following corollary:[1]

[1] A result for isotropic Gaussian LoRAs could be obtained via the quantiles of the Marchenko-Pastur Law.

Corollary 1. *Suppose (e.g., due to normalization) that the inputs to the joint diagonalization algorithm all have unit Frobenius norm, i.e.,* $\|B_iA_i\|_{\text{Fro}} = 1$. *Moreover, assume that the LoRAs are all orthogonal in the sense* $\text{tr}((B_iA_i)(B_jA_j)^\top) = 0$ *for* $i \neq j$. *Then, using the JD-Full method equation 2, we have* $1 \leq \sum_{i=1}^n \|\Sigma_i\|_{\text{Fro}}^2 \leq \min(r^2, n)$, *implying that the sum of squared Frobenius norms of the reconstructed LoRAs satisfies*

$$1-\frac{1}{n} \geq \frac{\sum_{i=1}^n \|U\Sigma_iV^\top - B_iA_i\|_{\text{Fro}}^2}{\sum_{i=1}^n \|B_iA_i\|_{\text{Fro}}^2} \leq 1-\min\left(\frac{r^2}{n}, 1\right).$$

This implies that for the common setting where $r^2 \ll n$, the reconstructed LoRAs will be significantly smaller than the original LoRAs, with significant reconstruction error.

Our analysis illustrates the tradeoffs of joint diagonalization. If the LoRAs are similar or well-clustered, reconstruction error will be low. On the other hand, if the LoRAs are random and orthogonal, reconstruction error will be high.

Since the loss space of transformers is highly complex, increasing reconstruction error does not necessarily degrade LLM performance. Interestingly, Figure 3 below shows that while large reconstruction error rapidly decreases performance, moderate (but still relatively large, at around 60%) reconstruction error does not damage performance and may even slightly outperform the zero-error setting. At the same reconstruction error, clustering outperforms non-clustering. This motivates our focus on minimizing reconstruction error, while also suggesting that our approach achieves something deeper than compression. Specifically, joint diagonalization finds subspaces that are shared among many LoRAs when r is large and *merges* subspaces when r is small. When r is particularly small, this tendency towards *averaging* all or some of the LoRAs connects to *merging* LoRAs, whose empirical success (Shah et al., 2023; Huang et al., 2024) could explain the procedure's success despite the nonlinearity of transformers.

Appendix H.11 explores this idea further, comparing reconstruction of real-world LoRAs to reconstruction of randomly sampled LoRAs. The reconstruction error is generally large, but significantly lower than the reconstruction error for random noise, indicating that a major shared component between the LoRAs is successfully retained.

That said, as the number of LoRAs grows, the shared component may not be significant enough to maintain sufficiently low reconstruction error with low rank r. This motivates the introduction of *clustering* in §3.2, since clustering seeks to find groups of LoRAs that are similar and better compressible by joint diagonalization. In particular, if the number of clusters k grows with n, the reconstruction error may no longer degrade with n even when r is fixed.

In the extreme case where $k = n$, each LoRA is compressed independently. By the Eckart-Young Theorem, JD applied to a single LoRA reduces to an SVD, replacing each rank-r_i LoRA adapter B_iA_i with a reduced rank-r approximation, where typically $r < \frac{1}{n}\sum_{i=1}^n r_i$:

$$\text{SVD}_r(B_iA_i) = U_i\Sigma_iV_i^\top, \quad \forall i = 1, \ldots, n. \tag{4}$$

As $\Sigma_iV_i^\top$ can be saved as a single matrix, this approach has $rn(d_A + d_B)$ parameters. We refer to this $k = n$ method as r-SVD and find that it underperforms our other methods while slightly outperforming the baseline uncompressed LoRAs. This result parallels Jiang et al. (2023b)'s observation that lowering LoRA ranks is beneficial for multi-task learning and model merging.

5. Training & Performance Evaluation

5.1. Training

We trained LoRA adapters on 1000 natural instruction tasks (Wang et al., 2022) using `Mistral-7B-Instruct-v0.2` (Jiang et al., 2023a) as the base. We set all LoRA adapter ranks to 16 (i.e., $\forall i, r_i = 16$), except for those in our ablation study (Appendix H.1), where we vary the LoRA rank.

We selected 10 diverse tasks (Table 2 in Appendix C) manually for consistent evaluation across experiments and randomly sampled an additional 990 tasks, resulting in a total of 1000 tasks (Table 3). The tasks went through a robust reviewing protocol to ensure high quality and diversity. Each task data was divided into training, validation, and test sets.

Hyperparameters, such as early stopping, were tuned using the validation sets. Table 1, Appendix C shows that on the test sets, LoRA consistently outperformed the base model in terms of Rouge scores and loss metrics.

5.2. Evaluation

We evaluated multiple metrics for the natural instruction tasks, including cross-entropy loss, Rouge-1, Rouge-L (Lin, 2004), exact match, and *agreement* between uncompressed and compressed LoRA. Here, *agreement* measures the exact match in task-generations between the uncompressed LoRA model and the compressed LoRA model, rather than comparing to ground truth data. While detailed results and discussions for all metrics are provided in Appendix H, our primary focus in the main text is on Rouge-L. We find that all metrics correlate, but Rouge-L correlates most strongly with downstream utility. This finding aligns with prior work (Wang et al., 2022), which demonstrates that Rouge-L correlates well with classification accuracy.

While cross-entropy is used for optimization during training, identical generation outputs across models can yield different cross-entropy losses. Exact match is too rigid and

does not account for the variability in task responses. Similarly, agreement does not capture the inexactness associated with most of our tasks, nor does it account for the performance gains or losses of the compressed LoRAs. Arguably, practitioners are primarily concerned with task performance in the settings for which the LoRA was designed, rather than exact generational agreement between models.

Joint diagonalization optimizes reconstruction error measured by the Frobenius norm, bounded by our theoretical analysis in §4. We empirically study the relation between the reconstruction error and downstream Rouge-L performance in Section 6.2.

Instead of listing absolute performance, we compute the performance difference between the base model and the LoRA model for each task via the ratio

$$\text{Performance relative to LoRA} := \frac{\text{method-performance}}{\text{LoRA-performance}}$$

for the method in question, highlighting relative improvement wrt the uncompressed LoRAs.

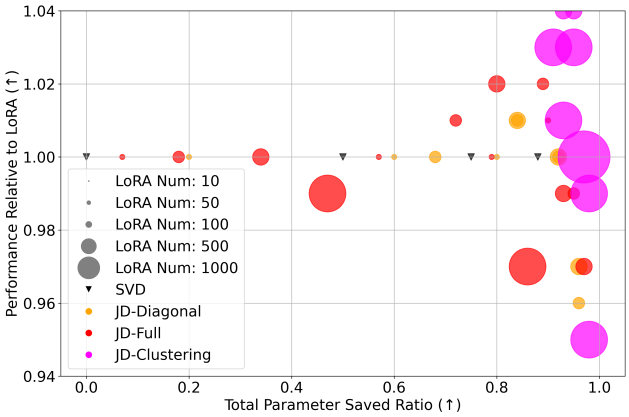

Figure 2: Performance after compression. We compare the performance of compressed LoRAs relative to uncompressed ones, with higher values on both axes reflecting better performance. The Total Parameter Saved Ratio depicts the number of parameters saved for a system with a large number n of different LoRAs. It is computed as: $r_{total} := 1 - \frac{\text{num. parameters after compression}}{\text{num. parameters before compression}}$.

6. Experiments

6.1. Task Performance

For each method, we vary the number n of compressed LoRAs and the compression rank r. We run each experiment three times with different random seeds and report the mean and standard deviation. See Table 7 for results evaluated on the same ten manually-selected tasks (Table 2) across settings. Every compressed collection of LoRAs contains these 10 tasks (i.e., in-distribution tasks), and each collection contains the smaller collections as subsets.

We normalize each LoRA adapter to have a Frobenius norm of one prior to running joint diagonalization. This normalization enhances performance and reduces the variance in reconstruction error. We restore the original norms of the LoRA adapters before reconstruction and testing.

Figure 2 illustrates the Rouge-L scores of the compressed LoRAs divided by the Rouge-L scores of the uncompressed LoRAs. JD variants often increase generalization and outperform the original LoRA. Notably, our JD methods approach the compression efficacy of a single LoRA, and with clustering, this aggressive reduction in size also maintains performance in larger collections. Appendix H includes tables of additional relative and absolute metrics.

For efficiency, we limited the JD methods to ten iterations instead of full convergence. While the alternating algorithm quickly reaches an approximate minimizer, squeezing out the last few digits of precision takes many more iterations with limited to no performance gain. Appendix H.12 also evaluates an alternative iterative algorithm that converges more rapidly once U, V are close to a minimizer, with minimal performance differences.

6.2. Performance and Reconstruction Error

Figure 3 relates reconstruction error and performance. The y-axis measures the mean performance improvement of Rouge-L relative to uncompressed LoRA, and the x-axis quantifies the mean relative reconstruction error between the compressed reconstruction of the product BA and the original product BA. Although performance and reconstruction error relate non-linearly, we see a decreasing, somewhat exponential trend. Notably, minimizing reconstruction error does not yield optimal performance, indicating that mild lossy reconstruction may enhance generalization. Interestingly, under the clustering approach, compared to non-clustering, even more aggressive lossy reconstruction can outperform less lossy reconstruction, suggesting that reconstruction error is even less critical for performance in the clustering scenario.

To select hyperparameters (compression rank and number of clusters) for the clustering experiments, we first assessed reconstruction error on a single LoRA module over a range of settings (see Appendix G). These preliminary experiments enabled efficient selection of cluster counts and rank values for compressing all LoRA modules.

6.3. Benefits of Compression

Compressing LoRAs reduces their parameter counts, thus lowering their overall memory footprint. While this offers

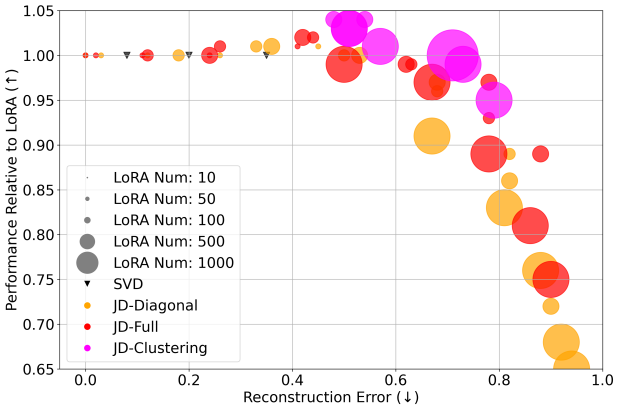

Figure 3: Reconstruction error vs. performance.

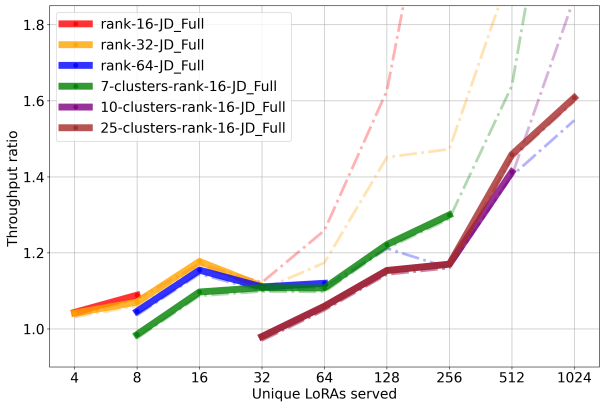

Figure 4: Throughput ratio when serving varying numbers of LoRAs with vLLM. Highlighted settings preserve at least 99% of the uncompressed LoRA performance.

many benefits, in both training and inference scenarios parameters are often transferred between different memory hierarchies (e.g., from CPU to GPU), and these transfers usually scale linearly with the amount of data moved. At the same time, compressed LoRAs alter the forward-pass formulation (unless merged with the base model weights). As shown in Figure 5 in the Appendix E, although compression greatly reduces memory usage and transfer time, it does not affect the forward-pass latency. Of the various ways to leverage these improvements, this work focuses on optimizing inference for multiple LoRAs using vLLM.

6.4. Throughput of Serving Compressed LoRAs

The previous sections demonstrate how to select an appropriate joint compression setting guided by the reconstruction error, such that the performance of the original LoRAs is preserved. Naturally, the rank and/or the number of clusters for the compression needs to increase as we compress larger LoRA collections to match LoRA performance.

Figure 4 studies how throughput with various compression settings compares to the vLLM multi-LoRA throughput with the matched GPU memory footprint. Specifically, for each number of unique LoRAs served and each compression setting, we compute the corresponding number of LoRAs to be placed on the GPU during serving and report the ratio of the two throughputs. For example, when serving 64 unique LoRAs and using rank 64 JD-Full compression, we report the ratio of throughputs of rank 64 JD-Full and vLLM multi-LoRA with 6 LoRAs allowed on the GPU at a time (see Appendix F for details). As the number of unique LoRAs increases, vLLM multi-LoRA throughput degrades as it needs to schedule the requests and load and offload the adapters. We note that vLLM multi-LoRA already employs advanced optimizations, such as efficient scheduling and non-blocking CPU-GPU communication when swapping LoRAs as well as techniques introduced in S-LoRA (Sheng et al., 2023; Kwon et al., 2023), but system optimization alone is insufficient to mitigate throughput degradation when serving many LoRAs.

Figure 4 shows that across LoRA collection sizes our compression techniques improve the throughput of vLLM multi-LoRA. Additionally, we highlight regions for each compression setting where compression is sufficiently moderate to achieve 99%+ of LoRA performance, according to the results in §6.2. Compression with a larger rank or too many clusters does not improve baseline throughput when serving a smaller number of LoRAs and should not be used in such cases. For example, rank 16 JD-Full improves baseline throughput with 4 and 8 LoRAs, but will underperform with more LoRAs, while 25 clusters rank 15 JD-Full does not improve throughput with 32 or fewer LoRAs, but when serving 1000+ LoRAs it improves the throughput significantly while maintaining the performance. Overall, an appropriate joint compression setting improves vLLM multi-LoRA throughput and preserves performance for LoRA collections of any size between 4 and 1024, as in Figure 1. Appendix F provides compression settings for each collection size.

vLLM extensively uses custom CUDA kernels. To accommodate our compression techniques, we minimally adjusted the vLLM code to generate additional kernels needed by the compressed LoRAs and used the Punica (Chen et al., 2023) kernel to further accelerate matrix multiplication. Pseudocode is given in §F.4 to show how we use the batch multiplication kernel. There likely is room for improvement to optimize the newly added kernels.

Additional details. In this experiment, we considered a varying number of rank-16 LoRAs, using a dataset of Shakespeare sonnets as inputs[2] arriving *asynchronously*.

[2] https://www.kaggle.com/datasets/shivamshinde123/william-shakespeares-sonnet/data

We measured throughput, i.e., the number of requests served per second when generating ten tokens per request. The base was Mistral 7B Instruct; we simulated random LoRAs and assigned inputs to LoRAs at random. Experiments were conducted on H100 80GB GPU capped at 40% memory consumption to reflect situations where a service provider might want to serve many LoRAs from cheaper hardware with lower memory than higher-end GPUs. This setting applies to the scenario where the LLM is large compared to the size of GPU and yet a provider may want to serve many LoRAs efficiently using one device.

6.5. Recommendations

JD-Full is generally preferred over JD-Diag, although for smaller numbers of LoRAs (less than 100), the performance difference is negligible. While JD-Full alone is effective up to 100 LoRAs, incorporating clustering at scales of 500-1000 LoRAs significantly enhances performance.

We recommend the following procedure for hyperparameter selection. For $\leq$ 100 LoRAs, JD-Full can be used without substantial degradation, using a rank $\approx$ (number of LoRAs$/2) + 7$. Beyond 100 LoRAs, clustering becomes increasingly critical. A robust method for any number of LoRAs up to 1000 uses JD-Full with clustering. Specifically, select a LoRA module from the middle of the network, apply a compression rank of 16, and experiment with an exponentially increasing number of clusters. Compute the reconstruction error for each setting on this module across all LoRAs—a computationally efficient process. Choose the minimal number of clusters that achieves a reconstruction loss below 0.6, and then use these settings across LoRA modules. Figure 6 in the Appendix illustrates this procedure applied to 500 LoRAs.

Tuning hyperparameters as discussed above using reconstruction loss as a validation metric is convenient since it can be done efficiently on CPU without expensive LLM evaluation. As our experiments demonstrate, compression settings that achieve below 0.6 reconstruction loss reliably preserve 99% or more of the LoRA performance, sometimes even outperforming the original LoRAs.

For inference, this procedure is executed as a preprocessing step before deploying our inference server. As new LoRAs are submitted, they are initially served uncompressed. A background CPU job can periodically re-run the compression algorithm and update the served LoRA parameters with the compressed versions.

7. Discussion

This study introduces approaches to LoRA compression, addressing significant challenges emerging as customization of foundation models such as LLMs and diffusion models becomes increasingly popular. Our contributions include theoretical formulations, empirical validation, and practical implementations that enhance the understanding and application of LLMs in scalable environments.

Our findings have several implications. Our theoretical bounds on reconstruction error not only increase confidence in the use of compressed models but also lay a groundwork for future explorations. Demonstrating that our compression techniques can preserve up to 100% of the original LoRAs' performance highlights the effectiveness of our methods. Furthermore, integrating LoRA compression into state-of-the-art LLM serving systems demonstrates potential for resource optimization, with throughput for thousands of LoRAs nearing that of a single LoRA.

Our promising results suggest several future research directions. First, further compression may be possible via quantization, since joint-diagonalization and quantization are independent compression strategies. Second, when scaling to hundreds of thousands of LoRAs, joint compression, while effective, will be insufficient to fit all LoRAs onto the GPU, thus requiring a procedure to schedule the requests. Clustering offers opportunities for efficient scheduling that incorporates the cluster assignments of LoRAs corresponding to the incoming requests.

Privacy presents another research direction, particularly regarding the possibility of information leakage during joint compression. As a preliminary study, Appendix H.2 investigates whether a base model with an adapter A for task T_A, after being jointly compressed alongside an adapter B for task T_B, inadvertently improves on T_B. Such an outcome would indicate that adapter A acquired information from adapter B. Our ablation study shows no performance gains on T_B, suggesting that the compressed adapter A remains independent and does not leak—or gain—information from adapter B. A more detailed investigation of the privacy properties of joint compression is an interesting next step.

In conclusion, our research advances LLM deployment by providing robust, scalable, and efficient compression. The ability of compressed LoRAs to maintain high performance while saving resources opens avenues for the broad application and adoption of LLMs across various industries. We encourage the community to build upon our findings and shared LoRAs to further enhance these technologies.

Impact Statement

This paper presents work whose goal is to advance machine learning. There are no societal consequences of our work that we feel must be specifically highlighted here.

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A. Joint Diagonalization Algorithms

A.1. Alternating Methods

Our goal is to derive algorithms that optimize equation 1. Common to both methods, we expand the objective functional:

$$\begin{aligned}\sum_i \|B_iA_i - U\Sigma_iV^\top\|_{\text{Fro}}^2 &= \sum_i \text{tr}((B_iA_i - U\Sigma_iV^\top)(B_iA_i - U\Sigma_iV^\top)^\top) \text{ by definition} \\ &= \sum_i \left[\text{tr}(B_iA_iA_i^\top B_i^\top) - 2\text{tr}(B_iA_iV\Sigma_i^\top U^\top) + \text{tr}(U\Sigma_iV^\top V\Sigma_i^\top U^\top)\right] \\ &= \text{const.} - 2\sum_i \text{tr}(B_iA_iV\Sigma_i^\top U^\top) + \sum_i \|U\Sigma_iV^\top\|_{\text{Fro}}^2. \end{aligned} \tag{5}$$

Using this expansion, we now consider the two settings discussed in §3.1.

Case 1: Non-diagonal Σ_i, orthogonal U, V. Setting the derivative of equation 5 with respect to Σ_i to zero, we find

$$\Sigma_i = \Sigma_i^*(U, V) = U^\top B_iA_iV. \tag{6}$$

We simplify our objective function after plugging in this expression:

$$\begin{aligned}\sum_i \|B_iA_i - U\Sigma_iV^\top\|_{\text{Fro}}^2 + \text{const.} &= \sum_i \left[\|\Sigma_i\|_{\text{Fro}}^2 - 2\text{tr}(B_iA_iV\Sigma_i^\top U^\top)\right] \text{ from equation 5} \\ &= \sum_i \left[\text{tr}(U^\top B_iA_iVV^\top A_i^\top B_i^\top U) - 2\text{tr}(B_iA_iVV^\top A_i^\top B_i^\top UU^\top)\right] \text{ from equation 6} \\ &= -\sum_i \text{tr}(B_iA_iVV^\top A_i^\top B_i^\top UU^\top).\end{aligned}$$

Substituting equation 6, we find

$$U_{opt}, V_{opt} = \arg\max_{\substack{U^\top U = I \\ VV^\top = I}} \sum_{i=1}^n \|U^\top B_iA_iV\|_{\text{Fro}}^2 = \arg\max_{\substack{U^\top U = I \\ VV^\top = I}} \sum_{i=1}^n \|\Sigma_i^*(U, V)\|_{\text{Fro}}^2. \tag{7}$$

Note that

$$\begin{aligned}\sum_{i=1}^n \|U^\top B_iA_iV\|_{\text{Fro}}^2 &= \text{tr}\left(\left(\sum_{i=1}^n B_iA_iVV^\top A_i^\top B_i^\top\right) UU^\top\right) \\ &= \text{tr}\left(\left(\sum_{i=1}^n B_i^\top A_i^\top UU^\top A_iB_i\right) VV^\top\right),\end{aligned}$$

by the identity $\|A\|_{\text{Fro}}^2 = \text{tr}(A^\top A)$. Hence, we optimize equation 7 by alternating between U and V:

- U **iteration:** Define $M \coloneqq \sum_i B_iA_iVV^\top A_i^\top B_i^\top$. Parenthesizing this expression properly requires only $O((m+n)r)$ storage/computation time. With this definition, we maximize $\text{tr}(MUU^\top)$ over U satisfying $U^\top U = I$. Since M is positive semidefinite, the optimum is to take U to be the r eigenvectors of M with largest eigenvalue, equivalent to an SVD problem.
- V **iteration:** Define $N \coloneqq \sum_i A_i^\top B_i^\top UU^\top B_iA_i$. Similarly to the previous step, we take V to contain the r eigenvectors of N with largest eigenvalue, again solvable using an SVD.

This method decreases the objective in each step.

Case 2: Diagonal Σ_i. If constrain Σ_i to be diagonal, we interpret our objective function equation 1 as a "triple least

squares" problem. We compute gradients:

$$\nabla_U \sum_i \|B_i A_i - U\Sigma_i V^\top\|_{\text{Fro}}^2 = 2\sum_i (U\Sigma_i V^\top - B_i A_i) V \Sigma_i^\top$$
$$\nabla_V \sum_i \|B_i A_i - U\Sigma_i V^\top\|_{\text{Fro}}^2 = 2\sum_i (V\Sigma_i^\top U^\top - A_i^\top B_i^\top) U \Sigma_i$$
$$\nabla_{\Sigma_i} \sum_i \|B_i A_i - U\Sigma_i V^\top\|_{\text{Fro}}^2 = 2U^\top (U\Sigma_i V^\top - B_i A_i) V$$

These expressions suggest efficient $r \times r$ linear systems to solve for U, V:

$$U = \left(\sum_i B_i A_i V \Sigma_i^\top\right)\left(\sum_i \Sigma_i V^\top V \Sigma_i^\top\right)^{-1}$$
$$V = \left(\sum_i A_i^\top B_i^\top U \Sigma_i\right)\left(\sum_i \Sigma_i^\top U^\top U \Sigma_i\right)^{-1}.$$

For Σ_i, we extract the diagonal from our gradient above:

$$\begin{aligned}
\text{diag}(U^\top U \Sigma_i V^\top V)_j &= (U^\top U \Sigma_i V^\top V)_{jj} \\
&= \sum_m (U^\top U)_{jm} \Sigma_{imm} (V^\top V)_{mj} \\
&= (U^\top U \circ V^\top V)\text{diag}(\Sigma_i) \\
\text{diag}(U^\top B_i A_i V)_j &= \sum_m (U^\top B_i)_{jm} (A_i V)_{mj} \\
&= \sum_m (U^\top B_i)_{jm} (V^\top A_i^\top)_{jm} \\
&= (U^\top B_i \circ V^\top A_i^\top)\mathbf{1} \\
\implies \text{diag}(\Sigma_i) &= (U^\top U \circ V^\top V)^{-1} (U^\top B_i \circ V^\top A_i^\top)\mathbf{1}
\end{aligned}$$

Here $\circ$ denotes the Hadamard product.

Combining these expressions, we use a simple coordinate descent algorithm cycling between the following three steps:

1. Solve for U
2. Solve for V
3. Solve for the Σ_i's
4. Optionally, normalize so $\sum_i \|\Sigma_i\|_{\text{Fro}}^2 = 1$

A.2. Additional Eigenvalue Iteration Algorithm

For the first case in §A.1, we introduce an alternative algorithm that eschews the use of SVD. This alternative is optimized for GPU execution, enabling tractable runs to convergence.

To derive this algorithm, we employ Lagrange multipliers to formulate the derived objective from equation 7:

$$U_{opt}, V_{opt} = \arg \max_{\substack{U^\top U = I \\ VV^\top = I}} \sum_{i=1}^{n} \|U^\top B_i A_i V\|_{\text{Fro}}^2, \tag{8}$$

yielding the expression

$$\Lambda = -\frac{1}{2}\|U^\top B_i A_i V\|_{\text{Fro}}^2 - \frac{1}{2}\text{tr}(X^\top (I - U^\top U)) - \frac{1}{2}\text{tr}(Y^\top (I - V^\top V)). \tag{9}$$

Taking the derivatives gives

$$\nabla_U \Lambda = -\sum_i B_i(A_iV)(V^\top A_i^\top)(B_i^\top U) + UX \tag{10}$$

$$\nabla_V \Lambda = -\sum_i A_i^\top(B_i^\top U)(U^\top B_i)(A_iV) + VY \tag{11}$$

Setting these derivatives to zero shows

$$\sum_i B_i(A_iV)(V^\top A_i^\top)(B_i^\top U) = UX \tag{12}$$

$$\sum_i A_i^\top(B_i^\top U)(U^\top B_i)(A_iV) = VY. \tag{13}$$

Here, one can show that the Lagrange multiplier matrices X and Y are diagonal and nonnegative, since the problem reduces to an eigenvalue problem when either U or V is fixed; this is essentially the argument behind the alternating algorithm in Appendix A. Hence, taking inspiration from classical eigenvalue iteration, we use the following updates to improve our estimates of U and V:

$$U_0^{(k+1)} \leftarrow \sum_i B_i(A_iV^{(k)})((V^{(k)})^\top A_i^\top)(B_i^\top U^{(k)}) \tag{14}$$

$$V_0^{(k+1)} \leftarrow \sum_i A_i^\top(B_i^\top U^{(k)})((U^{(k)})^\top B_i)(A_iV^{(k)}) \tag{15}$$

$$U^{(k+1)} \leftarrow \texttt{orthogonalize}(U_0^{(k+1)}) \tag{16}$$

$$V^{(k+1)} \leftarrow \texttt{orthogonalize}(V_0^{(k+1)}) \tag{17}$$

Here, the function `orthogonalize` orthogonalizes the columns of a matrix, e.g. by using the Q part of the reduced-size QR factorization. Although we lack a formal convergence proof, in practice we find that this method reliably reaches a local optimum of our problem.

By executing matrix operations in the specified sequence, these computations can be rapidly performed on GPUs. Note the expressions above are parenthesized to avoid constructing a large matrix product as an intermediate computation.

A.3. Clustering algorithm

Initialization: We run joint diagonalization with a single U, V then perform k-means with k clusters on the space of Σ_i's. This gives us our first clusters and we can use random initialization U_j, V_j for each cluster but the Σ_i can be maintained as initialization.

Step 1: Using the alternating JD algorithms from earlier in this section, we optimize the problem $\min_{U_j,V_j,\Sigma_i} \sum_{i\in C_j} ||B_iA_i - U_j\Sigma_iV_j^\top||_F^2$ for each j independently.

Step 2: New cluster assignment for $i : \min_j \min_{\Sigma_i} ||B_iA_i - U_j\Sigma_iV_j^\top||_F^2$. If any assignment changes we go to Step 1, else we have converged.

B. Proof of Theorem 1

Proof. For the lower bound, note that by Jensen's inequality,

$$\sum_{i=1}^n \|U^\top B_iA_iV\|_{\text{Fro}}^2 \geq \left\| U^\top \sum_{i=1}^n B_iA_iV \right\|_{\text{Fro}}^2,$$

for any U, V. Hence,

$$\sup_{U,V\in\text{St}(k,d)} \sum_{i=1}^n \|U^\top B_iA_iV\|_{\text{Fro}}^2 \geq \sup_{U,V\in\text{St}(k,d)} \left\| U^\top \sum_{i=1}^n B_iA_iV \right\|_{\text{Fro}}^2. \tag{18}$$

By the definition of singular value decomposition, the right hand side of equation 18 is maximized with U, V being the top r singular vectors of $\sum_{i=1}^{n} B_i A_i$, yielding $\left\|U^\top \sum_{i=1}^{n} B_i A_i V\right\|_{\text{Fro}}^2 = \sum_{i=1}^{r} \bar{\sigma}_i^2$. Recalling that $\Sigma_i = U^\top B_i A_i V$ yields the lower bound.

For the upper bound, recall that $\Sigma_i = U^\top B_i A_i V$. Rearranging,

$$\text{vec}(\Sigma_i) = (V^\top \otimes U^\top)\text{vec}(B_i A_i).$$

Define

$$\bar{\Sigma} := [\text{vec}(\Sigma_1), \ldots, \text{vec}(\Sigma_n)].$$

By our previous simplification,

$$\bar{\Sigma} = (V^\top \otimes U^\top)L.$$

Now

$$\sum_{i=1}^{n} \|\Sigma_i\|_{\text{Fro}}^2 = \|\bar{\Sigma}\|_{\text{Fro}}^2 = \text{tr}\left(((V \otimes U)(V \otimes U)^\top)(LL^\top)\right)$$

Since U, V are orthogonal and size $d \times r$, the top r^2 eigenvalues of the symmetric matrix $(V \otimes U)(V \otimes U)^\top$ will be equal to 1, and the rest will equal 0. The eigenvalues of the symmetric matrix $LL^\top$ will be equal to the squared singular values of L. We can then apply the Von Neumann trace inequality to obtain the upper bound.

The last statement follows from the Pythagorean theorem and the fact that the Σ_i is a projection of $B_i A_i$ to the U, V subspace. □

Note that we have only used the fact that the matrix $(V \otimes U)$ has singular values equal to 1; we have not used the fact that it has Kronecker product structure. On the other hand, each vector $\text{vec}(B_i A_i)$ is a sum of r_i Kronecker products and cannot be expressed as a Kronecker product. As a result, while the upper bound in the Von Neumann trace inequality is achieved if the eigenvectors of the two matrices align, the Kronecker product structure is a severe constraint and the upper bound we have provided is generous.

C. Training LoRAs

We trained LoRA adapters on 500 natural instruction tasks (Wang et al., 2022) using `Mistral-7B-Instruct-v0.2` (Jiang et al., 2023a) as the base model. All LoRA adapters were configured with a rank of 16, i.e., $\forall i, r_i = 16$. We selected 10 diverse tasks manually for consistent evaluation across experiments and randomly sampled an additional 490 tasks, resulting in a total of 500 tasks. These tasks were exclusively in English (both input and output), ensuring higher quality and thorough review (Wang et al., 2022). Each task dataset was divided into training, validation, and test sets (80-10-10). Hyperparameters, such as early stopping, were tuned using the validation sets; that is, we train for five epochs and take the best-performing epoch-checkpoint per validation loss. Evaluation on the test sets demonstrated that LoRA consistently outperformed the base model in terms of both Rouge scores and loss metrics (see Table 1).

In Table 1, we compare metrics between base model and LoRA finetuning.

Table 1: Comparison of metrics before and after LoRA training across 1000 tasks.

Metric	Base Model	LoRA
Loss	4.14 ± 3.07	0.56 ± 0.58
Exact Match	1.81 ± 6.56	51.38 ± 40.90
Rouge-1	21.70 ± 19.22	68.88 ± 29.73
Rouge-L	20.62 ± 18.21	67.80 ± 30.15

In Table 3 we include all 1000 tasks that were used.

We use Huggingface (Wolf et al., 2020) in our implementation. For the base model, we use quantization with configuration:

Table 2: Main Evaluation Tasks

Task Number	Name	Type	Domain
task280	stereoset_classification_stereotype_type	classification	stereoset
task190	snli_classification	snli	image captions
task391	causal_relationship	commonsense	cause and effect
task290	tellmewhy_question_answerability	answerability	story
task1391	winogrande_easy_answer_generation	commonsense	social and physical
task1342	amazon_us_reviews_title	title generation	amazon reviews
task442	com_qa_paraphrase_question_generation	question generation	wikipedia
task620	ohsumed_medical_subject_headings_answer_generation	keyword tagging	scientific
task1598	nyc_long_text_generation	data to text	restaurants
task039	qasc_find_overlapping_words	overlap extraction	natural science

```
BitsAndBytesConfig(
    load_in_4bit=True,
    bnb_4bit_use_double_quant=True,
    bnb_4bit_quant_type="nf4",
    bnb_4bit_compute_dtype=torch.bfloat16,
)
```

and LoRA configuration:

```
LoraConfig(
    r=16,
    lora_alpha=32,
    target_modules=["q_proj", "k_proj", "v_proj"],
    lora_dropout=0.05,
    bias="none",
    task_type="CAUSAL_LM",
    init_lora_weights=init_lora_weights,
)
```

D. Avoiding Batched Matrix Multiplication (BMM)

Fast LoRA (Wen & Chaudhuri, 2024) aims to alleviate the batched matrix multiplication (BMM) bottleneck when serving many LoRAs. They propose an adapter parameterization that replaces addition with elementwise multiplication, avoiding BMM and improving LoRA throughput at lower ranks. Our JD LoRA formulation also circumvents or heavily reduces the impact of BMM as discussed below, and both individual and joint compression methods can be applied to Fast LoRAs.

In the envisioned deployment scenario, a service provider hosts a large collection of LoRAs. Upon receiving a request, each user specifies both the input data and the desired LoRA identifier. The provider then processes the base model augmented with the specified LoRA for each user's data. As a provider is batching a collection of requests for GPU parallelization, they can expect to frequently have more than one unique LoRA identifier per batch.

Traditionally, a specific LoRA is integrated into the base model by transforming $W_0 \rightarrow W_0 + B_i A_i$. Serving multiple LoRAs conventionally would necessitate maintaining and executing a separate copy of the base model for each LoRA, bringing substantial computational overhead. Alternatively, the computation for $W_0 x$ and $B_i A_i x$ can be performed independently and subsequently merged. This strategy necessitates only a single instance of $W_0 x$ computation and storage of LoRA-specific parameters rather than the entire base model.

Consider the batch processing of $\mathbf{BAx}$, where boldface indicates that B_i, A_i are stacked into tensors of dimensions $(b \times$

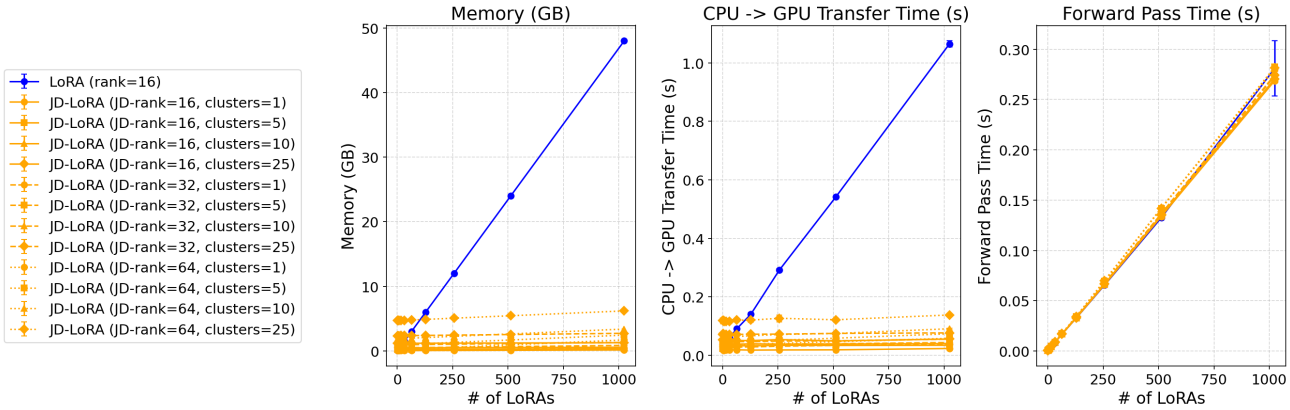


Figure 5: Memory load, transfer time, and forward-pass performance of LoRA and JD-LoRA.

$m \times r)$ and $(b \times r \times n)$ respectively, with batched data $\mathbf{x}$ shaped $(b \times l \times n)$:

$$\mathbf{A}\mathbf{x} \leftrightarrow (b \times r \times n) \times (b \times l \times n) \rightarrow (b \times l \times r) \text{ bmm}$$
$$\mathbf{B}(\mathbf{A}\mathbf{x}) \leftrightarrow (b \times m \times r) \times (b \times l \times r) \rightarrow (b \times l \times m) \text{ bmm.}$$

Here, "bmm" denotes batched matrix multiplication, a known bottleneck in both throughput and latency. Consider the corresponding operations for our joint compression scheme, $U\mathbf{\Sigma}V^\top x$:

$$V^\top \mathbf{x} \leftrightarrow (\tilde{r} \times n) \times (b \times l \times n) \rightarrow (b \times l \times \tilde{r}) \text{ broadcasted}$$
$$\mathbf{\Sigma}(V^\top \mathbf{x}) \leftrightarrow (b \times \tilde{r}) \times (b \times l \times \tilde{r}) \rightarrow (b \times l \times \tilde{r}) \text{ broadcasted}$$
$$U(\mathbf{\Sigma}V^\top \mathbf{x}) \leftrightarrow (m \times \tilde{r}) \times (b \times l \times \tilde{r}) \rightarrow (b \times l \times m) \text{ broadcasted}$$

In our optimized setup, batched matrix multiplications can be completely circumvented if the Σ_i matrices are diagonal. If not, given that $\tilde{r} \ll m, n$, any required batched matrix multiplication remains computationally inexpensive.

E. Simple Timing Experiments

In Figure 5, we present a set of simple experiments comparing the memory load, transfer time, and forward-pass performance of LoRA and JD-LoRA. These experiments were conducted across multiple clusters and various rank configurations. For memory usage, we measured all 96 LoRA modules in the Mistral model; however, for transfer time and forward-pass performance, we tested only a single LoRA module. We also implemented F-LoRA (Wen & Chaudhuri, 2024), but contrary to their reported results, we were unable to achieve faster forward-pass performance than standard LoRA.

F. GPU Memory Usage Computation for JD Compression.

The GPU memory consumption is primarily influenced by the number of parameters that need to be stored and processed during inference. In this section, we introduce the detail of how we compute the GPU consumption of our method, and how we find the number of vLLM multi-LoRA that share the same GPU utilization.

- D: Hidden dimension size (e.g., $D = 4098$).
- r: Rank of the shared basis matrices for compression (e.g., $r = 16, 32, 64$).
- N: Maximum number of LoRA modules being served simultaneously (max_lora_num).
- c: Number of clusters in our clustering method (e.g., $c = 7, 10, 25$).

In Figure 1, we use different JD-compression settings for serving different number of unique LoRAs. Specifically:

- **Serving 4 unique LoRAs:**
  Ours: rank 16 JD-Full.
  vLLM multiLoRA baseline: max-gpu-lora = 2.

- **Serving 8 unique LoRAs:**
  Ours: rank 16 JD-Full.
  vLLM multiLoRA baseline: max-gpu-lora = 2.
- **Serving 16 unique LoRAs:**
  Ours: rank 32 JD-Full.
  vLLM multiLoRA baseline: max-gpu-lora = 3.
- **Serving 32 unique LoRAs:**
  Ours: rank 64 JD-Full.
  vLLM multiLoRA baseline: max-gpu-lora = 5.
- **Serving 64 unique LoRAs:**
  Ours: rank 64 JD-Full.
  vLLM multiLoRA baseline: max-gpu-lora = 6.
- **Serving 128 unique LoRAs:**
  Ours: 7 clusters, rank 16 JD-Full.
  vLLM multiLoRA baseline: max-gpu-lora = 8.
- **Serving 256 unique LoRAs:**
  Ours: 10 clusters, rank 16 JD-Full.
  vLLM multiLoRA baseline: max-gpu-lora = 10.
- **Serving 512 unique LoRAs:**
  Ours: 25 clusters, rank 16 JD-Full.
  vLLM multiLoRA baseline: max-gpu-lora = 26.
- **Serving 1024 unique LoRAs:**
  Ours: 25 clusters, rank 16 JD-Full.
  vLLM multiLoRA baseline: max-gpu-lora = 28

F.1. Baseline GPU Memory Usage

The baseline for our comparison is the standard LoRA method with a rank of 16. The total parameter count for the baseline is given by:

$$\text{Params}_{\text{baseline}} = D \times 2 \times 16.$$

This accounts for the parameters in the LoRA-adapted layers, where the factor of 2 represents the weights and biases.

F.2. GPU Memory Usage for JD Full Method

For the Joint Decomposition (JD) Full method without clustering, the total parameter count is:

$$\text{Params}_{\text{JD\_Full}} = D \times 2 \times r + N \times r^2.$$

- $D \times 2 \times r$: Parameters for the base model adapted with rank-r LoRA.
- $N \times r^2$: Additional parameters introduced by each of the N LoRA modules, each of size $r \times r$.

The GPU memory usage ratio relative to the baseline is:

$$\text{GPU Usage Ratio}_{\text{JD\_Full}} = \frac{\text{Params}_{\text{JD\_Full}}}{\text{Params}_{\text{baseline}}} = \frac{D \times 2 \times r + N \times r^2}{D \times 2 \times 16}.$$

F.3. GPU Memory Usage for Clustering Method

When employing clustering, the parameter count changes due to the addition of cluster-specific parameters:

$$\text{Params}_{\text{Clustering}} = D \times 2 \times r \times c + N \times (r^2 + 1).$$

- $D \times 2 \times r \times c$: Parameters for the base model adapted with rank-r LoRA across c clusters.

- $N \times (r^2 + 1)$: Additional parameters for each LoRA module and cluster assignments.

The GPU memory usage ratio is:

$$\text{GPU Usage Ratio}_{\text{Clustering}} = \frac{\text{Params}_{\text{Clustering}}}{\text{Params}_{\text{baseline}}} = \frac{D \times 2 \times r \times c + N \times (r^2 + 1)}{D \times 2 \times 16}.$$

F.4. Punica

In our `vLLM` experiments, we specifically used the Punica kernel for implementing multi-LoRA, applying our approach in conjunction with Punica's capabilities. Our custom function, `add_lora_slice_with_sigma`, implements the following key steps:

1. **Initialize Buffers**: Creates temporary storage for intermediate calculations if not already provided.
2. **Apply Matrix A**: Transforms `x` using matrix `A`, storing the result in `buffer`.
3. **Apply Matrix Sigma**: Further transforms `buffer` using `Sigma`, storing the result in `buffer_sigma`.
4. **Apply Matrix B and Update `y`**: Finally, transforms `buffer_sigma` using `B`, applies scaling, and updates a slice of `y` in place.

Below is the pseudocode for `add_lora_slice_with_sigma`, illustrating the integration:

Listing 1: Pseudocode for 'add_lora_slice_with_sigma'

```
Function add_lora_slice_with_sigma(y, x, wa_t_all, wb_t_all, wsigma_t_all, indices,
    layer_idx, scale, y_offset, y_slice_size, buffer=None):
    # Initialize buffers if not provided
    if buffer is None:
        buffer = create_tensor(shape=(x.size(0), R), dtype=float32)
        buffer_sigma = create_tensor(shape=(buffer.size(0), R), dtype=float32)
    # Step 1: Apply matrix A
    dispatch_bgmv_low_level(buffer, x, wa_t_all, indices, layer_idx, scale=1.0)
    # Step 2: Apply matrix Sigma
    dispatch_bgmv_low_level(buffer_sigma, buffer, wsigma_t_all, indices, layer_idx, scale
        =1.0)
    # Step 3: Apply matrix B and update y slice
    dispatch_bgmv_low_level(y, buffer_sigma, wb_t_all, indices, layer_idx, scale, y_offset
        , y_slice_size)
End Function
```

G. Selecting Number of Clusters

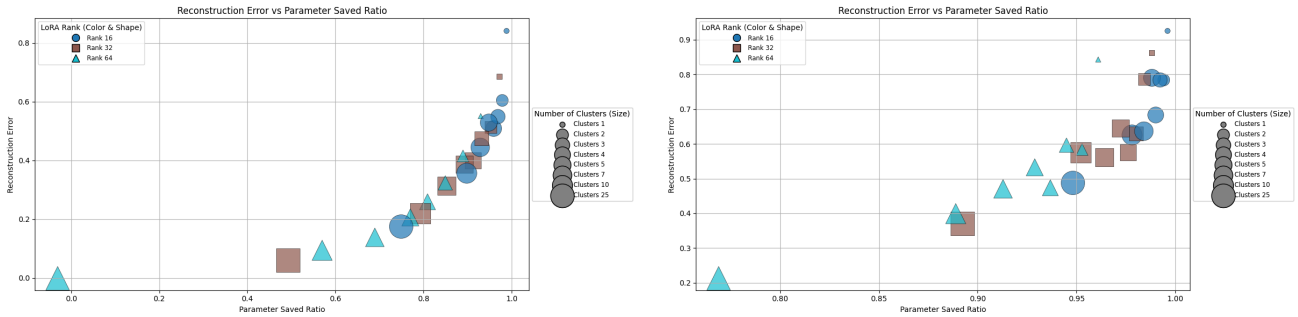
To identify optimal hyperparameters for the clusters compression method, we analyzed the relationship between reconstruction error and the parameter saved ratio for a single LoRA module, as shown in Figure 6. By comparing the results across different numbers of Low-Rank Adaptation (LoRA) configurations (100 and 500, depicted in subfigures 6a and 6b), we were able to observe the trade-off between model size reduction and reconstruction accuracy. Based on these findings, we selected the rank and number-of-clusters hyperparameters that effectively balance these two objectives. The chosen settings were then used to conduct full-scale experiments.

H. Additional Results

This section elaborates on the results that underpin the figures presented in the main text and showcases a consistent correlation across various evaluation metrics. Additionally, we assess the significance of achieving convergence and the performance of compression on new unseen LoRA models.

H.1. LoRAs of different ranks

In Table 4, we report the performance of our compression method on LoRAs with ranks uniformly sampled between 16 and 64 (mean rank of 43). In Table 5, we present compression results for LoRAs of rank 43, matching the average rank from Table 4. Because these same-rank LoRAs have an identical parameter count, they also exhibit identical parameter-saving



(a) Recon. Error vs Parameter Saved Ratio for 100 LoRAs

(b) Recon. Error vs Parameter Saved Ratio for 500 LoRAs

Figure 6: Comparison of reconstruction error against the parameter saved ratio for different numbers of LoRA configurations for a single LoRA module. The left subplot shows results for 100 LoRAs, while the right subplot displays results for 500 LoRAs. These plots illustrate the trade-off between reconstruction accuracy and compression efficiency, providing insights into optimal parameter settings for compression.

ratios. While performance declines slightly for LoRAs with varying ranks, our compression method still preserves over 99% of the original performance.

H.2. Privacy Ablation

We investigate whether jointly compressing certain tasks results in improved performance on tasks within that same compressed group, compared to tasks outside of the group. In other words, we examine whether information about which tasks were compressed together could be inferred from subsequent performance differences. Table 6 presents the results.

In the Compressed Together setting, tasks (task1391, task190, task280, task290, and task391) are compressed jointly with five other tasks. We then evaluate their cross-task performance within this group. In the Compressed Separately setting, the same set of tasks is each compressed alongside nine other tasks that are not part of the original group. We again evaluate the cross-task performance using the same sets of tasks, allowing us to compare and assess any differences in performance attributable to joint versus separate compression.

Table 6: Privacy Ablation: Performance on tasks compressed together vs. performance on tasks compressed separately

Source Task	Task Combination	Test Loss	Exact Match	Rouge1	RougeL
	Compressed Together				
task1391	task1391 on task190	2.320	29	29.00	29.00
	task1391 on task280	1.445	10	12.00	12.00
	task1391 on task290	2.662	0	0.00	0.00
	task1391 on task391	1.560	4	7.00	7.00
	Average	1.997	10.8	12.00	12.00
task190	task190 on task1391	1.392	0	0.00	0.00
	task190 on task280	1.647	2	2.00	2.00
	task190 on task290	1.838	0	0.00	0.00
	task190 on task391	2.622	2	2.00	2.00
	Average	1.875	1.0	1.00	1.00
task280	task280 on task1391	2.259	14	16.47	16.47
	task280 on task190	2.922	20	20.50	20.50
	task280 on task290	0.729	33	43.07	43.07
	task280 on task391	2.318	44	62.52	62.52
	Average	2.057	27.8	35.64	35.64
task290	task290 on task1391	0.867	36	46.58	46.58
	task290 on task190	1.553	36	36.00	36.00
	task290 on task280	1.036	43	43.40	43.40
	task290 on task391	0.461	59	86.33	86.33
	Average	0.979	43.5	53.08	53.08
task391	task391 on task1391	0.502	65	65.75	65.75

Table 6: Privacy Ablation: Performance on tasks compressed together vs. performance on tasks compressed separately (continued)

Source Task	Task Combination	Test Loss	Exact Match	Rouge1	RougeL
	task391 on task190	1.421	31	31.00	31.00
	task391 on task280	0.417	69	69.00	69.00
	task391 on task290	0.265	71	90.33	90.33
	Average	0.651	59.0	64.02	64.02
	Compressed Separately				
	task1391 on task190	2.347	30	30.00	30.00
	task1391 on task280	1.543	11	13.58	13.58
task1391	task1391 on task290	2.477	0	0.00	0.00
	task1391 on task391	1.253	10	18.00	18.00
	Average	1.905	12.8	15.40	15.40
	task190 on task1391	1.388	0	0.00	0.00
	task190 on task280	1.603	2	2.00	2.00
task190	task190 on task290	1.771	0	0.00	0.00
	task190 on task391	2.326	4	4.00	4.00
	Average	1.772	1.5	1.50	1.50
	task280 on task1391	3.111	2	5.13	5.13
	task280 on task190	2.711	19	19.00	19.00
task280	task280 on task290	0.948	16	22.09	22.09
	task280 on task391	2.449	39	53.11	53.11
	Average	2.305	19.0	24.83	24.83
	task290 on task1391	0.848	41	47.58	47.58
	task290 on task190	1.355	38	38.00	38.00
task290	task290 on task280	1.050	41	41.00	41.00
	task290 on task391	0.463	59	86.33	86.33
	Average	0.929	44.8	53.23	53.23
	task391 on task1391	0.428	68	68.74	68.74
	task391 on task190	1.507	31	31.00	31.00
task391	task391 on task280	0.368	70	70.00	70.00
	task391 on task290	0.269	73	91.00	91.00
	Average	0.643	60.5	65.18	65.18

H.3. Relative Rouge-L Performance and Compression Rate

Table 7 presents comprehensive results from the experiments underlying Figure 2 for each evaluation task. Additionally, we incorporate results using the Ties-merging benchmark (Yadav et al., 2023b), which consolidates all LoRA-adapters into a single adapter of identical configuration and parameter count; this integration significantly compromises performance.

H.4. Absolute Rouge-L Performance and Compression Rate

Table 8 provides the full results behind Table 7, but with Rouge-L scores instead of relative performance compared to LoRA.

H.5. Relative Rouge-1 Performance and Compression Rate

Table 9 provides full results for relative performance of Rouge-1, which shows the same trends as the results for relative performance of Rouge-L (Table 7).

H.6. Absolute Rouge-1 Performance and Compression Rate

Table 10 provides full results for absolute performance of Rouge-1, which shows the same trends as the results for absolute performance of Rouge-L (Table 8).

H.7. Relative Exact-Match Performance and Compression Rate

Table 11 provides full results for relative performance of exact-match, which shows the same trends as the results for relative performance of Rouge-L (Table 7).

H.8. Loss and Compression Rate

Table 12 provides full results for test loss (cross-entropy), which shows the same trends as the results for relative performance of Rouge-L (Table 7).

H.9. Agreement and Compression Rate

Table 13 provides full results for *agreement*, which shows the same trends as the results for relative performance of Rouge-L (Table 7). Note that *agreement* measures the exact match in task generations between the uncompressed LoRA model and the compressed LoRA model, rather than comparing to the task's ground truth data. The comparison is very strict and requires an exact match between the generations of the two models (LoRA and the compressed LoRA), comparing each sample one at a time.

H.10. Reconstruction Error and Compression Rate

Table 14 provides the full results of the experiments behind Figure 3 for every evaluation task.

H.11. Reconstruction Error: Trained vs. Random

Table 15 provides the reconstruction error on random (untrained) LoRA matrices. Comparing with Table 14, we find that reconstruction error is consistently higher on random (untrained LoRA) matrices than on trained LoRA matrices. This demonstrates that after training, LoRAs have a shared structure that JD exploits.

H.12. Convergence

Table 16 presents outcomes where the JD-Full algorithm is executed until convergence. Our convergence criterion is defined as follows:

$$\max\left(\|U_{t+1} - U_t U_t^\top U_{t+1}\|_{\text{Fro}}/\|U_{t+1}\|_{\text{Fro}}, \|V_{t+1} - V_t V_t^\top V_{t+1}\|_{\text{Fro}}/\|V_{t+1}\|_{\text{Fro}}\right) < \tau \tag{19}$$

where the tolerance threshold τ is set to 0.001. Due to the slow per-iteration computation times of the primary JD-Full algorithm, which quickly reaches an approximate optimum but then has a long tail of convergence for final digits of precision, we devised an alternative eigenvalue iteration algorithm (Appendix A.2) optimized for GPU acceleration. Our analysis indicates that adherence to this convergence criterion does not significantly alter the results.

H.13. Out-of-distribution Performance (LoRA-hub)

For completeness, we incorporate results using the protocol of LoRA-hub (Huang et al., 2024). That is, 100 LoRA-adapters are sampled, independent of the evaluation task, representing a measure of out-of-distribution performance. This also means that each result on a task is averaged across all 100 LoRA-adapters (as there is no *a priori* LoRA-to-task mapping). These results were obtained without normalizing the LoRA-adapters before applying the JD algorithms, a step we later identified as beneficial. We present performance comparison in Table 18. Table 17 presents the average agreement between uncompressed and compressed LoRA across 10 evaluation tasks. Results per task for JD-diagonal and JD-full are shown in Table 19 and Table 20, respectively.

From these tables, we find that the JD algorithms successfully maintain performance in this out-of-distribution context.

Table 3: List of all 1000 Tasks

Task ID	Description	Task ID	Description	Task ID	Description	Task ID	Description
task100	concatenate all elements from index i to j	task1009	pib translation bengali hindi	task101	reverse and concatenate all elements from index i to j	task102	commongen sentence generation
task1023	pib translation english hindi	task1024	pib translation hindi english	task104	semeval 2019 task10 closed vocabulary mathematical answer generation	task1047	pib translation english telugu
task1048	pib translation telugu english	task105	story cloze-rocstories sentence generation	task1053	pib translation hindi urdu	task1057	pib translation english urdu
task1062	pib translation marathi bengali	task107	splash question to sql	task1079	pib translation english gujarati	task108	contextualabusedetection classification
task1082	pib translation marathi hindi	task1083	pib translation marathi tamil	task1085	pib translation english marathi	task1087	two number sum
task1088	array of products	task1089	check monotonic array	task109	smsspamcollection spamsmsdetection	task1090	ted translation en gl
task1094	ted translation en pt	task1095	ted translation ja gl	task1097	ted translation ja pl	task1098	ted translation ja fa
task110	logic2text sentence generation	task1101	ted translation es it	task1102	ted translation es pl	task1105	ted translation ar gl
task1106	ted translation ar it	task1109	ted translation ar pt	task111	asset sentence simplification	task1110	ted translation he gl
task1111	ted translation he it	task1113	ted translation he fa	task1114	ted translation he pt	task1115	alt ja id translation
task1116	alt id ja translation	task1117	alt ja id answer generation	task1118	alt ja fil translation	task112	asset simple sentence identification
task1120	alt ja fil answer generation	task1122	alt khm ja translation	task1127	alt ja th translation	task1128	alt th ja translation
task113	count frequency of letter	task1130	xcsr vi commonsense mc classification	task1132	xcsr ur commonsense mc classification	task1135	xcsr en commonsense mc classification
task1139	xcsr ru commonsense mc classification	task114	is the given word longest	task1140	xcsr pl commonsense mc classification	task1141	xcsr zh commonsense mc classification
task1142	xcsr ar commonsense mc classification	task1144	xcsr sw commonsense mc classification	task1145	xcsr jap commonsense mc classification	task1146	country capital
task1147	country currency	task1148	maximum ascii value	task1149	item check edible	task115	help advice classification
task1150	delete max min	task1151	swap max min	task1152	bard analogical reasoning causation	task1154	bard analogical reasoning travel
task1156	bard analogical reasoning tools	task1157	bard analogical reasoning rooms for containers	task1158	bard analogical reasoning manipulating items	task116	com2sense commonsense reasoning
task1167	penn treebank coarse pos tagging	task1168	brown coarse pos tagging	task117	spl translation en de	task118	semeval 2019 task10 open vocabulary mathematical answer generation
task1186	nne hrngo classification	task1188	count max freq char	task1189	check char in string	task119	semeval 2019 task10 geometric mathematical answer generation
task1190	add integer to list	task1191	food veg nonveg	task1192	food flavor profile	task1193	food course classification
task1194	kth largest element	task1196	atomic classification oeffect	task1197	atomic classification oreact	task1198	atomic classification owant
task1199	atomic classification xattr	task1200	atomic classification xeffect	task1201	atomic classification xintent	task1202	atomic classification xneed
task1203	atomic classification xreact	task1204	atomic classification hinderedby	task1205	atomic classification isafter	task1206	atomic classification isbefore
task1207	atomic classification atlocation	task1208	atomic classification xreason	task1209	atomic classification objectuse	task121	zest text modification
task1210	atomic classification madeupof	task1211	atomic classification hassubevent	task1212	atomic classification hasproperty	task1213	atomic classification desires
task1214	atomic classification xwant	task1215	atomic classification capableof	task1216	atomic classification causes	task1217	atomic answer generation
task1219	ted translation en es	task122	conala list index addition	task1221	ted translation en he	task1222	ted translation ja en
task1224	ted translation ja ar	task1225	ted translation ja he	task1226	ted translation es en	task1227	ted translation es ja
task1228	ted translation es ar	task1229	ted translation es he	task123	conala sort dictionary	task1232	ted translation ar es
task1233	ted translation ar he	task1235	ted translation he ja	task1236	ted translation he es	task1237	ted translation he ar
task1238	ted translation gl en	task1239	ted translation gl ja	task124	conala pair averages	task1240	ted translation gl es
task1241	ted translation gl ar	task1242	ted translation gl he	task1244	ted translation gl pl	task1245	ted translation gl fa
task1246	ted translation gl pt	task1247	ted translation it en	task1248	ted translation it ja	task125	conala pair differences
task1252	ted translation it gl	task1254	ted translation it fa	task1255	ted translation it pt	task1257	ted translation pl ja
task1258	ted translation pl es	task126	scan structured text generation command action all	task1261	ted translation pl gl	task1262	ted translation pl it
task1263	ted translation pl fa	task1264	ted translation pl pt	task1266	ted translation fa ja	task1267	ted translation fa es
task127	scan long text generation action command all	task1270	ted translation fa gl	task1271	ted translation fa it	task1272	ted translation fa pl
task1273	ted translation fa pt	task1274	ted translation pt en	task1276	ted translation pt es	task1278	ted translation pt he
task1279	ted translation pt gl	task128	scan structured text generation command action short	task1280	ted translation pt it	task1281	ted translation pt pl
task1283	hrngo quality classification	task1284	hrngo informativeness classification	task1285	kpa keypoint matching	task1286	openbookqa question answering
task1288	glue mrpc paraphrasing	task1289	trec classification	task129	scan long text generation action command short	task1292	yelp review full text categorization
task1293	kilt tasks hotpotqa question answering	task1294	wiki qa answer verification	task130	scan structured text generation command action long	task1308	amazonreview category classification
task1309	amazonreview summary classification	task131	scan long text generation action command long	task1310	amazonreview rating classification	task1311	amazonreview rating classification
task1312	amazonreview polarity classification	task1313	amazonreview polarity classification	task1314	country abbreviation	task1315	find range array
task1316	remove duplicates string	task1317	country calling code	task1318	country national dish	task1319	country by barcode prefix
task132	dais text modification	task1320	country domain tld	task1321	country continent	task1322	country government type
task1323	open subtitles hi en translation	task1324	open subtitles te en translation	task1325	qa zre question generation on subject relation	task1326	qa zre question generation from answer
task1327	qa zre answer generation from question	task1328	qa zre relation generation from question	task1330	open subtitles en te translation	task1331	reverse array
task1332	check leap year	task1333	check validity date ddmmyyyy	task1336	peixian equity evaluation corpus gender classifier	task1338	peixian equity evaluation corpus sentiment classifier
task1339	peixian equity evaluation corpus text completion	task1340	msr text compression compression	task1341	msr text classification	task1342	amazon us reviews title
task1346	glue cola grammatical correctness classification	task1347	glue sts-b similarity classification	task1351	opus100 translation gu en	task1352	hind encorp translation hi en
task1353	hind encorp translation en hi	task1354	sent comp classification	task1355	sent comp summarization	task1359	numer sense answer generation
task1360	numer sense multiple choice qa generation	task1364	hans answer generation	task137	detoxifying-lms classification toxicity	task1370	newscomm classification
task1371	newscomm translation	task1373	newscomm translation	task1375	newscomm translation	task1377	newscomm translation
task1378	quarel correct answer generation	task1379	quarel incorrect answer generation	task138	detoxifying-lms classification fluency	task1380	quarel correct option generation
task1381	quarel incorrect option generation	task1382	quarel write correct answer	task1383	quarel write incorrect answer	task1384	deal or no dialog classification
task1385	anli r1 entailment	task1386	anli r2 entailment	task1387	anli r3 entailment	task1389	hellaswag completion
task139	detoxifying-lms classification topicality	task1390	wscfixed coreference	task1391	winogrande easy answer generation	task1393	superglue copa text completion
task1394	meta woz task classification	task1397	europa ecdc tm fr en translation	task1398	obqa question generation	task1399	obqa answer generation
task140	detoxifying-lms classification style	task1400	obqa incorrect answer generation	task1401	obqa sentence generation	task1403	check validity date mmddyyyy
task1404	date conversion	task1405	find median	task1406	kth smallest element	task1409	dart text generation
task141	odd-man-out classification category	task1412	web questions question answering	task1418	bless semantic relation classification	task1419	mathqa gain
task142	odd-man-out classification no category	task1420	mathqa general	task1421	mathqa other	task1422	mathqa physics
task1423	mathqa geometry	task1424	mathqa probability	task1425	country iso numeric	task1426	country independence year
task1427	country region in world	task1428	country surface area	task1429	evalution semantic relation classification	task143	odd-man-out classification generate category
task1431	head qa answer generation	task1432	head qa language translation en to es	task1433	head qa language translation es to en	task1434	head qa classification
task1435	ro sts parallel language translation ro to en	task144	subjqa question answering	task1443	string to number	task1444	round power of two
task1445	closest integers	task1446	farthest integers	task1447	drug extraction ade	task1448	disease entity extraction ncbi dataset
task1449	disease entity extraction bc5cdr dataset	task145	afs argument similarity death penalty	task1451	drug dose extraction	task1452	location entity extraction btc corpus
task1453	person entity extraction btc corpus	task146	afs argument similarity gun control	task147	afs argument similarity gay marriage	task1479	organization entity extraction btc corpus
task148	afs argument quality gay marriage	task1480	gene extraction jnlpba dataset	task1481	gene extraction bc2gm dataset	task1482	gene extraction chemprot dataset
task1483	chemical extraction chemprot dataset	task1484	gene extraction linnaeus dataset	task1485	organ extraction anem dataset	task1486	cell extraction anem dataset
task1487	organism substance extraction anem dataset	task1488	sarcasmdetection headline classification	task1489	sarcasmdetection tweet classification	task149	afs argument quality death penalty
task1490	bengali personal hate speech binary classification	task1491	bengali political hate speech binary classification	task1492	bengali religious hate speech binary classification	task1494	bengali hate speech classification
task1495	adverse drug event classification	task1496	bengali reviews sentiment classification	task1498	24hour to 12hour clock	task150	afs argument quality gun control
task1502	hatexplain classification	task1503	hatexplain classification	task1504	hatexplain answer generation	task1505	root09 semantic relation classification
task1506	celebrity minimal dob span	task1507	boolean temporal reasoning	task1508	wordnet antonyms	task1509	evalution antonyms
task151	tomqa find location easy clean	task1510	evalution relation extraction	task1517	limit classfication	task1518	limit answer generation
task1519	qa srl question generation	task152	tomqa find location easy noise	task1520	qa srl answer generation	task1529	scitail1.1 classification
task153	tomqa find location hard clean	task1533	daily dialog formal classification	task1534	daily dialog question classification	task1537	tamil offenseval dravidian classification
task1538	malayalam offenseval dravidian classification	task1541	agnews classification	task1542	every ith element from starting	task1544	conll2002 named entity recognition answer generation
task1545	conll2002 person name extraction answer generation	task1548	wiqa binary classification	task1549	wiqa answer generation missing step	task155	count nouns verbs
task1551	every ith element from kth element	task1557	jfleg answer generation	task1559	blimp binary classification	task156	codah classification adversarial
task1560	blimp binary classification	task1562	zest text modification	task1564	triviaqa answer generation	task1565	triviaqa classification
task1566	propara structured text generation	task1567	propara question generation	task1568	propara classification	task157	count vowels and consonants
task1572	samsum summary	task1573	samsum classification	task1574	amazon reviews multi language identification	task1575	amazon reviews multi sentiment classification
task1576	amazon reviews multi english language classification	task1577	amazon reviews multi japanese language classification	task158	count frequency of words	task1580	eqasc-perturbed question generation
task1581	eqasc-perturbed answer generation	task1582	bless hypernym generation	task1583	bless meronym classification	task1584	evalution meronym classification
task1585	root09 hypernym generation	task159	check frequency of words in sentence pair	task1590	diplomacy text generation	task1592	yahoo answers topics classfication
task1593	yahoo answers topics classification	task1594	yahoo answers topics question generation	task1595	event2mind text generation 1	task1596	event2mind text generation 2
task1598	nyc long text generation	task1599	smcalflow classification	task160	replace letter in a sentence	task1600	smcalflow sentence generation
task1601	webquestions answer generation	task1602	webquestion question genreation	task1603	smcalflow sentence generation	task1604	ethos text classification
task1605	ethos text classification	task1606	ethos text classification	task1607	ethos text classification	task1608	xquad en answer generation
task1609	xquad en question generation	task161	count words containing letter	task1610	xquad es answer generation	task1616	cc alligned translate eng tel
task1617	cc alligned translate tel eng	task1618	cc alligned classify tel eng	task162	count words starting with letter	task1622	disfl qa text modication
task163	count words ending with letter	task1631	openpi answer generation	task1645	medical question pair dataset text classification	task1646	dataset card for catalonia independence corpus text classification
task1648	opus books en-sv translation	task1649	opus books en-no translation	task1651	opus books en-es translation	task1652	opus books ca-en translation
task1655	mkb translation	task1656	gooaq answer generation	task1657	gooaq question generation	task166	clariq sentence generation
task1660	super glue question generation	task1661	super glue classification	task1662	cedr ru classification	task1665	trainglecopa question generation
task1669	md gender bias text modification	task167	strategyqa question generation	task1670	md gender bias text modification	task1676	xquad-ca translation
task1677	xquad-ca translation	task1678	mathqa answer selection	task168	strategyqa question decomposition	task1685	menyo20k translation
task1686	menyo20k translation	task1689	qed amara translation	task169	strategyqa sentence generation	task1690	qed amara translation
task1691	qed amara translation	task1703	ljspeech textmodification	task1704	ljspeech textmodification	task1705	ljspeech classification
task1706	ljspeech classification	task171	spl translation en es	task1711	poki text generation	task1712	poki classification
task1713	convai3 sentence generation	task1714	convai3 sentence generation	task172	spl translation en fa	task1720	civil comments toxicity classification
task1721	civil comments obscenity classification	task1722	civil comments threat classification	task1723	civil comments sexuallyexplicit classification	task1724	civil comments insult classification
task1725	civil comments severtoxicity classification	task1726	mathqa correct answer generation	task1727	wiqa what is the effect	task1728	web nlg data to text
task1729	personachat generate next	task1731	quartz question answering	task174	spl translation en ja	task175	spl translation en pl
task176	break decompose questions	task177	para-nmt paraphrasing	task178	quartz question answering	task179	participant extraction
task180	intervention extraction	task181	outcome extraction	task183	rhyme generation	task184	break generate question
task190	snli classification	task192	hotpotqa sentence generation	task195	sentiment140 classification	task196	sentiment140 answer generation
task201	mnli neutral classification	task202	mnli contradiction classification	task205	remove even elements	task206	collatz conjecture
task207	max element lists	task208	combinations of list	task209	stancedetection classification	task210	logic2text structured text generation
task211	logic2text classification	task219	rocstories title answer generation	task022	cosmosqa passage inappropriate binary	task223	quartz explanation generation
task227	clariq classification	task228	arc answer generation easy	task229	arc answer generation hard	task023	cosmosqa question generation
task024	cosmosqa answer generation	task243	count elements in set intersection	task244	count elements in set union	task245	check presence in set intersection
task246	dream question generation	task247	dream answer generation	task249	enhanced wsc pronoun disambiguation	task025	cosmosqa incorrect answer generation
task250	spl translation en ar	task251	spl translation en fi	task254	spl translation fi en	task255	spl translation it en
task257	spl translation ar en	task260	spl translation zh en	task261	spl translation es en	task262	spl translation ja en
task263	spl translation pl en	task265	paper reviews language identification	task266	paper reviews reviewer perspective classification	task267	concatenate and reverse all elements from index i to j
task269	csrg counterfactual story generation	task270	csrg counterfactual context generation	task272	europarl translation	task273	europarl classification
task274	overruling legal classification	task275	enhanced wsc paraphrase generation	task276	enhanced wsc classification	task277	stereoset sentence generation stereotype
task278	stereoset sentence generation antistereotype	task279	stereoset classification stereotype	task280	stereoset classification stereotype type	task283	dream incorrect answer generation
task284	imdb classification	task285	imdb answer generation	task286	olid offense judgment	task288	gigaword summarization
task290	tellmewhy question answerability	task291	semeval 2020 task4 commonsense validation	task292	storycommonsense character text generation	task293	storycommonsense emotion text generation
task294	storycommonsense motiv text generation	task295	semeval 2020 task4 commonsense reasoning	task296	storycloze correct end classification	task297	storycloze incorrect end classification
task298	storycloze correct end classification	task299	storycloze sentence generation	task300	storycloze order generation	task304	numeric fused head resolution
task305	jeopardy answer generation normal	task306	jeopardy answer generation double	task307	jeopardy answer generation final	task308	jeopardy answer generation all
task312	europarl sv en translation	task315	europarl sv-en language identification	task316	crows-pairs classification stereotype	task317	crows-pairs classification stereotype type
task318	stereoset classification gender	task319	stereoset classification profession	task320	stereoset classification race	task321	stereoset classification religion
task322	jigsaw classification threat	task323	jigsaw classification sexually explicit	task324	jigsaw classification disagree	task325	jigsaw classification identity attack
task326	jigsaw classification obscene	task327	jigsaw classification toxic	task328	jigsaw classification insult	task329	gap classification
task033	winogrande answer generation	task330	gap answer generation	task333	hateeval classification hate en	task334	hateeval classification hate es
task335	hateeval classification aggresive en	task336	hateeval classification aggresive es	task337	hateeval classification individual en	task034	winogrande question modification object
task340	winomt classification gender pro	task341	winomt classification gender anti	task342	winomt classification profession pro	task343	winomt classification profession anti
task344	hybridqa answer generation	task345	hybridqa answer generation	task346	hybridqa classification	task347	hybridqa incorrect answer generation
task035	winogrande question modification person	task350	winomt classification gender identifiability pro	task351	winomt classification gender identifiability anti	task353	casino classification negotiation elicit pref
task354	casino classification negotiation no need	task355	casino classification negotiation other need	task356	casino classification negotiation self need	task357	casino classification negotiation small talk
task358	casino classification negotiation uv part	task359	casino classification negotiation vouch fair	task362	spolin yesand prompt response sub classification	task363	sst2 polarity classification
task364	regard social impact classification	task365	synthetic remove vowels	task366	synthetic return primes	task367	synthetic remove floats
task368	synthetic even or odd calculation	task369	synthetic remove odds	task370	synthetic remove divisible by 3	task371	synthetic product of list
task372	synthetic palindrome numbers	task373	synthetic round tens place	task374	synthetic pos or neg calculation	task375	classify type of sentence in debate
task376	reverse order of words	task377	remove words of given length	task378	reverse words of given length	task379	agnews topic classification
task380	boolq yes no question	task381	boolq question generation	task382	hybridqa answer generation	task383	matres classification
task384	socialiqa question classification	task385	socialiqa incorrect answer generation	task386	semeval 2018 task3 irony detection	task387	semeval 2018 task3 irony classification
task388	torque token classification	task389	torque generate temporal question	task039	qasc find overlapping words	task390	torque text span selection
task391	causal relationship	task393	plausible result generation	task397	semeval 2018 task1 tweet anger detection	task398	semeval 2018 task1 tweet joy detection
task399	semeval 2018 task1 tweet sadness detection	task400	paws paraphrase classification	task403	creak commonsense inference	task406	mickey fr sentence perturbation generation
task408	mickey it sentence perturbation generation	task409	mickey nl sentence perturbation generation	task410	mickey ru sentence perturbation generation	task411	mickey vi sentence perturbation generation
task413	mickey en sentence perturbation generation	task414	mickey ar sentence perturbation generation	task415	mickey bg sentence perturbation generation	task416	mickey de sentence perturbation generation
task417	mickey es sentence perturbation generation	task424	hindienglish corpora hi en translation	task425	hindienglish corpora en hi translation	task426	hindienglish corpora hi-en classification
task428	senteval inversion	task429	senteval tense	task043	essential terms answering incomplete questions	task430	senteval subject count
task431	senteval object count	task434	alt en hi answer generation	task435	alt en ja translation	task436	alt ja en translation
task437	alt en ja answer generation	task438	eng guj parallel corpus en gu translation	task439	eng guj parallel corpus gu en translation	task044	essential terms identifying essential words
task440	eng guj parallel corpus gu-en classification	task441	eng guj parallel corpus gu-en language identification	task442	com qa paraphrase question generation	task446	opus paracrawl en so translation
task448	opus paracrawl en tl translation	task449	opus paracrawl ig en translation	task045	miscellaneous sentence paraphrasing	task450	opus paracrawl so en translation
task452	opus paracrawl en ig translation	task453	swag answer generation	task454	swag incorrect answer generation	task455	swag context generation
task456	matres intention classification	task457	matres conditional classification	task458	matres negation classification	task459	matres static classification
task046	miscellaneous question typing	task460	qasper answer generation	task461	qasper question generation	task462	qasper classification
task047	miscellaneous answering science questions	task471	haspart answer generation	task472	haspart classification	task475	yelp polarity classification
task477	cls english dvd classification	task483	cls french dvd classification	task488	extract all alphabetical elements from list in order	task489	mwsc question generation
task490	mwsc options generation	task491	mwsc answer generation	task492	mwsc incorrect answer generation	task493	review polarity classification
task494	review polarity answer generation	task495	semeval headline classification	task496	semeval answer generation	task497	extract all numbers from list in order
task499	extract and add all numbers from list	task050	multirc answerability	task504	count all alphabetical elements in list	task505	count all numerical elements in list
task506	position of all alphabetical elements in list	task507	position of all numerical elements in list	task509	collate of all alphabetical and numerical elements in list separately	task512	twitter emotion classification
task513	argument stance classification	task514	argument consequence classification	task515	senteval odd word out	task516	senteval conjoints inversion
task517	emo classify emotion of dialogue	task518	emo different dialogue emotions	task521	trivia question classification	task523	find if numbers or alphabets are more in list
task530	europarl en es translation	task531	europarl es en translation	task532	europarl en-es classification	task533	europarl es-en language identification
task535	alt translation ch en	task537	alt translation th en	task539	alt translation ma en	task541	alt translation kh en
task542	alt translation ja en	task543	alt translation bh en	task547	alt translation entk en	task550	discofuse sentence generation
task551	alt translation en th	task560	alt translation en entk	task561	alt translation en bg	task563	discofuse answer generation
task564	discofuse classification	task565	circa answer generation	task566	circa classification	task567	circa text generation
task568	circa question generation	task573	air dialogue classification	task574	air dialogue sentence generation	task575	air dialogue classification
task576	curiosity dialogs answer generation	task577	curiosity dialogs classification	task579	socialiqa classification	task580	socialiqa answer generation
task581	socialiqa question generation	task582	naturalquestion answer generation	task583	udeps eng coarse pos tagging	task584	udeps eng fine pos tagging
task585	preposition classification	task586	amazonfood polarity classification	task587	amazonfood polarity correction classification	task588	amazonfood rating classification
task589	amazonfood summary text generation	task059	ropes story generation	task590	amazonfood summary correction classification	task591	sciq answer generation
task592	sciq incorrect answer generation	task593	sciq explanation generation	task594	sciq question generation	task595	mocha answer generation
task596	mocha question generation	task600	find the longest common substring in two strings	task601	flores translation sntoen	task604	flores translation entosn
task605	find the longest common subsequence in two lists	task606	sum of all numbers in list between positions i and j	task607	sbic intentional offense binary classification	task608	sbic sexual offense binary classification
task609	sbic potentially offense binary classification	task610	conllpp ner	task614	glucose cause event detection	task615	moviesqa answer generation
task616	cola classification	task617	amazonreview category text generation	task618	amazonreview summary text generation	task619	ohsumed abstract title generation
task062	bigbench repeat copy logic	task620	ohsumed medical subject headings answer generation	task622	replace alphabets in a list by their position in english alphabet	task625	xlwic true or false answer generation
task626	xlwic sentence based on given word sentence generation	task627	xlwic word with same meaning sentence generation	task628	xlwic word with different meaning sentence generation	task629	dbpedia 14 classification
task063	first i elements	task630	dbpedia 14 classification	task631	dbpedia 14 incorrect answer generation	task632	dbpedia 14 classification
task633	dbpedia 14 answer generation	task636	extract and sort unique alphabets in a list	task637	extract and sort unique digits in a list	task638	multi woz classification
task639	multi woz user utterance generation	task064	all elements except first i	task640	esnli classification	task641	esnli classification
task642	esnli classification	task643	refresd classification	task645	summarization	task648	answer generation
task065	timetravel consistent sentence classification	task651	opus100 en ar translation	task652	parsinlu en fa translation	task653	parsinlu fa en translation
task654	bible fa en translation	task657	quran fa en translation	task659	tep fa en translation	task066	timetravel binary consistency classification
task660	mizan fa en translation	task662	global voices fa en translation	task664	mmmlu answer generation abstract algebra	task665	mmmlu answer generation anatomy
task666	mmmlu answer generation astronomy	task667	mmmlu answer generation business ethics	task067	abductivenli answer generation	task670	ambigqa question generation
task671	ambigqa text generation	task672	nummersense	task673	google wellformed query classification	task674	google wellformed query sentence generation
task675	google wellformed query sentence generation	task679	hope edi english text classification	task068	abductivenli incorrect answer generation	task680	hope edi tamil text classification
task681	hope edi malayalam text classification	task682	online privacy policy text classification	task683	online privacy policy text purpose answer generation	task684	online privacy policy text information type generation
task685	mmmlu answer generation clinical knowledge	task686	mmmlu answer generation college biology	task687	mmmlu answer generation college chemistry	task688	mmmlu answer generation college computer science
task689	mmmlu answer generation college mathematics	task069	abductivenli classification	task691	mmmlu answer generation college physics	task692	mmmlu answer generation computer security
task693	mmmlu answer generation conceptual physics	task694	mmmlu answer generation econometrics	task695	mmmlu answer generation electrical engineering	task696	mmmlu answer generation elementary mathematics
task697	mmmlu answer generation formal logic	task698	mmmlu answer generation global facts	task699	mmmlu answer generation high school biology	task070	abductivenli incorrect classification
task700	mmmlu answer generation high school chemistry	task701	mmmlu answer generation high school computer science	task703	mmmlu answer generation high school geography	task704	mmmlu answer generation high school government and politics
task705	mmmlu answer generation high school macroeconomics	task706	mmmlu answer generation high school mathematics	task707	mmmlu answer generation high school microeconomics	task708	mmmlu answer generation high school physics
task709	mmmlu answer generation high school psychology	task071	abductivenli answer generation	task710	mmmlu answer generation high school statistics	task713	mmmlu answer generation human aging
task714	mmmlu answer generation human sexuality	task715	mmmlu answer generation international law	task716	mmmlu answer generation jurisprudence	task717	mmmlu answer generation logical fallacies
task718	mmmlu answer generation machine learning	task719	mmmlu answer generation management	task072	abductivenli answer generation	task720	mmmlu answer generation marketing
task721	mmmlu answer generation medical genetics	task722	mmmlu answer generation random topic	task723	mmmlu answer generation moral disputes	task724	mmmlu answer generation moral scenarios
task725	mmmlu answer generation nutrition	task726	mmmlu answer generation philosophy	task727	mmmlu answer generation prehistory	task728	mmmlu answer generation professional accounting
task073	commonsenseqa answer generation	task731	mmmlu answer generation professional psychology	task732	mmmlu answer generation public relations	task733	mmmlu answer generation security studies
task734	mmmlu answer generation sociology	task735	mmmlu answer generation us foreign policy	task736	mmmlu answer generation virology	task737	mmmlu answer generation world religions
task739	lhoestq question generation	task074	squad1.1 question generation	task740	lhoestq answer generation quantity	task741	lhoestq answer generation place
task742	lhoestq answer generation frequency	task745	ai2 arithmetic questions arithmetic	task746	yelp restaurant review classification	task075	squad1.1 answer generation
task750	aqua multiple choice answering	task751	svamp subtraction question answering	task752	svamp multiplication question answering	task753	svamp addition question answering
task754	svamp common-division question answering	task755	find longest substring and replace its sorted lowercase version in both lists	task756	find longert substring and return all unique alphabets in it	task076	splash correcting sql mistake
task761	app review classification	task762	emea fr sk translation	task764	emea bg el classification	task769	qed summarization
task077	splash explanation to sql	task770	pawsx english text modification	task771	pawsx korean text modification	task772	pawsx french text modification
task774	pawsx german text modification	task775	pawsx chinese text modification	task776	pawsx japanese text modification	task777	pawsx english korean translation
task778	pawsx english french translation	task779	pawsx english spanish translation	task078	all elements except last i	task784	pawsx korean french translation
task785	pawsx korean spanish translation	task787	pawsx korean chinese translation	task788	pawsx korean japanese translation	task789	pawsx french english translation
task079	conala concat strings	task790	pawsx french korean translation	task792	pawsx french german translation	task794	pawsx french japanese translation
task795	pawsx spanish english translation	task796	pawsx spanish korean translation	task797	pawsx spanish french translation	task799	pawsx spanish chinese translation
task080	piqa answer generation	task802	pawsx german korean translation	task803	pawsx german french translation	task804	pawsx german spanish translation
task805	pawsx german chinese translation	task808	pawsx chinese korean translation	task809	pawsx chinese french translation	task081	piqa wrong answer generation
task810	pawsx chinese spanish translation	task811	pawsx chinese german translation	task813	pawsx japanese english translation	task814	pawsx japanese korean translation
task815	pawsx japanese french translation	task817	pawsx japanese german translation	task819	pec sentiment classification	task082	babi t1 single supporting fact question generation
task820	protoqa answer generation	task821	protoqa question generation	task823	peixian-rtgender sentiment analysis	task828	copa commonsense cause effect
task083	babi t1 single supporting fact answer generation	task830	poleval2019 mt translation	task831	giga fren classification	task832	poleval2019 mt classification
task833	poem sentiment classification	task834	mathdataset classification	task835	mathdataset answer generation	task838	cdt classification
task839	cdt classification	task084	babi t1 single supporting fact identify relevant fact	task840	para pdt en es translation	task841	para pdt de en translation
task843	financial phrasebank classification	task844	financial phrasebank classification	task846	pubmedqa classification	task085	unnatural addsub arithmetic
task850	synthetic longest palindrome	task851	synthetic multiply evens	task852	synthetic multiply odds	task856	conv ai 2 classification
task857	inquisitive question generation	task858	inquisitive span detection	task859	prost question generation	task086	translated symbol arithmetic
task860	prost mcq generation	task861	prost mcq answers generation	task862	asdiv multidiv question answering	task863	asdiv multiop question answering
task864	asdiv singleop question answering	task865	mawps addsub question answering	task866	mawps multidiv question answering	task867	mawps multiop question answering
task868	mawps singleop question answering	task087	new operator addsub arithmetic	task872	opus xhosanavy translation eng xhosa	task873	opus xhosanavy translation xhosa eng
task874	opus xhosanavy sr	task875	emotion classification	task877	kde4 translation	task878	kde4 translation
task879	schema guided dstc8 classification	task088	identify typo verification	task888	reviews classification	task889	goemotions classification
task089	swap words verification	task890	gcwd classification	task891	gap coreference resolution	task892	gap reverse coreference resolution
task893	gap fill the blank coreference resolution	task896	miam language classification	task897	freebase qa topic question generation	task898	freebase qa answer generation
task899	freebase qa topic generation	task090	equation learner algebra	task900	freebase qa category classification	task901	freebase qa category question generation
task902	deceptive opinion spam classification	task903	deceptive opinion spam classification	task904	hate speech offensive classification	task905	hate speech offensive classification
task908	dialogre identify familial relationships	task909	dialogre prevalent speakers	task091	all elements from index i to j	task910	bianet classification
task911	bianet translation	task913	bianet translation	task914	bianet translation	task092	check prime classification
task922	event2mind word generation	task923	event2mind classifier	task924	event2mind word generation	task925	coached conv pref classifier
task926	coached conv pref word generation	task927	yelp negative to positive style transfer	task928	yelp positive to negative style transfer	task929	products reviews classification
task093	conala normalize lists	task933	wiki auto style transfer	task934	turk simplification	task936	defeasible nli snli classification
task094	conala calculate mean	task095	conala max absolute value	task955	wiki auto style transfer	task956	leetcode 420 strong password check
task096	conala list index subtraction	task961	ancora-ca-ner text auto completion	task962	ancora-ca-ner missing word prediction	task963	librispeech asr next word prediction
task964	librispeech asr text auto completion	task965	librispeech asr missing word prediction	task966	ruletaker fact checking based on given context	task967	ruletaker incorrect fact generation based on given paragraph
task097	conala remove duplicates	task098	conala list intersection	task982	pib translation tamil bengali	task099	reverse elements between index i and j
task990	pib translation urdu marathi	task991	pib translation english tamil	task995	pib translation bengali english	task996	pib translation english bengali

Table 4: Results for Different Numbers of Clusters (different LoRA-ranks)

Metric	Uncompressed	1 Cluster	2 Clusters	4 Clusters	8 Clusters
Loss	0.5009	1.6023	0.9986	0.5603	0.5000
Exact match	69.6103	37.3392	60.1071	66.6474	69.6289
ROUGE-1	79.5755	50.3598	71.5644	76.2930	79.0576
ROUGE-L	78.9355	49.7491	71.0193	75.6152	78.3958
Recon. error	0	0.8311	0.6990	0.4846	0.3246
Agreement	1.0	36.6706	62.6699	72.2231	80.5545
Exact match ratio	1.0	0.5724	0.8079	0.9219	0.9541
Relative ROUGE-1	1.0	0.6220	0.8533	0.9406	0.9917
Relative ROUGE-L	1.0	0.6221	0.8550	0.9404	0.9915
Param. saved ratio	1.0	0.99	0.98	0.97	0.94

Table 5: Results for Different Numbers of Clusters (same LoRA-ranks of 43)

Metric	Uncompressed	1 Cluster	2 Clusters	4 Clusters	8 Clusters
Loss	0.5764	0.7767	0.6277	0.5841	0.5659
Exact match	61.7746	53.2000	60.2000	61.5000	61.3000
ROUGE-1	79.9322	74.7695	79.4621	80.6950	80.4298
ROUGE-L	77.7961	72.4257	77.2141	78.3369	78.1883
Recon. error	0	0.7333	0.5594	0.3999	0.2502
Agreement	1.0	0.0000	6.6667	6.6667	6.6667
Exact match ratio	1.0	0.8175	0.9261	0.9618	1.0206
Relative ROUGE-1	1.0	0.9560	1.0153	1.0310	1.0241
Relative ROUGE-L	1.0	0.9501	1.0118	1.0268	1.0231
Param. saved ratio	1.0	0.99	0.98	0.97	0.94

Model Type	Method Type	Tasks										Average	Para. Saved
		task039	task190	task280	task290	task391	task442	task620	task1342	task1391	task1598		
	base	0.26 ± 0.00	0.02 ± 0.00	0.19 ± 0.00	0.42 ± 0.00	0.11 ± 0.00	0.47 ± 0.00	0.11 ± 0.00	0.23 ± 0.00	0.19 ± 0.00	0.77 ± 0.00	0.28 ± 0.21	1.00/1.00
	lora	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	0.00/0.00
TIES	10	0.81 ± 0.00	0.57 ± 0.02	0.45 ± 0.04	0.10 ± 0.01	0.83 ± 0.01	0.47 ± 0.00	0.69 ± 0.01	0.57 ± 0.00	0.82 ± 0.01	0.85 ± 0.00	0.62 ± 0.23	1.00 / 1.00
	50	0.59 ± 0.00	0.41 ± 0.00	0.18 ± 0.05	0.03 ± 0.01	0.91 ± 0.01	0.31 ± 0.00	0.65 ± 0.00	0.62 ± 0.00	0.32 ± 0.04	0.84 ± 0.00	0.48 ± 0.28	1.00 / 1.00
	100	0.55 ± 0.00	0.40 ± 0.00	0.20 ± 0.05	0.01 ± 0.02	0.88 ± 0.00	0.33 ± 0.00	0.64 ± 0.00	0.57 ± 0.02	0.01 ± 0.00	0.82 ± 0.00	0.44 ± 0.30	1.00 / 1.00
	500	0.37 ± 0.00	0.26 ± 0.00	0.01 ± 0.00	0.00 ± 0.00	0.83 ± 0.00	0.29 ± 0.00	0.57 ± 0.00	0.37 ± 0.00	0.01 ± 0.00	0.43 ± 0.00	0.31 ± 0.26	1.00 / 1.00
SVD	SVD 2	0.98 ± 0.03	1.07 ± 0.02	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	0.98 ± 0.01	1.00 ± 0.01	1.00 ± 0.10	1.00 ± 0.01	1.00 ± 0.01	1.00 ± 0.04	0.88 / 0.88
	SVD 4	0.99 ± 0.04	1.04 ± 0.01	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.01	1.00 ± 0.00	0.99 ± 0.02	0.99 ± 0.08	0.99 ± 0.01	1.00 ± 0.01	1.00 ± 0.03	0.75 / 0.75
	SVD 8	1.00 ± 0.00	1.02 ± 0.01	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.01 ± 0.00	1.00 ± 0.01	1.01 ± 0.01	1.01 ± 0.00	1.00 ± 0.01	0.50 / 0.50
	SVD 16	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	0.00 / 0.00
10 diagonal (D)	16 D	1.02 ± 0.01	1.01 ± 0.01	1.00 ± 0.00	1.00 ± 0.01	0.99 ± 0.00	0.96 ± 0.00	1.02 ± 0.02	1.13 ± 0.03	0.99 ± 0.02	0.98 ± 0.01	1.01 ± 0.05	1.00 / 0.90
	32 D	1.01 ± 0.01	1.05 ± 0.01	1.00 ± 0.00	0.99 ± 0.00	1.01 ± 0.01	0.99 ± 0.00	0.97 ± 0.01	1.05 ± 0.03	1.00 ± 0.01	1.00 ± 0.01	1.00 ± 0.03	1.00 / 0.80
	64 D	1.00 ± 0.00	1.03 ± 0.01	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.01 ± 0.01	0.99 ± 0.01	1.01 ± 0.00	1.01 ± 0.00	1.00 ± 0.01	1.00 / 0.60
	128 D	1.00 ± 0.00	1.01 ± 0.01	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.01 ± 0.01	0.99 ± 0.01	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.01	1.00 / 0.20
	256 D	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 / -0.60
10 full (F)	16 F	1.02 ± 0.00	1.06 ± 0.01	1.00 ± 0.00	1.00 ± 0.00	0.99 ± 0.01	0.98 ± 0.00	1.01 ± 0.02	1.07 ± 0.00	1.01 ± 0.01	1.00 ± 0.00	1.01 ± 0.03	1.00/0.90
	32 F	1.02 ± 0.01	1.04 ± 0.01	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	0.99 ± 0.00	0.96 ± 0.01	1.00 ± 0.02	1.00 ± 0.01	1.01 ± 0.00	1.00 ± 0.02	0.99/0.79
	64 F	1.00 ± 0.00	1.03 ± 0.01	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.01 ± 0.01	0.98 ± 0.01	1.01 ± 0.00	1.01 ± 0.00	1.00 ± 0.01	0.97/0.57
	128 F	1.00 ± 0.00	1.01 ± 0.01	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	0.99 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	0.88/0.07
	256 F	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	0.50/-1.10
50 diagonal (D)	16 D	0.98 ± 0.04	0.98 ± 0.01	1.00 ± 0.00	0.92 ± 0.06	0.84 ± 0.07	0.92 ± 0.02	0.68 ± 0.05	0.87 ± 0.10	0.88 ± 0.07	0.83 ± 0.02	0.89 ± 0.10	1.00 / 0.98
	32 D	1.00 ± 0.02	1.02 ± 0.02	1.00 ± 0.00	0.99 ± 0.00	0.96 ± 0.01	0.95 ± 0.02	0.84 ± 0.02	1.00 ± 0.13	0.98 ± 0.01	0.88 ± 0.01	0.96 ± 0.07	1.00 / 0.96
	64 D	1.02 ± 0.00	1.05 ± 0.02	1.00 ± 0.00	1.00 ± 0.00	0.99 ± 0.01	0.97 ± 0.00	0.99 ± 0.01	1.09 ± 0.03	1.01 ± 0.01	0.90 ± 0.01	1.00 ± 0.05	1.00 / 0.92
	128 D	1.01 ± 0.01	1.08 ± 0.01	1.00 ± 0.00	1.00 ± 0.00	0.99 ± 0.01	0.98 ± 0.00	0.98 ± 0.01	1.11 ± 0.03	1.00 ± 0.00	1.00 ± 0.01	1.01 ± 0.04	1.00 / 0.84
	256 D	1.01 ± 0.01	1.03 ± 0.01	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.01	1.00 ± 0.00	0.97 ± 0.03	1.01 ± 0.03	1.00 ± 0.01	1.01 ± 0.01	1.00 ± 0.02	1.00 / 0.68
50 full (F)	16 F	0.99 ± 0.04	1.00 ± 0.01	1.00 ± 0.01	0.96 ± 0.01	0.95 ± 0.02	0.94 ± 0.01	0.64 ± 0.10	1.01 ± 0.15	0.97 ± 0.02	0.87 ± 0.00	0.93 ± 0.12	1.00/0.98
	32 F	1.02 ± 0.00	1.00 ± 0.02	1.00 ± 0.00	1.00 ± 0.00	0.98 ± 0.01	0.96 ± 0.00	0.95 ± 0.01	1.09 ± 0.02	1.01 ± 0.02	0.89 ± 0.01	0.99 ± 0.05	0.99/0.95
	64 F	1.02 ± 0.01	1.06 ± 0.02	1.00 ± 0.00	1.00 ± 0.00	0.99 ± 0.01	0.98 ± 0.01	1.03 ± 0.01	1.11 ± 0.00	1.00 ± 0.01	0.98 ± 0.02	1.02 ± 0.04	0.97/0.89
	128 F	1.02 ± 0.00	1.06 ± 0.01	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.01	0.98 ± 0.00	0.98 ± 0.01	1.03 ± 0.04	1.00 ± 0.01	1.00 ± 0.00	1.01 ± 0.03	0.88/0.72
	256 F	1.00 ± 0.00	1.02 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	0.99 ± 0.00	1.01 ± 0.01	1.00 ± 0.00	1.01 ± 0.00	1.01 ± 0.00	1.00 ± 0.01	0.50/0.18
100 diagonal (D)	16 D	0.80 ± 0.07	0.89 ± 0.06	0.93 ± 0.03	0.96 ± 0.01	0.50 ± 0.09	0.78 ± 0.01	0.28 ± 0.07	0.52 ± 0.10	0.78 ± 0.03	0.81 ± 0.02	0.72 ± 0.22	1.00 / 0.99
	32 D	0.95 ± 0.06	0.98 ± 0.01	1.00 ± 0.00	0.91 ± 0.06	0.80 ± 0.14	0.89 ± 0.06	0.60 ± 0.10	0.77 ± 0.26	0.91 ± 0.02	0.83 ± 0.02	0.86 ± 0.14	1.00 / 0.98
	64 D	1.01 ± 0.03	1.01 ± 0.01	1.00 ± 0.00	0.98 ± 0.02	0.96 ± 0.01	0.94 ± 0.01	0.88 ± 0.05	1.11 ± 0.08	0.96 ± 0.02	0.87 ± 0.03	0.97 ± 0.07	1.00 / 0.96
	128 D	1.01 ± 0.00	1.02 ± 0.01	1.00 ± 0.00	1.00 ± 0.00	0.99 ± 0.01	0.97 ± 0.00	1.00 ± 0.03	1.11 ± 0.02	0.99 ± 0.01	0.89 ± 0.02	1.00 ± 0.05	1.00 / 0.92
	256 D	1.00 ± 0.00	1.06 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	0.99 ± 0.00	0.98 ± 0.00	1.00 ± 0.01	1.11 ± 0.03	1.00 ± 0.01	0.98 ± 0.01	1.01 ± 0.04	1.00 / 0.84
100 full (F)	16 F	0.95 ± 0.01	0.97 ± 0.03	0.97 ± 0.03	0.97 ± 0.03	0.93 ± 0.01	0.92 ± 0.01	0.64 ± 0.03	0.89 ± 0.16	0.87 ± 0.02	0.83 ± 0.01	0.89 ± 0.11	1.00/0.99
	32 F	1.00 ± 0.02	0.99 ± 0.01	1.00 ± 0.00	1.00 ± 0.00	0.97 ± 0.01	0.95 ± 0.00	0.86 ± 0.03	1.12 ± 0.03	0.96 ± 0.01	0.87 ± 0.00	0.97 ± 0.07	0.99/0.97
	64 F	1.02 ± 0.00	1.00 ± 0.02	1.00 ± 0.00	1.00 ± 0.00	0.98 ± 0.00	0.96 ± 0.00	0.99 ± 0.01	1.09 ± 0.01	0.99 ± 0.02	0.89 ± 0.00	0.99 ± 0.05	0.97/0.93
	128 F	1.01 ± 0.01	1.05 ± 0.01	1.00 ± 0.00	0.99 ± 0.00	1.00 ± 0.00	0.98 ± 0.00	1.03 ± 0.01	1.10 ± 0.01	1.01 ± 0.00	0.99 ± 0.01	1.02 ± 0.04	0.88/0.80
	256 F	1.01 ± 0.01	1.03 ± 0.01	1.00 ± 0.00	1.00 ± 0.00	1.01 ± 0.00	0.99 ± 0.00	0.98 ± 0.00	1.00 ± 0.03	1.01 ± 0.00	1.01 ± 0.00	1.00 ± 0.01	0.50/0.34
100 w/clusters (C)	16 C 5	1.13 ± 0.01	1.03 ± 0.01	1.00 ± 0.00	1.00 ± 0.00	0.99 ± 0.01	0.96 ± 0.00	1.01 ± 0.02	1.23 ± 0.02	1.05 ± 0.01	0.99 ± 0.06	1.04 ± 0.08	1.00/0.95
	16 C 7	1.12 ± 0.01	1.01 ± 0.01	1.00 ± 0.00	1.00 ± 0.00	0.99 ± 0.01	0.96 ± 0.01	1.02 ± 0.02	1.24 ± 0.05	1.03 ± 0.01	0.99 ± 0.05	1.04 ± 0.08	1.00/0.93
500 diagonal (D)	16 D	0.57 ± 0.07	0.55 ± 0.03	0.83 ± 0.04	0.78 ± 0.16	0.85 ± 0.04	0.68 ± 0.07	0.24 ± 0.01	0.43 ± 0.01	0.76 ± 0.06	0.79 ± 0.01	0.65 ± 0.20	1.00 / 1.00
	32 D	0.61 ± 0.12	0.55 ± 0.08	0.83 ± 0.02	0.84 ± 0.12	0.91 ± 0.02	0.71 ± 0.05	0.29 ± 0.05	0.47 ± 0.08	0.79 ± 0.04	0.79 ± 0.01	0.68 ± 0.20	1.00 / 1.00
	64 D	0.73 ± 0.02	0.63 ± 0.11	0.89 ± 0.04	0.97 ± 0.00	0.94 ± 0.00	0.83 ± 0.05	0.45 ± 0.09	0.50 ± 0.07	0.82 ± 0.02	0.80 ± 0.02	0.76 ± 0.18	1.00 / 0.99
	128 D	0.84 ± 0.00	0.92 ± 0.02	0.97 ± 0.03	0.98 ± 0.01	0.94 ± 0.00	0.88 ± 0.02	0.60 ± 0.15	0.53 ± 0.01	0.85 ± 0.05	0.80 ± 0.02	0.83 ± 0.15	1.00 / 0.98
	256 D	0.99 ± 0.03	0.99 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	0.96 ± 0.00	0.92 ± 0.03	0.66 ± 0.06	0.84 ± 0.14	0.92 ± 0.02	0.84 ± 0.01	0.91 ± 0.11	1.00 / 0.97
500 full (F)	16 F	0.57 ± 0.01	0.43 ± 0.07	0.78 ± 0.01	0.97 ± 0.00	0.96 ± 0.00	0.83 ± 0.01	0.64 ± 0.00	0.53 ± 0.03	0.83 ± 0.01	0.83 ± 0.00	0.75 ± 0.17	1.00/1.00
	32 F	0.79 ± 0.05	0.54 ± 0.04	0.93 ± 0.02	0.98 ± 0.00	0.97 ± 0.00	0.90 ± 0.01	0.69 ± 0.01	0.50 ± 0.00	0.86 ± 0.02	0.83 ± 0.01	0.81 ± 0.16	0.99/0.99
	64 F	1.02 ± 0.00	0.96 ± 0.01	0.94 ± 0.01	1.00 ± 0.01	0.96 ± 0.00	0.97 ± 0.01	0.73 ± 0.01	0.54 ± 0.01	0.91 ± 0.01	0.86 ± 0.00	0.89 ± 0.14	0.97/0.96
	128 F	1.03 ± 0.01	0.97 ± 0.02	0.99 ± 0.00	1.00 ± 0.00	0.98 ± 0.00	0.96 ± 0.00	0.87 ± 0.01	1.07 ± 0.02	0.98 ± 0.00	0.87 ± 0.00	0.97 ± 0.06	0.88/0.86
	256 F	1.03 ± 0.00	1.03 ± 0.01	1.00 ± 0.00	1.00 ± 0.00	0.99 ± 0.01	0.97 ± 0.01	0.99 ± 0.02	1.03 ± 0.01	1.00 ± 0.01	0.87 ± 0.00	0.99 ± 0.05	0.50/0.47
500 w/clusters (C)	16 C 7	1.09	1.00	0.99	1.00	0.98	0.95	0.72	0.87	0.98	0.90	0.95	1.00/0.98
	16 C 10	1.10	1.01	1.00	0.99	0.97	0.93	0.70	1.30	1.02	0.88	0.99	1.00/0.98
	16 C 25	1.10	1.00	1.00	0.99	0.99	0.96	0.98	1.31	1.03	0.91	1.03	1.00/0.95
	64 C 5	1.09	0.98	1.00	1.00	0.99	0.96	0.99	1.18	1.04	0.87	1.01	0.97/0.93
	64 C 7	1.12	1.02	1.00	1.00	1.00	0.96	0.99	1.22	1.04	0.93	1.03	0.97/0.91
1000 w/clusters (C)	16 C 25	1.09	0.98	1.00	1.00	0.97	0.96	0.72	1.30	1.05	0.91	1.00	1.00/0.97

Table 7: **Relative** In-Distribution *ROUGE-L* scores for various tasks and methods

Model Type	Method Type	Tasks										Average	Para. Saved
		task039	task190	task280	task290	task391	task442	task620	task1342	task1391	task1598		
	base	24.44 ± 0.00	1.60 ± 0.00	19.13 ± 0.00	39.22 ± 0.00	10.27 ± 0.00	35.46 ± 0.00	7.85 ± 0.00	6.22 ± 0.00	17.82 ± 0.00	38.87 ± 0.00	20.24 ± 13.27	1.00/1.00
	lora	95.00 ± 0.00	86.00 ± 0.00	99.00 ± 0.00	93.67 ± 0.00	94.33 ± 0.00	74.88 ± 0.00	74.40 ± 0.00	26.68 ± 0.00	95.00 ± 0.00	50.32 ± 0.00	78.87 ± 22.56	0.00/0.00
TIES	10	76.50 ± 0.00	49.00 ± 1.73	44.33 ± 4.04	9.80 ± 0.58	78.56 ± 0.96	35.24 ± 0.00	51.37 ± 0.67	15.26 ± 0.12	77.67 ± 1.15	42.72 ± 0.01	48.05 ± 23.61	1.00 / 1.00
	50	55.80 ± 0.00	35.00 ± 0.00	18.00 ± 5.20	2.42 ± 0.50	85.78 ± 0.96	23.03 ± 0.00	48.03 ± 0.00	16.50 ± 0.00	30.00 ± 3.46	42.47 ± 0.02	35.70 ± 23.01	1.00 / 1.00
	100	52.43 ± 0.00	34.00 ± 0.00	19.67 ± 4.62	1.09 ± 1.66	83.33 ± 0.00	24.89 ± 0.00	47.52 ± 0.00	15.18 ± 0.42	1.00 ± 0.00	41.19 ± 0.03	32.03 ± 24.50	1.00 / 1.00
	500	35.18 ± 0.00	22.00 ± 0.00	1.00 ± 0.00	0.00 ± 0.00	78.00 ± 0.00	21.46 ± 0.00	42.22 ± 0.04	9.93 ± 0.13	1.00 ± 0.00	21.50 ± 0.03	23.27 ± 23.64	1.00 / 1.00
SVD	SVD 2	93.15 ± 2.77	92.24 ± 1.85	99.09 ± 0.18	93.44 ± 0.14	93.89 ± 0.35	73.74 ± 0.51	74.55 ± 0.98	26.80 ± 2.79	95.06 ± 1.35	50.21 ± 0.44	79.11 ± 22.72	0.88 / 0.88
	SVD 4	94.01 ± 3.60	89.21 ± 0.71	99.05 ± 0.09	93.65 ± 0.03	94.66 ± 0.63	74.89 ± 0.33	73.61 ± 1.15	26.34 ± 2.13	93.98 ± 0.77	50.47 ± 0.54	78.90 ± 22.68	0.75 / 0.75
	SVD 8	95.00 ± 0.00	87.40 ± 0.59	99.05 ± 0.09	93.65 ± 0.03	94.36 ± 0.38	74.58 ± 0.12	75.07 ± 0.00	26.71 ± 0.27	95.51 ± 1.09	50.89 ± 0.07	81.01 ± 21.74	0.50 / 0.50
	SVD 16	95.00 ± 0.00	86.00 ± 0.00	99.00 ± 0.00	93.67 ± 0.00	94.33 ± 0.00	74.90 ± 0.03	74.23 ± 0.18	26.68 ± 0.00	95.00 ± 0.00	50.30 ± 0.02	78.36 ± 22.97	0.00 / 0.00
10 diagonal (D)	16 D	96.67 ± 0.58	87.00 ± 1.00	99.00 ± 0.00	94.00 ± 0.67	93.11 ± 0.38	72.08 ± 0.06	76.26 ± 1.19	30.11 ± 0.79	94.00 ± 1.73	49.30 ± 0.46	79.15 ± 22.18	1.00 / 0.90
	32 D	95.67 ± 0.58	90.00 ± 1.00	99.00 ± 0.00	93.00 ± 0.33	94.89 ± 0.51	73.86 ± 0.31	71.92 ± 0.84	27.89 ± 0.70	94.67 ± 0.58	50.36 ± 0.26	79.13 ± 22.75	1.00 / 0.80
	64 D	95.00 ± 0.00	88.33 ± 0.58	99.00 ± 0.00	93.67 ± 0.00	94.78 ± 0.38	74.61 ± 0.13	74.97 ± 0.58	26.35 ± 0.25	96.00 ± 0.00	50.99 ± 0.06	79.37 ± 22.94	1.00 / 0.60
	128 D	95.00 ± 0.00	86.67 ± 0.58	99.00 ± 0.00	93.67 ± 0.00	94.33 ± 0.00	74.92 ± 0.13	74.96 ± 0.51	26.45 ± 0.23	95.00 ± 0.00	50.21 ± 0.12	79.02 ± 22.84	1.00 / 0.20
	256 D	95.00 ± 0.00	86.00 ± 0.00	99.00 ± 0.00	93.67 ± 0.00	94.33 ± 0.00	74.88 ± 0.00	74.40 ± 0.00	26.68 ± 0.00	95.00 ± 0.00	50.27 ± 0.02	78.92 ± 22.77	1.00 / -0.60
10 full (F)	16 F	97.00 ± 0.00	91.00 ± 1.00	99.00 ± 0.00	93.56 ± 0.19	93.56 ± 0.69	73.60 ± 0.36	74.94 ± 1.25	28.66 ± 0.03	96.00 ± 1.00	50.15 ± 0.20	79.75 ± 22.72	1.00/0.90
	32 F	96.67 ± 0.58	89.33 ± 0.58	99.00 ± 0.00	93.22 ± 0.19	94.44 ± 0.19	74.11 ± 0.19	71.74 ± 0.59	26.74 ± 0.50	94.67 ± 0.58	50.63 ± 0.24	79.06 ± 23.01	0.99/0.79
	64 F	95.00 ± 0.00	88.67 ± 0.58	99.00 ± 0.00	93.67 ± 0.00	94.56 ± 0.38	74.56 ± 0.13	75.47 ± 0.58	26.26 ± 0.34	96.00 ± 0.00	50.89 ± 0.17	79.41 ± 22.97	0.97/0.57
	128 F	95.00 ± 0.00	86.67 ± 0.58	99.00 ± 0.00	93.67 ± 0.00	94.33 ± 0.00	75.04 ± 0.03	74.40 ± 0.00	26.53 ± 0.13	95.00 ± 0.00	50.36 ± 0.03	79.00 ± 22.81	0.88/0.07
	256 F	95.00 ± 0.00	86.00 ± 0.00	99.00 ± 0.00	93.67 ± 0.00	94.33 ± 0.00	74.90 ± 0.03	74.29 ± 0.19	26.68 ± 0.00	95.00 ± 0.00	50.30 ± 0.03	78.92 ± 22.77	0.50/-1.10
50 diagonal (D)	16 D	92.76 ± 3.53	84.67 ± 1.15	99.00 ± 0.00	86.17 ± 5.81	79.68 ± 6.21	69.07 ± 1.54	50.65 ± 3.97	23.27 ± 2.60	83.90 ± 6.43	41.86 ± 0.96	71.10 ± 23.99	1.00 / 0.98
	32 D	95.33 ± 2.08	87.33 ± 2.08	99.00 ± 0.00	92.60 ± 0.29	90.32 ± 1.04	71.16 ± 1.47	62.51 ± 1.64	26.60 ± 3.54	93.33 ± 1.15	44.35 ± 0.41	76.25 ± 23.81	1.00 / 0.96
	64 D	97.00 ± 0.00	90.33 ± 1.53	99.00 ± 0.00	93.78 ± 0.19	93.00 ± 0.58	72.37 ± 0.35	73.39 ± 0.93	29.06 ± 0.80	95.67 ± 0.58	45.43 ± 0.34	78.90 ± 23.29	1.00 / 0.92
	128 D	96.33 ± 0.58	92.67 ± 0.58	99.00 ± 0.00	93.56 ± 0.19	93.00 ± 0.58	73.32 ± 0.24	73.03 ± 1.09	29.51 ± 0.93	95.00 ± 0.00	50.16 ± 0.74	79.56 ± 22.51	1.00 / 0.84
	256 D	95.67 ± 0.58	88.33 ± 0.58	99.00 ± 0.00	93.56 ± 0.19	94.67 ± 0.67	74.82 ± 0.24	72.36 ± 2.07	26.90 ± 0.75	95.33 ± 0.58	50.73 ± 0.46	79.14 ± 22.90	1.00 / 0.68
50 full (F)	16 F	94.06 ± 3.54	85.67 ± 1.15	98.67 ± 0.58	90.35 ± 1.37	89.90 ± 1.91	70.32 ± 0.66	47.62 ± 7.28	26.88 ± 3.96	92.33 ± 1.53	43.68 ± 0.24	73.95 ± 24.73	1.00/0.98
	32 F	97.00 ± 0.00	85.67 ± 1.53	99.00 ± 0.00	93.67 ± 0.00	92.22 ± 0.69	71.88 ± 0.30	71.01 ± 1.02	29.07 ± 0.65	95.67 ± 1.53	44.97 ± 0.41	78.02 ± 23.18	0.99/0.95
	64 F	96.67 ± 0.58	91.00 ± 2.00	99.00 ± 0.00	93.56 ± 0.19	93.22 ± 0.51	73.16 ± 0.41	76.28 ± 0.51	29.67 ± 0.12	95.33 ± 0.58	49.31 ± 1.00	79.72 ± 22.50	0.97/0.89
	128 F	97.00 ± 0.00	91.00 ± 1.00	99.00 ± 0.00	93.33 ± 0.00	94.11 ± 0.51	73.51 ± 0.23	73.17 ± 0.58	27.53 ± 1.12	95.00 ± 1.00	50.56 ± 0.06	79.42 ± 22.93	0.88/0.72
	256 F	95.00 ± 0.00	88.00 ± 0.00	99.00 ± 0.00	93.67 ± 0.00	94.44 ± 0.19	74.25 ± 0.21	74.97 ± 0.58	26.79 ± 0.09	96.00 ± 0.00	50.86 ± 0.19	79.30 ± 22.82	0.50/0.18
100 diagonal (D)	16 D	76.43 ± 7.07	76.67 ± 4.93	91.61 ± 2.75	89.99 ± 1.07	47.55 ± 8.56	58.08 ± 0.72	20.77 ± 5.50	13.90 ± 2.79	73.93 ± 3.13	40.74 ± 0.85	58.97 ± 26.83	1.00 / 0.99
	32 D	90.10 ± 5.85	84.00 ± 1.00	99.00 ± 0.00	85.52 ± 5.34	75.69 ± 12.75	66.62 ± 4.18	44.66 ± 7.26	20.49 ± 7.07	86.67 ± 1.86	42.01 ± 0.94	69.48 ± 25.14	1.00 / 0.98
	64 D	95.56 ± 2.49	86.67 ± 0.58	99.00 ± 0.00	92.24 ± 1.68	90.89 ± 1.17	70.35 ± 0.45	65.62 ± 4.03	29.58 ± 2.02	91.67 ± 2.31	43.64 ± 1.36	76.52 ± 23.02	1.00 / 0.96
	128 D	96.00 ± 0.00	87.33 ± 1.15	99.00 ± 0.00	93.89 ± 0.19	93.00 ± 0.58	72.70 ± 0.30	74.34 ± 2.07	29.66 ± 0.54	93.67 ± 0.58	44.82 ± 0.89	78.44 ± 22.87	1.00 / 0.92
	256 D	95.00 ± 0.00	91.00 ± 0.00	99.00 ± 0.00	93.56 ± 0.19	93.11 ± 0.19	73.05 ± 0.20	74.52 ± 0.95	29.67 ± 0.67	95.33 ± 0.58	49.42 ± 0.65	79.37 ± 22.38	1.00 / 0.84
100 full (F)	16 F	90.70 ± 1.07	83.00 ± 2.65	96.00 ± 3.00	91.22 ± 2.94	87.94 ± 0.54	68.72 ± 1.05	47.57 ± 2.54	23.75 ± 4.33	82.33 ± 2.08	41.51 ± 0.67	71.27 ± 24.23	1.00/0.99
	32 F	95.33 ± 1.53	85.00 ± 1.00	99.00 ± 0.00	93.50 ± 0.22	91.44 ± 0.84	70.94 ± 0.02	63.64 ± 1.98	29.82 ± 0.81	91.67 ± 0.58	43.94 ± 0.18	76.43 ± 23.01	0.99/0.97
	64 F	97.00 ± 0.00	85.67 ± 1.53	99.00 ± 0.00	93.78 ± 0.19	92.56 ± 0.19	72.11 ± 0.08	73.29 ± 0.64	29.15 ± 0.24	94.33 ± 1.53	44.97 ± 0.05	78.18 ± 23.03	0.97/0.93
	128 F	96.33 ± 0.58	90.33 ± 0.58	99.00 ± 0.00	93.00 ± 0.00	93.89 ± 0.19	73.11 ± 0.36	76.50 ± 1.01	29.45 ± 0.35	96.00 ± 0.00	49.81 ± 0.34	79.74 ± 22.47	0.88/0.80
	256 F	96.33 ± 0.58	88.67 ± 0.58	99.00 ± 0.00	93.67 ± 0.00	94.89 ± 0.19	74.40 ± 0.16	72.90 ± 0.12	26.77 ± 0.68	96.00 ± 0.00	50.83 ± 0.09	79.35 ± 23.04	0.50/0.34
100 w/clusters (C)	16 C 5	98.33 ± 0.47	89.00 ± 0.82	99.00 ± 0.00	93.25 ± 0.40	92.89 ± 0.87	72.32 ± 0.36	77.08 ± 1.67	28.26 ± 0.38	96.67 ± 0.47	68.30 ± 15.72	81.51 ± 20.63	1.00/0.95
	16 C 7	97.67 ± 0.47	87.00 ± 0.82	99.00 ± 0.00	93.46 ± 0.29	93.11 ± 0.68	72.52 ± 0.43	77.66 ± 1.30	28.51 ± 1.26	95.33 ± 0.47	68.46 ± 14.94	81.27 ± 20.35	1.00/0.93
500 diagonal (D)	16 D	54.44 ± 6.87	47.00 ± 2.83	82.21 ± 3.59	73.38 ± 14.97	80.08 ± 3.71	51.02 ± 5.31	17.49 ± 1.10	11.58 ± 0.21	72.67 ± 6.03	39.65 ± 0.28	53.16 ± 24.97	1.00 / 1.00
	32 D	58.08 ± 11.52	47.00 ± 7.07	82.06 ± 1.69	78.62 ± 11.23	85.57 ± 1.48	52.98 ± 3.81	21.73 ± 3.95	12.53 ± 2.26	75.33 ± 4.04	39.78 ± 0.42	55.66 ± 25.48	1.00 / 1.00
	64 D	69.21 ± 2.03	54.50 ± 9.19	88.33 ± 4.04	91.11 ± 0.38	88.78 ± 0.38	62.36 ± 3.52	33.36 ± 6.69	13.34 ± 1.86	77.67 ± 2.31	40.42 ± 0.98	62.16 ± 26.05	1.00 / 0.99
	128 D	79.77 ± 0.37	79.50 ± 2.12	95.89 ± 2.83	91.89 ± 1.39	88.67 ± 0.00	65.92 ± 1.79	44.98 ± 10.98	14.14 ± 0.19	81.00 ± 5.00	40.34 ± 0.80	67.82 ± 26.35	1.00 / 0.98
	256 D	93.83 ± 2.52	85.00 ± 0.00	99.00 ± 0.00	93.78 ± 0.19	90.56 ± 0.38	68.95 ± 1.92	49.39 ± 4.36	22.33 ± 3.78	87.33 ± 2.31	42.15 ± 0.73	72.83 ± 25.93	1.00 / 0.97
500 full (F)	16 F	54.30 ± 1.13	37.00 ± 5.66	77.67 ± 0.58	91.00 ± 0.00	90.56 ± 0.19	62.47 ± 0.79	47.56 ± 0.29	14.18 ± 0.67	79.00 ± 1.00	41.58 ± 0.23	60.31 ± 24.42	1.00/1.00
	32 F	75.10 ± 4.92	46.50 ± 3.54	91.67 ± 1.53	91.56 ± 0.19	91.56 ± 0.38	67.37 ± 0.83	51.17 ± 0.81	13.44 ± 0.02	81.67 ± 1.53	41.92 ± 0.42	65.84 ± 25.64	0.99/0.99
	64 F	96.94 ± 0.42	82.50 ± 0.71	93.33 ± 0.58	93.89 ± 0.69	90.67 ± 0.00	72.30 ± 0.71	54.63 ± 0.79	14.49 ± 0.27	86.33 ± 0.58	43.16 ± 0.08	72.49 ± 26.64	0.97/0.96
	128 F	97.67 ± 0.58	83.50 ± 2.12	98.00 ± 0.00	93.56 ± 0.19	92.00 ± 0.00	71.92 ± 0.19	65.02 ± 0.81	28.49 ± 0.55	93.00 ± 0.00	43.85 ± 0.12	76.47 ± 23.77	0.88/0.86
	256 F	98.00 ± 0.00	88.50 ± 0.71	99.00 ± 0.00	93.78 ± 0.19	93.00 ± 0.88	72.45 ± 0.38	73.77 ± 1.21	27.59 ± 0.39	95.33 ± 0.58	43.81 ± 0.17	78.18 ± 24.16	0.50/0.47
500 w/clusters (C)	16 C 7	95.00	86.00	98.00	93.67	91.67	71.19	54.69	20.03	90.00	46.34	74.66	1.00/0.98
	16 C 10	96.00	87.00	99.00	93.00	91.33	69.93	53.48	30.09	94.00	44.89	75.87	1.00/0.98
	16 C 25	96.00	86.00	99.00	92.71	93.00	72.13	74.59	30.21	95.00	46.66	78.53	1.00/0.95
	64 C 5	95.00	84.00	99.00	93.67	92.67	72.32	75.60	27.17	96.00	44.43	77.99	0.97/0.93
	64 C 7	98.00	88.00	99.00	94.00	93.33	72.18	75.83	28.14	96.00	47.68	79.22	0.97/0.91
1000 w/clusters (C)	16 C 25	95.00	84.00	99.00	93.67	90.67	72.20	55.04	29.97	97.00	46.86	76.34	1.00/0.97

Table 8: **Absolute** In-Distribution ROUGE-L scores for various tasks and methods

Model Type	Method Type	Tasks										Average	Para. Saved
		task039	task190	task280	task290	task391	task442	task620	task1342	task1391	task1598		
	base	0.26 ± 0.00	0.02 ± 0.00	0.19 ± 0.00	0.42 ± 0.00	0.11 ± 0.00	0.51 ± 0.00	0.11 ± 0.00	0.26 ± 0.00	0.19 ± 0.00	0.80 ± 0.00	0.29 ± 0.22	1.00/1.00
	lora	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	0.00/0.00
TIES	10	0.81 ± 0.00	0.57 ± 0.02	0.45 ± 0.04	0.10 ± 0.01	0.83 ± 0.01	0.52 ± 0.00	0.71 ± 0.01	0.58 ± 0.00	0.82 ± 0.01	0.80 ± 0.00	0.62 ± 0.22	1.00 / 1.00
	50	0.59 ± 0.00	0.41 ± 0.00	0.18 ± 0.05	0.03 ± 0.01	0.91 ± 0.01	0.34 ± 0.00	0.67 ± 0.00	0.62 ± 0.00	0.32 ± 0.04	0.78 ± 0.00	0.48 ± 0.27	1.00 / 1.00
	100	0.55 ± 0.00	0.40 ± 0.00	0.20 ± 0.05	0.01 ± 0.02	0.88 ± 0.00	0.36 ± 0.00	0.65 ± 0.00	0.57 ± 0.02	0.01 ± 0.00	0.78 ± 0.00	0.44 ± 0.29	1.00 / 1.00
	500	0.37 ± 0.00	0.26 ± 0.00	0.01 ± 0.00	0.00 ± 0.00	0.83 ± 0.00	0.31 ± 0.00	0.58 ± 0.00	0.37 ± 0.00	0.01 ± 0.00	0.41 ± 0.00	0.32 ± 0.26	1.00 / 1.00
SVD	SVD 2	0.98 ± 0.03	1.07 ± 0.02	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	0.99 ± 0.00	1.01 ± 0.01	1.00 ± 0.10	1.00 ± 0.01	0.99 ± 0.01	1.00 ± 0.04	0.88 / 0.88
	SVD 4	0.99 ± 0.04	1.04 ± 0.01	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.01	1.00 ± 0.00	0.99 ± 0.01	0.99 ± 0.08	0.99 ± 0.01	1.01 ± 0.00	1.00 ± 0.03	0.75 / 0.75
	SVD 8	1.00 ± 0.00	1.02 ± 0.01	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.01 ± 0.00	1.00 ± 0.01	1.01 ± 0.01	1.01 ± 0.00	1.00 ± 0.01	0.50 / 0.50
	SVD 16	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	0.00 / 0.00
10 diagonal (D)	16 D	1.02 ± 0.01	1.01 ± 0.01	1.00 ± 0.00	1.00 ± 0.01	0.99 ± 0.00	0.97 ± 0.00	1.03 ± 0.02	1.12 ± 0.03	0.99 ± 0.02	0.99 ± 0.00	1.01 ± 0.04	1.00 / 0.90
	32 D	1.01 ± 0.01	1.05 ± 0.01	1.00 ± 0.00	0.99 ± 0.00	1.01 ± 0.01	0.99 ± 0.00	0.97 ± 0.01	1.04 ± 0.03	1.00 ± 0.01	1.01 ± 0.01	1.01 ± 0.02	1.00 / 0.80
	64 D	1.00 ± 0.00	1.03 ± 0.01	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.01 ± 0.01	0.99 ± 0.01	1.01 ± 0.00	1.01 ± 0.00	1.00 ± 0.01	1.00 / 0.60
	128 D	1.00 ± 0.00	1.01 ± 0.01	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.01 ± 0.01	0.99 ± 0.01	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.01	1.00 / 0.20
	256 D	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 / -0.60
10 full (F)	16 F	1.02 ± 0.00	1.06 ± 0.01	1.00 ± 0.00	1.00 ± 0.00	0.99 ± 0.01	0.99 ± 0.00	1.01 ± 0.02	1.07 ± 0.00	1.01 ± 0.01	1.00 ± 0.00	1.02 ± 0.03	1.00/0.90
	32 F	1.02 ± 0.01	1.04 ± 0.01	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	0.99 ± 0.00	0.96 ± 0.01	1.00 ± 0.02	1.00 ± 0.01	1.01 ± 0.00	1.00 ± 0.02	0.99/0.79
	64 F	1.00 ± 0.00	1.03 ± 0.01	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.01 ± 0.01	0.98 ± 0.01	1.01 ± 0.00	1.01 ± 0.00	1.00 ± 0.01	0.97/0.57
	128 F	1.00 ± 0.00	1.01 ± 0.01	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	0.99 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	0.88/0.07
	256 F	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	0.50/-1.10
50 diagonal (D)	16 D	0.98 ± 0.04	0.98 ± 0.01	1.00 ± 0.00	0.92 ± 0.06	0.85 ± 0.06	0.94 ± 0.02	0.69 ± 0.05	0.88 ± 0.10	0.88 ± 0.07	0.86 ± 0.01	0.90 ± 0.10	1.00 / 0.98
	32 D	1.00 ± 0.02	1.02 ± 0.02	1.00 ± 0.00	0.99 ± 0.00	0.96 ± 0.01	0.96 ± 0.02	0.85 ± 0.02	1.00 ± 0.12	0.98 ± 0.01	0.90 ± 0.00	0.97 ± 0.06	1.00 / 0.96
	64 D	1.02 ± 0.00	1.05 ± 0.02	1.00 ± 0.00	1.00 ± 0.00	0.99 ± 0.01	0.97 ± 0.01	0.99 ± 0.01	1.09 ± 0.03	1.01 ± 0.01	0.94 ± 0.00	1.01 ± 0.04	1.00 / 0.92
	128 D	1.01 ± 0.01	1.08 ± 0.01	1.00 ± 0.00	1.00 ± 0.00	0.99 ± 0.01	0.98 ± 0.00	0.98 ± 0.02	1.10 ± 0.03	1.00 ± 0.00	1.01 ± 0.01	1.02 ± 0.04	1.00 / 0.84
	256 D	1.01 ± 0.01	1.03 ± 0.01	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.01	1.00 ± 0.00	0.97 ± 0.03	1.00 ± 0.03	1.00 ± 0.01	1.01 ± 0.00	1.00 ± 0.02	1.00 / 0.68
50 full (F)	16 F	0.99 ± 0.04	1.00 ± 0.01	1.00 ± 0.01	0.96 ± 0.01	0.95 ± 0.02	0.95 ± 0.01	0.65 ± 0.09	1.01 ± 0.15	0.97 ± 0.02	0.88 ± 0.01	0.94 ± 0.11	1.00/0.98
	32 F	1.02 ± 0.00	1.00 ± 0.02	1.00 ± 0.00	1.00 ± 0.00	0.98 ± 0.01	0.97 ± 0.00	0.96 ± 0.01	1.09 ± 0.03	1.01 ± 0.02	0.93 ± 0.00	0.99 ± 0.04	0.99/0.95
	64 F	1.02 ± 0.01	1.06 ± 0.02	1.00 ± 0.00	1.00 ± 0.00	0.99 ± 0.01	0.98 ± 0.00	1.03 ± 0.01	1.11 ± 0.00	1.00 ± 0.01	0.99 ± 0.01	1.02 ± 0.04	0.97/0.89
	128 F	1.02 ± 0.00	1.06 ± 0.01	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.01	0.98 ± 0.00	0.98 ± 0.01	1.03 ± 0.04	1.00 ± 0.01	1.01 ± 0.00	1.01 ± 0.02	0.88/0.72
	256 F	1.00 ± 0.00	1.02 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	0.99 ± 0.00	1.01 ± 0.01	1.00 ± 0.00	1.01 ± 0.00	1.01 ± 0.00	1.00 ± 0.01	0.50/0.18
100 diagonal (D)	16 D	0.80 ± 0.07	0.89 ± 0.06	0.93 ± 0.03	0.96 ± 0.01	0.51 ± 0.09	0.81 ± 0.02	0.30 ± 0.07	0.54 ± 0.11	0.78 ± 0.03	0.83 ± 0.02	0.73 ± 0.21	1.00 / 0.99
	32 D	0.95 ± 0.06	0.98 ± 0.01	1.00 ± 0.00	0.91 ± 0.06	0.80 ± 0.13	0.91 ± 0.05	0.62 ± 0.10	0.78 ± 0.25	0.91 ± 0.02	0.85 ± 0.01	0.87 ± 0.14	1.00 / 0.98
	64 D	1.01 ± 0.03	1.01 ± 0.01	1.00 ± 0.00	0.98 ± 0.02	0.96 ± 0.01	0.95 ± 0.01	0.90 ± 0.05	1.11 ± 0.07	0.96 ± 0.02	0.88 ± 0.02	0.98 ± 0.07	1.00 / 0.96
	128 D	1.01 ± 0.00	1.02 ± 0.01	1.00 ± 0.00	1.00 ± 0.00	0.99 ± 0.01	0.98 ± 0.00	1.00 ± 0.03	1.11 ± 0.02	0.99 ± 0.01	0.92 ± 0.00	1.00 ± 0.05	1.00 / 0.92
	256 D	1.00 ± 0.00	1.06 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	0.99 ± 0.00	0.98 ± 0.00	1.00 ± 0.01	1.11 ± 0.03	1.00 ± 0.01	0.99 ± 0.02	1.01 ± 0.04	1.00 / 0.84
100 full (F)	16 F	0.95 ± 0.01	0.97 ± 0.03	0.97 ± 0.03	0.97 ± 0.03	0.93 ± 0.01	0.93 ± 0.01	0.66 ± 0.03	0.90 ± 0.16	0.87 ± 0.02	0.85 ± 0.01	0.90 ± 0.10	1.00/0.99
	32 F	1.00 ± 0.02	0.99 ± 0.01	1.00 ± 0.00	1.00 ± 0.00	0.97 ± 0.01	0.96 ± 0.00	0.87 ± 0.03	1.12 ± 0.03	0.96 ± 0.01	0.89 ± 0.00	0.98 ± 0.07	0.99/0.97
	64 F	1.02 ± 0.00	1.00 ± 0.02	1.00 ± 0.00	1.00 ± 0.00	0.98 ± 0.00	0.97 ± 0.00	0.99 ± 0.01	1.10 ± 0.01	0.99 ± 0.02	0.93 ± 0.01	1.00 ± 0.04	0.97/0.93
	128 F	1.01 ± 0.01	1.05 ± 0.01	1.00 ± 0.00	0.99 ± 0.00	1.00 ± 0.00	0.98 ± 0.00	1.03 ± 0.01	1.10 ± 0.01	1.01 ± 0.00	1.00 ± 0.00	1.02 ± 0.03	0.88/0.80
	256 F	1.01 ± 0.01	1.03 ± 0.01	1.00 ± 0.00	1.00 ± 0.00	1.01 ± 0.00	1.00 ± 0.00	0.98 ± 0.00	1.00 ± 0.03	1.01 ± 0.00	1.01 ± 0.00	1.00 ± 0.01	0.50/0.34
100 w/clusters (C)	16 C 5	1.13 ± 0.01	1.03 ± 0.01	1.00 ± 0.00	1.00 ± 0.00	0.99 ± 0.01	0.97 ± 0.00	1.01 ± 0.02	1.22 ± 0.02	1.05 ± 0.01	1.00 ± 0.05	1.04 ± 0.07	1.00/0.95
	16 C 7	1.12 ± 0.01	1.01 ± 0.01	1.00 ± 0.00	1.00 ± 0.00	0.99 ± 0.01	0.97 ± 0.00	1.02 ± 0.02	1.22 ± 0.05	1.03 ± 0.01	1.01 ± 0.03	1.04 ± 0.07	1.00/0.93
500 diagonal (D)	16 D	0.57 ± 0.07	0.55 ± 0.03	0.83 ± 0.04	0.78 ± 0.16	0.85 ± 0.04	0.73 ± 0.07	0.24 ± 0.02	0.45 ± 0.01	0.76 ± 0.06	0.81 ± 0.00	0.66 ± 0.20	1.00 / 1.00
	32 D	0.61 ± 0.12	0.55 ± 0.08	0.83 ± 0.02	0.84 ± 0.12	0.91 ± 0.02	0.75 ± 0.05	0.30 ± 0.05	0.49 ± 0.07	0.79 ± 0.04	0.82 ± 0.01	0.69 ± 0.20	1.00 / 1.00
	64 D	0.73 ± 0.02	0.63 ± 0.11	0.89 ± 0.04	0.97 ± 0.00	0.94 ± 0.00	0.86 ± 0.03	0.46 ± 0.09	0.51 ± 0.07	0.82 ± 0.02	0.83 ± 0.01	0.77 ± 0.18	1.00 / 0.99
	128 D	0.84 ± 0.00	0.92 ± 0.02	0.97 ± 0.03	0.98 ± 0.01	0.94 ± 0.00	0.90 ± 0.02	0.62 ± 0.14	0.54 ± 0.01	0.85 ± 0.05	0.83 ± 0.01	0.84 ± 0.15	1.00 / 0.98
	256 D	0.99 ± 0.03	0.99 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	0.96 ± 0.00	0.93 ± 0.02	0.68 ± 0.05	0.85 ± 0.14	0.92 ± 0.02	0.85 ± 0.00	0.92 ± 0.11	1.00 / 0.97
500 full (F)	16 F	0.57 ± 0.01	0.43 ± 0.07	0.78 ± 0.01	0.97 ± 0.00	0.96 ± 0.00	0.86 ± 0.01	0.65 ± 0.00	0.55 ± 0.02	0.83 ± 0.01	0.84 ± 0.00	0.76 ± 0.17	1.00/1.00
	32 F	0.79 ± 0.05	0.54 ± 0.04	0.93 ± 0.02	0.98 ± 0.00	0.97 ± 0.00	0.92 ± 0.00	0.70 ± 0.01	0.52 ± 0.00	0.86 ± 0.02	0.85 ± 0.00	0.81 ± 0.16	0.99/0.99
	64 F	1.02 ± 0.00	0.96 ± 0.01	0.94 ± 0.01	1.00 ± 0.01	0.96 ± 0.00	0.97 ± 0.01	0.74 ± 0.01	0.55 ± 0.01	0.91 ± 0.01	0.87 ± 0.00	0.89 ± 0.14	0.97/0.96
	128 F	1.03 ± 0.01	0.97 ± 0.02	0.99 ± 0.00	1.00 ± 0.00	0.98 ± 0.00	0.97 ± 0.00	0.88 ± 0.01	1.07 ± 0.02	0.98 ± 0.00	0.90 ± 0.00	0.98 ± 0.05	0.88/0.86
	256 F	1.03 ± 0.00	1.03 ± 0.01	1.00 ± 0.00	1.00 ± 0.00	0.99 ± 0.01	0.97 ± 0.00	1.00 ± 0.02	1.04 ± 0.02	1.00 ± 0.01	0.93 ± 0.00	1.00 ± 0.03	0.50/0.47
500 w/clusters (C)	16 C 7	1.09	1.00	0.99	1.00	0.98	0.96	0.72	0.88	0.98	0.93	0.95	1.00/0.98
	16 C 10	1.10	1.01	1.00	0.99	0.97	0.94	0.72	1.29	1.02	0.92	1.00	1.00/0.98
	16 C 25	1.10	1.00	1.00	0.99	0.99	0.97	0.98	1.30	1.03	0.96	1.03	1.00/0.95
	64 C 5	1.09	0.98	1.00	1.00	0.99	0.97	0.99	1.17	1.04	0.93	1.02	0.97/0.93
	64 C 7	1.12	1.02	1.00	1.00	1.00	0.97	1.00	1.22	1.04	0.99	1.04	0.97/0.91
1000 w/clusters (C)	16 C 25	1.09	0.98	1.00	1.00	0.97	0.97	0.74	1.29	1.05	0.94	1.00	1.00/0.97

Table 9: **Relative** In-Distribution ROUGE-1 scores for various tasks and methods

Model Type	Method Type	Tasks										Average	Para. Saved
		task039	task190	task280	task290	task391	task442	task620	task1342	task1391	task1598		
	base	24.44 ± 0.00	1.60 ± 0.00	19.13 ± 0.00	39.22 ± 0.00	10.42 ± 0.00	39.88 ± 0.00	8.05 ± 0.00	6.96 ± 0.00	17.82 ± 0.00	55.03 ± 0.00	22.43 ± 16.49	1.00 / 1.00
	lora	95.00 ± 0.00	86.00 ± 0.00	99.00 ± 0.00	93.67 ± 0.00	94.33 ± 0.00	78.43 ± 0.00	74.90 ± 0.00	26.87 ± 0.00	95.00 ± 0.00	68.66 ± 0.00	81.14 ± 20.67	0.00/0.00
TIES	10	76.50 ± 0.00	49.00 ± 1.73	44.33 ± 4.04	9.80 ± 0.58	78.56 ± 0.96	40.44 ± 0.00	53.10 ± 0.67	15.48 ± 0.12	77.67 ± 1.15	54.89 ± 0.06	49.98 ± 23.33	1.00 / 1.00
	50	55.80 ± 0.00	35.00 ± 0.00	18.00 ± 5.20	2.42 ± 0.50	85.78 ± 0.96	26.75 ± 0.00	49.96 ± 0.00	16.73 ± 0.00	30.00 ± 3.46	53.87 ± 0.02	37.43 ± 23.49	1.00 / 1.00
	100	52.43 ± 0.00	34.00 ± 0.00	19.67 ± 4.62	1.09 ± 1.66	83.33 ± 0.00	28.57 ± 0.00	48.89 ± 0.00	15.18 ± 0.42	1.00 ± 0.00	53.44 ± 0.02	33.76 ± 25.22	1.00 / 1.00
	500	35.18 ± 0.00	22.00 ± 0.00	1.00 ± 0.00	0.00 ± 0.00	78.00 ± 0.00	24.32 ± 0.00	43.80 ± 0.04	9.96 ± 0.13	1.00 ± 0.00	27.90 ± 0.03	24.40 ± 23.79	1.00 / 1.00
SVD	SVD 2	93.15 ± 2.77	92.24 ± 1.85	99.09 ± 0.18	93.44 ± 0.14	93.89 ± 0.35	77.33 ± 0.29	75.40 ± 1.01	26.90 ± 2.68	95.06 ± 1.35	67.71 ± 0.49	81.33 ± 20.85	0.88 / 0.88
	SVD 4	94.01 ± 3.60	89.21 ± 0.71	99.05 ± 0.09	93.65 ± 0.03	94.66 ± 0.63	78.42 ± 0.23	74.09 ± 1.12	26.47 ± 2.06	93.98 ± 0.77	69.37 ± 0.21	81.22 ± 20.80	0.75 / 0.75
	SVD 8	95.00 ± 0.00	87.40 ± 0.59	99.05 ± 0.09	93.65 ± 0.03	94.36 ± 0.38	78.21 ± 0.03	75.57 ± 0.00	26.88 ± 0.27	95.51 ± 1.09	69.33 ± 0.08	83.02 ± 19.87	0.50 / 0.50
	SVD 16	95.00 ± 0.00	86.00 ± 0.00	99.00 ± 0.00	93.67 ± 0.00	94.33 ± 0.00	78.44 ± 0.03	74.73 ± 0.18	26.87 ± 0.00	95.00 ± 0.00	68.62 ± 0.04	80.76 ± 21.05	0.00 / 0.00
10 diagonal (D)	16 D	96.67 ± 0.58	87.00 ± 1.00	99.00 ± 0.00	94.00 ± 0.67	93.11 ± 0.38	76.08 ± 0.17	77.26 ± 1.47	30.15 ± 0.72	94.00 ± 1.73	68.25 ± 0.18	81.55 ± 20.03	1.00 / 0.90
	32 D	95.67 ± 0.58	90.00 ± 1.00	99.00 ± 0.00	93.00 ± 0.33	94.89 ± 0.51	77.46 ± 0.24	72.53 ± 1.00	27.98 ± 0.71	94.67 ± 0.58	69.16 ± 0.41	81.44 ± 20.80	1.00 / 0.80
	64 D	95.00 ± 0.00	88.33 ± 0.58	99.00 ± 0.00	93.67 ± 0.00	94.78 ± 0.38	78.28 ± 0.07	75.47 ± 0.58	26.53 ± 0.25	96.00 ± 0.00	69.36 ± 0.05	81.64 ± 21.06	1.00 / 0.60
	128 D	95.00 ± 0.00	86.67 ± 0.58	99.00 ± 0.00	93.67 ± 0.00	94.33 ± 0.00	78.45 ± 0.16	75.46 ± 0.51	26.64 ± 0.23	95.00 ± 0.00	68.70 ± 0.14	81.29 ± 20.92	1.00 / 0.20
	256 D	95.00 ± 0.00	86.00 ± 0.00	99.00 ± 0.00	93.67 ± 0.00	94.33 ± 0.00	78.43 ± 0.00	74.90 ± 0.00	26.87 ± 0.00	95.00 ± 0.00	68.59 ± 0.03	81.18 ± 20.86	1.00 / -0.60
10 full (F)	16 F	97.00 ± 0.00	91.00 ± 1.00	99.00 ± 0.00	93.56 ± 0.19	93.56 ± 0.69	77.64 ± 0.25	75.78 ± 1.25	28.71 ± 0.09	96.00 ± 1.00	68.69 ± 0.08	82.09 ± 20.68	1.00/0.90
	32 F	96.67 ± 0.58	89.33 ± 0.58	99.00 ± 0.00	93.22 ± 0.19	94.44 ± 0.19	77.84 ± 0.21	72.24 ± 0.59	26.84 ± 0.50	94.67 ± 0.58	69.55 ± 0.08	81.38 ± 21.11	0.99/0.79
	64 F	95.00 ± 0.00	88.67 ± 0.58	99.00 ± 0.00	93.67 ± 0.00	94.56 ± 0.38	78.19 ± 0.08	75.97 ± 0.58	26.43 ± 0.34	96.00 ± 0.00	69.38 ± 0.11	81.69 ± 21.07	0.97/0.57
	128 F	95.00 ± 0.00	86.67 ± 0.58	99.00 ± 0.00	93.67 ± 0.00	94.33 ± 0.00	78.46 ± 0.03	74.90 ± 0.00	26.72 ± 0.13	95.00 ± 0.00	68.65 ± 0.03	81.24 ± 20.91	0.88/0.07
	256 F	95.00 ± 0.00	86.00 ± 0.00	99.00 ± 0.00	93.67 ± 0.00	94.33 ± 0.00	78.44 ± 0.03	74.79 ± 0.19	26.87 ± 0.00	95.00 ± 0.00	68.64 ± 0.03	81.17 ± 20.86	0.50/-1.10
50 diagonal (D)	16 D	92.76 ± 3.53	84.67 ± 1.15	99.00 ± 0.00	86.17 ± 5.81	79.83 ± 6.08	73.55 ± 1.39	51.72 ± 3.78	23.75 ± 2.66	83.90 ± 6.43	59.05 ± 0.94	73.44 ± 22.08	1.00 / 0.98
	32 D	95.33 ± 2.08	87.33 ± 2.08	99.00 ± 0.00	92.60 ± 0.29	90.35 ± 1.00	75.43 ± 1.33	63.84 ± 1.64	26.97 ± 3.21	93.33 ± 1.15	61.94 ± 0.32	78.61 ± 21.60	1.00 / 0.96
	64 D	97.00 ± 0.00	90.33 ± 1.53	99.00 ± 0.00	93.78 ± 0.19	93.00 ± 0.58	76.27 ± 0.49	74.39 ± 0.90	29.28 ± 0.81	95.67 ± 0.58	64.84 ± 0.27	81.36 ± 20.83	1.00 / 0.92
	128 D	96.33 ± 0.58	92.67 ± 0.58	99.00 ± 0.00	93.56 ± 0.19	93.00 ± 0.58	77.24 ± 0.19	73.76 ± 1.25	29.58 ± 0.93	95.00 ± 0.00	69.04 ± 0.54	81.92 ± 20.44	1.00 / 0.84
	256 D	95.67 ± 0.58	88.33 ± 0.58	99.00 ± 0.00	93.56 ± 0.19	94.67 ± 0.67	78.45 ± 0.14	72.86 ± 2.07	27.00 ± 0.77	95.33 ± 0.58	69.61 ± 0.18	81.45 ± 21.00	1.00 / 0.68
50 full (F)	16 F	94.06 ± 3.54	85.67 ± 1.15	98.67 ± 0.58	90.35 ± 1.37	89.97 ± 1.78	74.46 ± 0.58	49.03 ± 7.07	27.14 ± 3.94	92.33 ± 1.53	60.26 ± 1.03	76.19 ± 22.80	1.00/0.98
	32 F	97.00 ± 0.00	85.67 ± 1.53	99.00 ± 0.00	93.67 ± 0.00	92.22 ± 0.69	75.86 ± 0.22	71.68 ± 0.65	29.26 ± 0.70	95.67 ± 1.53	63.88 ± 0.10	80.39 ± 20.81	0.99/0.95
	64 F	96.67 ± 0.58	91.00 ± 2.00	99.00 ± 0.00	93.56 ± 0.19	93.22 ± 0.51	77.17 ± 0.38	77.11 ± 0.51	29.75 ± 0.03	95.33 ± 0.58	68.13 ± 0.75	82.09 ± 20.33	0.97/0.89
	128 F	97.00 ± 0.00	91.00 ± 1.00	99.00 ± 0.00	93.33 ± 0.00	94.11 ± 0.51	77.23 ± 0.17	73.67 ± 0.58	27.62 ± 1.12	95.00 ± 1.00	69.40 ± 0.16	81.74 ± 20.97	0.88/0.72
	256 F	95.00 ± 0.00	88.00 ± 0.00	99.00 ± 0.00	93.67 ± 0.00	94.44 ± 0.19	77.97 ± 0.24	75.47 ± 0.58	26.96 ± 0.09	96.00 ± 0.00	69.28 ± 0.05	81.58 ± 20.92	0.50/0.18
100 diagonal (D)	16 D	76.43 ± 7.07	76.67 ± 4.93	91.61 ± 2.75	89.99 ± 1.07	47.89 ± 8.62	63.17 ± 1.31	22.23 ± 5.27	14.46 ± 2.89	73.93 ± 3.13	57.17 ± 1.05	61.35 ± 25.78	1.00 / 0.99
	32 D	90.10 ± 5.85	84.00 ± 1.00	99.00 ± 0.00	85.52 ± 5.34	75.88 ± 12.57	71.15 ± 3.61	46.10 ± 7.39	21.04 ± 6.76	86.67 ± 1.86	58.64 ± 1.02	71.81 ± 23.39	1.00 / 0.98
	64 D	95.56 ± 2.49	86.67 ± 0.58	99.00 ± 0.00	92.24 ± 1.68	90.89 ± 1.17	74.57 ± 0.50	67.07 ± 3.81	29.78 ± 1.92	91.67 ± 2.31	60.28 ± 1.51	78.77 ± 20.77	1.00 / 0.96
	128 D	96.00 ± 0.00	87.33 ± 1.15	99.00 ± 0.00	93.89 ± 0.19	93.00 ± 0.58	76.68 ± 0.18	74.84 ± 2.23	29.79 ± 0.50	93.67 ± 0.58	63.49 ± 0.34	80.77 ± 20.47	1.00 / 0.92
	256 D	95.00 ± 0.00	91.00 ± 0.00	99.00 ± 0.00	93.56 ± 0.19	93.11 ± 0.19	76.93 ± 0.23	75.13 ± 0.84	29.75 ± 0.73	95.33 ± 0.58	67.89 ± 1.34	81.67 ± 20.28	1.00 / 0.84
100 full (F)	16 F	90.70 ± 1.07	83.00 ± 2.65	96.00 ± 3.00	91.22 ± 2.94	87.94 ± 0.54	73.07 ± 0.93	49.41 ± 2.04	24.17 ± 4.22	82.33 ± 2.08	58.18 ± 0.44	73.60 ± 22.23	1.00/0.99
	32 F	95.33 ± 1.53	85.00 ± 1.00	99.00 ± 0.00	93.50 ± 0.22	91.44 ± 0.84	75.00 ± 0.19	65.09 ± 2.23	30.20 ± 0.81	91.67 ± 0.58	60.92 ± 0.26	78.72 ± 20.72	0.99/0.97
	64 F	97.00 ± 0.00	85.67 ± 1.53	99.00 ± 0.00	93.78 ± 0.19	92.56 ± 0.19	76.01 ± 0.13	73.96 ± 0.89	29.46 ± 0.21	94.33 ± 1.53	64.07 ± 0.37	80.58 ± 20.59	0.97/0.93
	128 F	96.33 ± 0.58	90.33 ± 0.58	99.00 ± 0.00	93.00 ± 0.00	93.89 ± 0.19	77.04 ± 0.30	77.33 ± 1.01	29.49 ± 0.35	96.00 ± 0.00	68.76 ± 0.25	82.12 ± 20.35	0.88/0.80
	256 F	96.33 ± 0.58	88.67 ± 0.58	99.00 ± 0.00	93.67 ± 0.00	94.89 ± 0.19	78.16 ± 0.18	73.40 ± 0.12	26.86 ± 0.68	96.00 ± 0.00	69.47 ± 0.23	81.64 ± 21.15	0.50/0.34
100 w/clusters (C)	16 C 5	98.33 ± 0.47	89.00 ± 0.82	99.00 ± 0.00	93.25 ± 0.40	92.89 ± 0.87	76.33 ± 0.28	78.24 ± 2.26	28.45 ± 0.40	96.67 ± 0.47	75.93 ± 8.31	82.81 ± 20.02	1.00/0.95
	16 C 7	97.67 ± 0.47	87.00 ± 0.82	99.00 ± 0.00	93.46 ± 0.29	93.11 ± 0.68	76.55 ± 0.29	79.03 ± 2.05	28.62 ± 1.29	95.33 ± 0.47	76.55 ± 6.91	82.63 ± 19.76	1.00/0.93
500 diagonal (D)	16 D	54.44 ± 6.87	47.00 ± 2.83	82.21 ± 3.59	73.38 ± 14.97	80.13 ± 3.68	57.42 ± 5.29	18.33 ± 1.33	12.19 ± 0.30	72.67 ± 6.03	55.79 ± 0.20	55.64 ± 24.25	1.00 / 1.00
	32 D	58.08 ± 11.52	47.00 ± 7.07	82.06 ± 1.69	78.62 ± 11.23	85.57 ± 1.48	59.19 ± 3.70	22.76 ± 3.95	13.15 ± 1.94	75.33 ± 4.04	56.07 ± 0.52	58.16 ± 24.56	1.00 / 1.00
	64 D	69.21 ± 2.03	54.50 ± 9.19	88.33 ± 4.04	91.11 ± 0.38	88.78 ± 0.38	67.71 ± 2.59	34.79 ± 6.86	13.80 ± 1.95	77.67 ± 2.31	56.78 ± 0.73	64.61 ± 24.79	1.00 / 0.99
	128 D	79.77 ± 0.37	79.50 ± 2.12	95.89 ± 2.83	91.89 ± 1.39	88.67 ± 0.00	70.27 ± 1.73	46.64 ± 10.58	14.63 ± 0.25	81.00 ± 5.00	56.88 ± 0.55	70.20 ± 24.63	1.00 / 0.98
	256 D	93.83 ± 2.52	85.00 ± 0.00	99.00 ± 0.00	93.78 ± 0.19	90.56 ± 0.38	73.25 ± 1.86	51.14 ± 3.86	22.93 ± 3.86	87.33 ± 2.31	58.48 ± 0.20	75.20 ± 23.90	1.00 / 0.97
500 full (F)	16 F	54.30 ± 1.13	37.00 ± 5.66	77.67 ± 0.58	91.00 ± 0.00	90.56 ± 0.19	67.63 ± 0.45	48.81 ± 0.35	14.70 ± 0.65	79.00 ± 1.00	57.66 ± 0.19	62.69 ± 23.46	1.00/1.00
	32 F	75.10 ± 4.92	46.50 ± 3.54	91.67 ± 1.53	91.56 ± 0.19	91.56 ± 0.38	72.03 ± 0.15	52.63 ± 0.86	13.93 ± 0.02	81.67 ± 1.53	58.50 ± 0.20	68.24 ± 24.29	0.99/0.99
	64 F	96.94 ± 0.42	82.50 ± 0.71	93.33 ± 0.58	93.89 ± 0.69	90.67 ± 0.00	75.99 ± 0.64	55.63 ± 1.07	14.74 ± 0.27	86.33 ± 0.58	59.43 ± 0.05	74.69 ± 25.01	0.97/0.96
	128 F	97.67 ± 0.58	83.50 ± 2.12	98.00 ± 0.00	93.56 ± 0.19	92.00 ± 0.00	75.80 ± 0.16	66.19 ± 0.81	28.67 ± 0.49	93.00 ± 0.00	61.53 ± 0.13	78.84 ± 21.50	0.88/0.86
	256 F	98.00 ± 0.00	88.50 ± 0.71	99.00 ± 0.00	93.78 ± 0.19	93.00 ± 0.88	76.33 ± 0.29	74.60 ± 1.21	27.82 ± 0.42	95.33 ± 0.58	63.70 ± 0.14	80.75 ± 21.60	0.50/0.47
500 w/clusters (C)	16 C 7	95.00	86.00	98.00	93.67	91.67	75.10	55.52	20.50	90.00	63.57	76.90	1.00/0.98
	16 C 10	96.00	87.00	99.00	93.00	91.33	74.17	55.14	30.29	94.00	63.09	78.30	1.00/0.98
	16 C 25	96.00	86.00	99.00	92.71	93.00	76.42	75.42	30.40	95.00	66.07	81.00	1.00/0.95
	64 C 5	95.00	84.00	99.00	93.67	92.67	76.45	76.43	27.49	96.00	64.10	80.48	0.97/0.93
	64 C 7	98.00	88.00	99.00	94.00	93.33	76.42	76.67	28.48	96.00	68.00	81.79	0.97/0.91
1000 w/clusters (C)	16 C 25	95.00	84.00	99.00	93.67	90.67	76.43	56.71	30.20	97.00	64.61	78.73	1.00/0.97

Table 10: **Absolute** In-Distribution ROUGE-1 scores for various tasks and methods

Model Type	Method Type	Tasks										Average	Para. Saved
		task039	task190	task280	task290	task391	task442	task620	task1342	task1391	task1598		
	base	0.00 ± 0.00	0.00 ± 0.00	0.02 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.01	1.00 / 1.00
	lora	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	0.00 / 0.00
TIES	10	0.69 ± 0.00	0.57 ± 0.02	0.45 ± 0.04	0.10 ± 0.01	0.57 ± 0.03	0.00 ± 0.00	0.39 ± 0.01	0.21 ± 0.00	0.82 ± 0.01	0.00 ± 0.00	0.38 ± 0.28	1.00 / 1.00
	50	0.45 ± 0.00	0.41 ± 0.00	0.18 ± 0.05	0.03 ± 0.01	0.70 ± 0.02	0.00 ± 0.00	0.36 ± 0.00	0.21 ± 0.00	0.32 ± 0.04	0.00 ± 0.00	0.27 ± 0.22	1.00 / 1.00
	100	0.41 ± 0.00	0.40 ± 0.00	0.20 ± 0.05	0.01 ± 0.02	0.65 ± 0.00	0.00 ± 0.00	0.36 ± 0.00	0.21 ± 0.00	0.01 ± 0.00	0.00 ± 0.00	0.23 ± 0.22	1.00 / 1.00
	500	0.22 ± 0.00	0.26 ± 0.00	0.01 ± 0.00	0.00 ± 0.00	0.60 ± 0.00	0.00 ± 0.00	0.32 ± 0.00	0.07 ± 0.00	0.01 ± 0.00	0.00 ± 0.00	0.15 ± 0.20	1.00 / 1.00
SVD	SVD 2	0.98 ± 0.03	1.07 ± 0.02	1.00 ± 0.00	0.99 ± 0.01	0.98 ± 0.01	0.98 ± 0.03	0.94 ± 0.01	1.03 ± 0.17	1.00 ± 0.01	0.15 ± 0.29	0.91 ± 0.28	0.88 / 0.88
	SVD 4	0.99 ± 0.04	1.04 ± 0.01	1.00 ± 0.00	1.00 ± 0.00	1.01 ± 0.02	1.11 ± 0.00	0.97 ± 0.02	0.99 ± 0.13	0.99 ± 0.01	0.90 ± 0.17	1.00 ± 0.08	0.75 / 0.75
	SVD 8	1.00 ± 0.00	1.02 ± 0.01	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.01	1.02 ± 0.05	1.00 ± 0.00	1.00 ± 0.00	1.01 ± 0.01	1.00 ± 0.00	1.00 ± 0.02	0.50 / 0.50
	SVD 16	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	0.99 ± 0.01	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	0.00 / 0.00
10 diagonal (D)	16 D	1.02 ± 0.01	1.01 ± 0.01	1.00 ± 0.00	1.01 ± 0.02	0.96 ± 0.01	1.11 ± 0.11	0.89 ± 0.03	1.19 ± 0.04	0.99 ± 0.02	0.33 ± 0.58	0.95 ± 0.27	1.00 / 0.90
	32 D	1.01 ± 0.01	1.05 ± 0.01	1.00 ± 0.00	0.98 ± 0.01	1.02 ± 0.02	1.11 ± 0.00	0.93 ± 0.01	1.10 ± 0.04	1.00 ± 0.01	0.67 ± 0.58	0.98 ± 0.19	1.00 / 0.80
	64 D	1.00 ± 0.00	1.03 ± 0.01	1.00 ± 0.00	1.00 ± 0.00	1.02 ± 0.01	1.11 ± 0.00	0.99 ± 0.01	1.00 ± 0.00	1.01 ± 0.00	0.67 ± 0.58	0.98 ± 0.19	1.00 / 0.60
	128 D	1.00 ± 0.00	1.01 ± 0.01	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.01	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 / 0.20
	256 D	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 / -0.60
10 full (F)	16 F	1.02 ± 0.00	1.06 ± 0.01	1.00 ± 0.00	1.00 ± 0.01	0.97 ± 0.03	1.15 ± 0.06	0.92 ± 0.02	1.17 ± 0.04	1.01 ± 0.01	0.67 ± 0.58	1.00 ± 0.20	1.00/0.90
	32 F	1.02 ± 0.01	1.04 ± 0.01	1.00 ± 0.00	0.98 ± 0.01	1.00 ± 0.01	1.11 ± 0.00	0.92 ± 0.01	1.02 ± 0.04	1.00 ± 0.01	1.00 ± 0.00	1.01 ± 0.05	0.99/0.79
	64 F	1.00 ± 0.00	1.03 ± 0.01	1.00 ± 0.00	1.00 ± 0.00	1.01 ± 0.01	1.07 ± 0.06	1.01 ± 0.01	1.00 ± 0.00	1.01 ± 0.00	1.00 ± 0.00	1.01 ± 0.03	0.97/0.57
	128 F	1.00 ± 0.00	1.01 ± 0.01	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	0.88/0.07
	256 F	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.01	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	0.50/-1.10
50 diagonal (D)	16 D	0.91 ± 0.06	0.98 ± 0.01	1.00 ± 0.00	0.91 ± 0.09	0.78 ± 0.05	0.89 ± 0.29	0.34 ± 0.06	0.50 ± 0.45	0.86 ± 0.07	0.00 ± 0.00	0.72 ± 0.35	1.00 / 0.98
	32 D	1.00 ± 0.02	1.02 ± 0.02	1.00 ± 0.00	1.00 ± 0.01	0.90 ± 0.03	0.85 ± 0.42	0.56 ± 0.04	0.98 ± 0.23	0.98 ± 0.01	0.00 ± 0.00	0.83 ± 0.34	1.00 / 0.96
	64 D	1.02 ± 0.00	1.05 ± 0.02	1.00 ± 0.00	1.00 ± 0.01	0.95 ± 0.02	1.15 ± 0.17	0.81 ± 0.03	1.14 ± 0.00	1.01 ± 0.01	0.00 ± 0.00	0.91 ± 0.33	1.00 / 0.92
	128 D	1.01 ± 0.01	1.08 ± 0.01	1.00 ± 0.00	1.00 ± 0.01	0.95 ± 0.02	1.04 ± 0.06	0.92 ± 0.03	1.21 ± 0.07	1.00 ± 0.00	0.67 ± 0.58	0.99 ± 0.20	1.00 / 0.84
	256 D	1.01 ± 0.01	1.03 ± 0.01	1.00 ± 0.00	1.00 ± 0.01	1.01 ± 0.02	1.11 ± 0.00	0.95 ± 0.04	1.02 ± 0.04	1.00 ± 0.01	1.00 ± 0.00	1.01 ± 0.04	1.00 / 0.68
50 full (F)	16 F	0.96 ± 0.05	1.00 ± 0.01	1.00 ± 0.01	0.95 ± 0.04	0.87 ± 0.01	1.04 ± 0.06	0.31 ± 0.08	0.98 ± 0.23	0.97 ± 0.02	0.00 ± 0.00	0.81 ± 0.35	1.00/0.98
	32 F	1.02 ± 0.00	1.00 ± 0.02	1.00 ± 0.00	1.00 ± 0.00	0.92 ± 0.03	1.15 ± 0.06	0.73 ± 0.04	1.17 ± 0.04	1.01 ± 0.02	0.00 ± 0.00	0.90 ± 0.33	0.99/0.95
	64 F	1.02 ± 0.01	1.06 ± 0.02	1.00 ± 0.00	1.00 ± 0.01	0.96 ± 0.02	1.22 ± 0.00	0.94 ± 0.01	1.17 ± 0.04	1.00 ± 0.01	0.00 ± 0.00	0.94 ± 0.33	0.97/0.89
	128 F	1.02 ± 0.00	1.06 ± 0.01	1.00 ± 0.00	0.99 ± 0.00	0.99 ± 0.02	1.15 ± 0.06	0.92 ± 0.01	1.10 ± 0.08	1.00 ± 0.01	1.00 ± 0.00	1.02 ± 0.07	0.88/0.72
	256 F	1.00 ± 0.00	1.02 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.01	1.04 ± 0.06	0.99 ± 0.00	1.00 ± 0.00	1.01 ± 0.00	1.00 ± 0.00	1.01 ± 0.02	0.50/0.18
100 diagonal (D)	16 D	0.54 ± 0.16	0.89 ± 0.06	0.90 ± 0.04	0.89 ± 0.05	0.42 ± 0.08	0.44 ± 0.00	0.08 ± 0.02	0.00 ± 0.00	0.76 ± 0.05	0.00 ± 0.00	0.49 ± 0.36	1.00 / 0.99
	32 D	0.85 ± 0.15	0.98 ± 0.01	1.00 ± 0.00	0.86 ± 0.13	0.70 ± 0.14	0.74 ± 0.28	0.28 ± 0.07	0.48 ± 0.55	0.91 ± 0.02	0.00 ± 0.00	0.68 ± 0.36	1.00 / 0.98
	64 D	1.00 ± 0.04	1.01 ± 0.01	1.00 ± 0.00	0.98 ± 0.02	0.88 ± 0.04	1.07 ± 0.06	0.58 ± 0.09	1.10 ± 0.04	0.96 ± 0.02	0.00 ± 0.00	0.86 ± 0.32	1.00 / 0.96
	128 D	1.01 ± 0.00	1.02 ± 0.01	1.00 ± 0.00	1.01 ± 0.01	0.95 ± 0.02	1.11 ± 0.00	0.81 ± 0.06	1.21 ± 0.00	0.99 ± 0.01	0.00 ± 0.00	0.91 ± 0.33	1.00 / 0.92
	256 D	1.00 ± 0.00	1.06 ± 0.00	1.00 ± 0.00	1.00 ± 0.01	0.96 ± 0.01	1.11 ± 0.11	0.92 ± 0.02	1.21 ± 0.07	1.00 ± 0.01	0.00 ± 0.00	0.93 ± 0.33	1.00 / 0.84
100 full (F)	16 F	0.85 ± 0.03	0.97 ± 0.03	0.97 ± 0.03	0.95 ± 0.06	0.80 ± 0.02	0.81 ± 0.17	0.29 ± 0.04	0.60 ± 0.34	0.87 ± 0.02	0.00 ± 0.00	0.71 ± 0.33	1.00/0.99
	32 F	0.99 ± 0.02	0.99 ± 0.01	1.00 ± 0.00	1.00 ± 0.01	0.90 ± 0.03	1.04 ± 0.06	0.55 ± 0.04	1.07 ± 0.07	0.96 ± 0.01	0.00 ± 0.00	0.85 ± 0.32	0.99/0.97
	64 F	1.02 ± 0.00	1.00 ± 0.02	1.00 ± 0.00	1.00 ± 0.01	0.94 ± 0.01	1.04 ± 0.06	0.78 ± 0.01	1.14 ± 0.00	0.99 ± 0.02	0.00 ± 0.00	0.89 ± 0.31	0.97/0.93
	128 F	1.01 ± 0.01	1.05 ± 0.01	1.00 ± 0.00	0.98 ± 0.00	0.98 ± 0.01	1.15 ± 0.06	0.94 ± 0.01	1.21 ± 0.00	1.01 ± 0.00	0.33 ± 0.58	0.97 ± 0.28	0.88/0.80
	256 F	1.01 ± 0.01	1.03 ± 0.01	1.00 ± 0.00	1.00 ± 0.00	1.02 ± 0.01	1.19 ± 0.06	0.93 ± 0.01	1.02 ± 0.04	1.01 ± 0.00	1.00 ± 0.00	1.02 ± 0.06	0.50/0.34
100 w/clusters (C)	16 C 5	1.13 ± 0.01	1.03 ± 0.01	1.00 ± 0.00	1.00 ± 0.01	0.97 ± 0.03	1.24 ± 0.09	0.88 ± 0.06	1.42 ± 0.07	1.05 ± 0.01	0.65 ± 0.46	1.04 ± 0.19	1.00/0.95
	16 C 7	1.12 ± 0.01	1.01 ± 0.01	1.00 ± 0.00	1.00 ± 0.00	0.98 ± 0.03	1.16 ± 0.11	0.92 ± 0.05	1.45 ± 0.04	1.03 ± 0.01	0.69 ± 0.49	1.04 ± 0.18	1.00/0.93
500 diagonal (D)	16 D	0.22 ± 0.10	0.55 ± 0.03	0.81 ± 0.05	0.33 ± 0.49	0.70 ± 0.03	0.15 ± 0.17	0.03 ± 0.01	0.00 ± 0.00	0.76 ± 0.06	0.00 ± 0.00	0.35 ± 0.35	1.00 / 1.00
	32 D	0.27 ± 0.18	0.55 ± 0.08	0.82 ± 0.02	0.49 ± 0.37	0.75 ± 0.01	0.22 ± 0.11	0.05 ± 0.05	0.02 ± 0.04	0.79 ± 0.04	0.00 ± 0.00	0.39 ± 0.34	1.00 / 1.00
	64 D	0.40 ± 0.04	0.63 ± 0.11	0.89 ± 0.04	0.91 ± 0.01	0.80 ± 0.01	0.48 ± 0.06	0.13 ± 0.04	0.05 ± 0.08	0.82 ± 0.02	0.00 ± 0.00	0.51 ± 0.35	1.00 / 0.99
	128 D	0.61 ± 0.04	0.92 ± 0.02	0.97 ± 0.03	0.93 ± 0.05	0.80 ± 0.00	0.74 ± 0.17	0.22 ± 0.11	0.12 ± 0.08	0.85 ± 0.05	0.00 ± 0.00	0.61 ± 0.36	1.00 / 0.98
	256 D	0.95 ± 0.02	0.99 ± 0.00	1.00 ± 0.00	1.00 ± 0.01	0.86 ± 0.01	0.85 ± 0.28	0.28 ± 0.06	0.55 ± 0.39	0.92 ± 0.02	0.00 ± 0.00	0.73 ± 0.36	1.00 / 0.97
500 full (F)	16 F	0.21 ± 0.02	0.43 ± 0.07	0.78 ± 0.01	0.90 ± 0.00	0.86 ± 0.01	0.59 ± 0.06	0.21 ± 0.01	0.12 ± 0.04	0.83 ± 0.01	0.00 ± 0.00	0.50 ± 0.34	1.00/1.00
	32 F	0.54 ± 0.08	0.54 ± 0.04	0.93 ± 0.02	0.92 ± 0.01	0.90 ± 0.01	0.63 ± 0.13	0.26 ± 0.02	0.14 ± 0.00	0.86 ± 0.02	0.00 ± 0.00	0.57 ± 0.34	0.99/0.99
	64 F	0.99 ± 0.03	0.96 ± 0.01	0.94 ± 0.01	1.01 ± 0.03	0.87 ± 0.00	1.04 ± 0.17	0.36 ± 0.00	0.14 ± 0.00	0.91 ± 0.01	0.00 ± 0.00	0.71 ± 0.39	0.97/0.96
	128 F	1.02 ± 0.01	0.97 ± 0.02	0.99 ± 0.00	1.00 ± 0.01	0.92 ± 0.00	1.15 ± 0.06	0.61 ± 0.01	1.07 ± 0.00	0.98 ± 0.00	0.00 ± 0.00	0.87 ± 0.33	0.88/0.86
	256 F	1.03 ± 0.00	1.03 ± 0.01	1.00 ± 0.00	1.00 ± 0.01	0.95 ± 0.03	1.00 ± 0.00	0.78 ± 0.01	1.07 ± 0.00	1.00 ± 0.01	0.00 ± 0.00	0.88 ± 0.31	0.50/0.47
500 w/clusters (C)	16 C 7	1.08	1.00	0.99	1.00	0.92	1.01	0.39	0.62	0.98	0.00	0.80	1.00/0.98
	16 C 10	1.10	1.01	1.00	0.98	0.91	1.01	0.37	1.51	1.02	0.00	0.89	1.00/0.98
	16 C 25	1.10	1.00	1.00	0.99	0.97	1.12	0.81	1.42	1.03	0.00	0.95	1.00/0.95
	64 C 5	1.09	0.98	1.00	1.00	0.96	1.12	0.83	1.33	1.04	0.00	0.94	0.97/0.93
	64 C 7	1.13	1.02	1.00	1.01	0.98	1.12	0.90	1.42	1.04	0.00	0.96	0.97/0.91
1000 w/clusters (C)	16 C 25	1.09	0.98	1.00	1.00	0.89	1.01	0.39	1.42	1.05	0.00	0.88	1.00/0.97

Table 11: **Relative** In-Distribution exact match scores for various tasks and methods

Model Type	Method Type	Tasks										Average	Para. Saved
		task039	task190	task280	task290	task391	task442	task620	task1342	task1391	task1598		
	base	8.59 ± 0.08	9.15 ± 0.00	2.55 ± 0.00	2.88 ± 0.00	2.34 ± 0.00	3.46 ± 0.04	6.40 ± 0.18	5.55 ± 0.00	8.60 ± 0.00	2.67 ± 0.00	5.19 ± 2.65	1.00 / 1.00
	lora	0.36 ± 0.01	0.17 ± 0.00	0.01 ± 0.00	0.12 ± 0.00	0.11 ± 0.00	0.76 ± 0.02	1.17 ± 0.07	1.94 ± 0.00	0.16 ± 0.00	0.85 ± 0.00	0.57 ± 0.59	0.00 / 0.00
SVD	SVD 2	0.32 ± 0.01	0.15 ± 0.00	0.01 ± 0.00	0.12 ± 0.00	0.10 ± 0.00	0.76 ± 0.02	1.13 ± 0.08	1.94 ± 0.00	0.13 ± 0.00	0.97 ± 0.00	0.57 ± 0.60	0.88 / 0.88
	SVD 4	0.33 ± 0.01	0.16 ± 0.00	0.01 ± 0.00	0.12 ± 0.00	0.11 ± 0.00	0.76 ± 0.02	1.14 ± 0.08	1.94 ± 0.00	0.14 ± 0.00	0.86 ± 0.00	0.56 ± 0.59	0.75 / 0.75
	SVD 8	0.35 ± 0.01	0.17 ± 0.00	0.01 ± 0.00	0.12 ± 0.00	0.11 ± 0.00	0.77 ± 0.02	1.16 ± 0.07	1.94 ± 0.00	0.15 ± 0.00	0.84 ± 0.00	0.51 ± 0.58	0.50 / 0.50
	SVD 16	0.36 ± 0.01	0.17 ± 0.00	0.01 ± 0.00	0.12 ± 0.00	0.11 ± 0.00	0.76 ± 0.02	1.14 ± 0.06	1.94 ± 0.00	0.16 ± 0.00	0.85 ± 0.00	0.56 ± 0.59	0.00 / 0.00
10 diagonal (D)	16 D	0.33 ± 0.01	0.15 ± 0.01	0.01 ± 0.00	0.12 ± 0.00	0.10 ± 0.00	0.76 ± 0.03	1.13 ± 0.08	1.95 ± 0.01	0.14 ± 0.00	1.00 ± 0.02	0.57 ± 0.61	1.00 / 0.90
	32 D	0.33 ± 0.01	0.16 ± 0.00	0.01 ± 0.00	0.12 ± 0.00	0.10 ± 0.00	0.75 ± 0.02	1.11 ± 0.07	1.93 ± 0.00	0.14 ± 0.01	0.88 ± 0.00	0.55 ± 0.60	1.00 / 0.80
	64 D	0.35 ± 0.01	0.17 ± 0.00	0.01 ± 0.00	0.12 ± 0.00	0.11 ± 0.00	0.75 ± 0.02	1.11 ± 0.07	1.94 ± 0.00	0.15 ± 0.00	0.84 ± 0.00	0.55 ± 0.59	1.00 / 0.60
	128 D	0.35 ± 0.01	0.17 ± 0.00	0.01 ± 0.00	0.12 ± 0.00	0.11 ± 0.00	0.75 ± 0.02	1.11 ± 0.07	1.94 ± 0.00	0.16 ± 0.00	0.84 ± 0.00	0.56 ± 0.59	1.00 / 0.20
	256 D	0.36 ± 0.01	0.17 ± 0.00	0.01 ± 0.00	0.12 ± 0.00	0.11 ± 0.00	0.75 ± 0.02	1.12 ± 0.07	1.94 ± 0.00	0.16 ± 0.00	0.85 ± 0.00	0.56 ± 0.59	1.00 / -0.60
10 full (F)	16 F	0.33 ± 0.00	0.15 ± 0.00	0.01 ± 0.00	0.12 ± 0.00	0.10 ± 0.00	0.76 ± 0.02	1.20 ± 0.02	1.95 ± 0.00	0.13 ± 0.00	0.97 ± 0.00	0.57 ± 0.61	1.00/0.90
	32 F	0.33 ± 0.01	0.16 ± 0.00	0.01 ± 0.00	0.12 ± 0.00	0.10 ± 0.00	0.75 ± 0.02	1.11 ± 0.07	1.94 ± 0.00	0.14 ± 0.00	0.86 ± 0.00	0.55 ± 0.60	0.99/0.79
	64 F	0.34 ± 0.01	0.16 ± 0.00	0.01 ± 0.00	0.12 ± 0.00	0.11 ± 0.00	0.75 ± 0.02	1.11 ± 0.07	1.94 ± 0.00	0.15 ± 0.00	0.84 ± 0.00	0.55 ± 0.59	0.97/0.57
	128 F	0.35 ± 0.01	0.17 ± 0.00	0.01 ± 0.00	0.12 ± 0.00	0.11 ± 0.00	0.75 ± 0.02	1.12 ± 0.07	1.94 ± 0.00	0.16 ± 0.00	0.84 ± 0.00	0.56 ± 0.59	0.88/0.07
	256 F	0.36 ± 0.01	0.17 ± 0.00	0.01 ± 0.00	0.12 ± 0.00	0.11 ± 0.00	0.75 ± 0.02	1.12 ± 0.07	1.94 ± 0.00	0.16 ± 0.00	0.85 ± 0.00	0.56 ± 0.59	0.50/-1.10
50 diagonal (D)	16 D	0.61 ± 0.06	0.19 ± 0.02	0.03 ± 0.01	0.29 ± 0.04	0.36 ± 0.04	0.95 ± 0.05	1.73 ± 0.21	2.66 ± 0.22	0.32 ± 0.11	1.98 ± 0.01	0.91 ± 0.88	1.00 / 0.98
	32 D	0.37 ± 0.02	0.16 ± 0.00	0.01 ± 0.00	0.19 ± 0.03	0.18 ± 0.01	0.85 ± 0.05	1.37 ± 0.14	2.12 ± 0.05	0.16 ± 0.00	1.65 ± 0.03	0.71 ± 0.73	1.00 / 0.96
	64 D	0.33 ± 0.02	0.15 ± 0.00	0.01 ± 0.00	0.12 ± 0.00	0.10 ± 0.00	0.79 ± 0.02	1.12 ± 0.08	1.97 ± 0.01	0.13 ± 0.01	1.13 ± 0.03	0.59 ± 0.63	1.00 / 0.92
	128 D	0.33 ± 0.01	0.15 ± 0.00	0.01 ± 0.00	0.12 ± 0.00	0.10 ± 0.00	0.76 ± 0.03	1.10 ± 0.05	1.93 ± 0.01	0.14 ± 0.00	0.93 ± 0.01	0.56 ± 0.60	1.00 / 0.84
	256 D	0.34 ± 0.01	0.16 ± 0.00	0.01 ± 0.00	0.12 ± 0.00	0.10 ± 0.00	0.76 ± 0.03	1.11 ± 0.05	1.93 ± 0.00	0.15 ± 0.00	0.85 ± 0.00	0.55 ± 0.59	1.00 / 0.68
50 full (F)	16 F	0.47 ± 0.06	0.17 ± 0.00	0.02 ± 0.00	0.20 ± 0.02	0.19 ± 0.04	0.86 ± 0.03	1.71 ± 0.10	2.20 ± 0.04	0.17 ± 0.01	1.84 ± 0.07	0.78 ± 0.80	1.00/0.98
	32 F	0.36 ± 0.02	0.16 ± 0.00	0.01 ± 0.00	0.14 ± 0.00	0.11 ± 0.00	0.80 ± 0.03	1.14 ± 0.08	2.00 ± 0.01	0.14 ± 0.00	1.32 ± 0.02	0.62 ± 0.65	0.99/0.95
	64 F	0.33 ± 0.01	0.15 ± 0.00	0.01 ± 0.00	0.12 ± 0.00	0.10 ± 0.00	0.77 ± 0.03	1.10 ± 0.06	1.94 ± 0.00	0.13 ± 0.00	1.02 ± 0.00	0.57 ± 0.61	0.97/0.89
	128 F	0.33 ± 0.01	0.16 ± 0.00	0.00 ± 0.00	0.12 ± 0.00	0.10 ± 0.00	0.76 ± 0.03	1.11 ± 0.05	1.93 ± 0.00	0.14 ± 0.00	0.87 ± 0.00	0.55 ± 0.60	0.88/0.72
	256 F	0.35 ± 0.01	0.16 ± 0.00	0.01 ± 0.00	0.12 ± 0.00	0.11 ± 0.00	0.76 ± 0.03	1.11 ± 0.05	1.94 ± 0.00	0.15 ± 0.00	0.84 ± 0.00	0.55 ± 0.59	0.50/0.18
100 diagonal (D)	16 D	1.69 ± 0.49	0.26 ± 0.04	0.18 ± 0.07	0.34 ± 0.02	1.01 ± 0.20	1.45 ± 0.10	3.59 ± 0.25	3.72 ± 0.72	0.44 ± 0.20	2.37 ± 0.09	1.51 ± 1.32	1.00 / 0.99
	32 D	0.67 ± 0.24	0.18 ± 0.01	0.06 ± 0.05	0.31 ± 0.06	0.35 ± 0.08	1.04 ± 0.15	1.97 ± 0.13	2.88 ± 0.70	0.22 ± 0.01	2.12 ± 0.07	0.98 ± 0.98	1.00 / 0.98
	64 D	0.39 ± 0.06	0.16 ± 0.00	0.01 ± 0.00	0.18 ± 0.02	0.14 ± 0.01	0.86 ± 0.02	1.39 ± 0.07	2.18 ± 0.04	0.17 ± 0.00	1.79 ± 0.02	0.73 ± 0.76	1.00 / 0.96
	128 D	0.32 ± 0.00	0.15 ± 0.00	0.01 ± 0.00	0.12 ± 0.00	0.10 ± 0.00	0.79 ± 0.02	1.19 ± 0.02	2.00 ± 0.01	0.14 ± 0.01	1.24 ± 0.04	0.61 ± 0.65	1.00 / 0.92
	256 D	0.32 ± 0.00	0.15 ± 0.00	0.01 ± 0.00	0.12 ± 0.00	0.10 ± 0.00	0.77 ± 0.02	1.16 ± 0.00	1.94 ± 0.00	0.13 ± 0.00	0.96 ± 0.01	0.56 ± 0.61	1.00 / 0.84
100 full (F)	16 F	0.66 ± 0.07	0.19 ± 0.01	0.03 ± 0.01	0.25 ± 0.02	0.29 ± 0.02	0.99 ± 0.07	2.50 ± 0.51	2.63 ± 0.03	0.24 ± 0.02	2.21 ± 0.08	1.00 ± 1.01	1.00/0.99
	32 F	0.40 ± 0.00	0.17 ± 0.00	0.01 ± 0.00	0.15 ± 0.01	0.13 ± 0.01	0.85 ± 0.02	1.53 ± 0.12	2.17 ± 0.06	0.15 ± 0.01	1.93 ± 0.04	0.75 ± 0.80	0.99/0.97
	64 F	0.34 ± 0.01	0.15 ± 0.00	0.01 ± 0.00	0.12 ± 0.00	0.11 ± 0.00	0.79 ± 0.01	1.23 ± 0.07	1.98 ± 0.01	0.15 ± 0.00	1.26 ± 0.01	0.61 ± 0.65	0.97/0.93
	128 F	0.32 ± 0.00	0.15 ± 0.00	0.01 ± 0.00	0.12 ± 0.00	0.10 ± 0.00	0.77 ± 0.02	1.16 ± 0.01	1.94 ± 0.00	0.13 ± 0.00	0.99 ± 0.01	0.57 ± 0.61	0.88/0.80
	256 F	0.33 ± 0.00	0.16 ± 0.00	0.00 ± 0.00	0.12 ± 0.00	0.10 ± 0.00	0.76 ± 0.02	1.15 ± 0.01	1.93 ± 0.00	0.14 ± 0.00	0.86 ± 0.00	0.56 ± 0.60	0.50/0.34
100 w/clusters (C)	16 C 5	0.34 ± 0.01	0.15 ± 0.00	0.01 ± 0.00	0.14 ± 0.01	0.11 ± 0.00	0.79 ± 0.02	0.97 ± 0.27	1.97 ± 0.00	0.13 ± 0.00	0.77 ± 0.25	0.54 ± 0.58	1.00/0.95
	16 C 7	0.34 ± 0.01	0.15 ± 0.00	0.01 ± 0.00	0.13 ± 0.00	0.10 ± 0.00	0.78 ± 0.02	0.96 ± 0.27	1.96 ± 0.00	0.14 ± 0.00	0.74 ± 0.22	0.53 ± 0.57	1.00/0.93
500 diagonal (D)	16 D	2.95 ± 0.28	0.73 ± 0.29	0.27 ± 0.09	0.67 ± 0.28	0.52 ± 0.07	2.06 ± 0.30	4.85 ± 0.31	3.94 ± 0.42	0.50 ± 0.05	2.50 ± 0.03	1.94 ± 1.59	1.00 / 1.00
	32 D	2.33 ± 0.30	0.62 ± 0.17	0.24 ± 0.05	0.50 ± 0.16	0.37 ± 0.07	1.86 ± 0.25	4.73 ± 0.35	3.81 ± 0.59	0.39 ± 0.04	2.46 ± 0.05	1.77 ± 1.57	1.00 / 1.00
	64 D	1.67 ± 0.18	0.43 ± 0.16	0.13 ± 0.04	0.29 ± 0.02	0.23 ± 0.02	1.32 ± 0.28	3.99 ± 0.36	3.41 ± 0.28	0.32 ± 0.05	2.35 ± 0.11	1.45 ± 1.39	1.00 / 0.99
	128 D	1.12 ± 0.02	0.23 ± 0.00	0.04 ± 0.03	0.21 ± 0.04	0.22 ± 0.03	1.08 ± 0.06	3.05 ± 0.87	3.09 ± 0.37	0.26 ± 0.03	2.31 ± 0.04	1.19 ± 1.21	1.00 / 0.98
	256 D	0.54 ± 0.03	0.18 ± 0.01	0.01 ± 0.00	0.16 ± 0.01	0.15 ± 0.01	0.92 ± 0.08	2.42 ± 0.14	2.51 ± 0.13	0.19 ± 0.01	2.09 ± 0.02	0.94 ± 0.99	1.00 / 0.97
500 full (F)	16 F	2.14 ± 0.06	0.70 ± 0.04	0.28 ± 0.00	0.27 ± 0.01	0.21 ± 0.00	1.14 ± 0.04	3.06 ± 0.27	2.71 ± 0.01	0.34 ± 0.01	2.21 ± 0.01	1.33 ± 1.09	1.00/1.00
	32 F	1.17 ± 0.07	0.48 ± 0.03	0.08 ± 0.04	0.21 ± 0.01	0.17 ± 0.00	0.99 ± 0.04	2.69 ± 0.10	2.47 ± 0.02	0.25 ± 0.02	2.11 ± 0.04	1.08 ± 0.99	0.99/0.99
	64 F	0.51 ± 0.03	0.21 ± 0.00	0.02 ± 0.00	0.17 ± 0.01	0.14 ± 0.00	0.88 ± 0.04	2.19 ± 0.14	2.34 ± 0.03	0.20 ± 0.00	1.97 ± 0.02	0.89 ± 0.91	0.97/0.96
	128 F	0.39 ± 0.01	0.16 ± 0.00	0.01 ± 0.00	0.13 ± 0.00	0.11 ± 0.00	0.81 ± 0.03	1.42 ± 0.07	2.03 ± 0.01	0.16 ± 0.00	1.71 ± 0.01	0.71 ± 0.74	0.88/0.86
	256 F	0.32 ± 0.01	0.15 ± 0.00	0.01 ± 0.00	0.12 ± 0.00	0.10 ± 0.00	0.77 ± 0.01	1.18 ± 0.04	1.96 ± 0.00	0.14 ± 0.01	1.25 ± 0.00	0.61 ± 0.65	0.50/0.47
500 w/clusters (C)	16 C 7	0.40	0.18	0.01	0.15	0.13	0.90	2.03	2.21	0.16	1.50	0.77	1.00/0.98
	16 C 10	0.36	0.16	0.01	0.14	0.13	0.87	2.19	2.04	0.15	1.38	0.74	1.00/0.98
	16 C 25	0.32	0.16	0.01	0.13	0.10	0.81	1.28	1.96	0.12	1.07	0.60	1.00/0.95
	64 C 5	0.36	0.16	0.01	0.12	0.10	0.80	1.17	1.98	0.14	1.17	0.60	0.97/0.93
	64 C 7	0.34	0.15	0.01	0.12	0.10	0.79	1.14	1.96	0.13	1.08	0.58	0.97/0.91
1000 w/clusters (C)	16 C 25	0.37	0.16	0.01	0.13	0.13	0.86	2.12	2.04	0.14	1.35	0.73	1.00/0.97

Table 12: **Absolute** In-Distribution test loss for various tasks and methods

Model Type	Method Type	Tasks										Average	Para. Saved
		task039	task190	task280	task290	task391	task442	task620	task1342	task1391	task1598		
	base	0.00 ± 0.00	0.00 ± 0.00	1.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.10 ± 0.30	1.00 / 1.00
	lora	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	0.00 / 0.00
TIES	10	41.00 ± 0.00	53.67 ± 0.58	44.33 ± 4.04	10.33 ± 0.58	46.33 ± 4.04	1.00 ± 0.00	8.00 ± 0.00	8.00 ± 0.00	76.67 ± 1.15	1.00 ± 0.00	29.03 ± 25.69	1.00 / 1.00
	50	24.00 ± 0.00	38.67 ± 0.58	17.67 ± 4.62	2.00 ± 0.00	56.33 ± 0.58	1.00 ± 0.00	8.00 ± 0.00	8.00 ± 0.00	29.67 ± 2.89	0.00 ± 0.00	18.53 ± 18.07	1.00 / 1.00
	100	22.00 ± 0.00	38.00 ± 0.00	18.67 ± 4.62	1.00 ± 1.73	51.67 ± 4.62	1.00 ± 0.00	8.00 ± 0.00	7.33 ± 0.58	2.00 ± 0.00	0.00 ± 0.00	14.97 ± 17.20	1.00 / 1.00
	500	8.00 ± 0.00	25.00 ± 0.00	1.00 ± 0.00	0.00 ± 0.00	59.00 ± 0.00	0.00 ± 0.00	3.00 ± 0.00	6.00 ± 0.00	2.00 ± 0.00	0.00 ± 0.00	9.90 ± 18.12	1.00 / 1.00
SVD	SVD 2	88.33 ± 0.65	91.91 ± 0.94	100.00 ± 0.00	97.25 ± 0.45	92.83 ± 0.39	76.50 ± 1.51	66.00 ± 1.41	58.08 ± 1.16	98.67 ± 0.49	5.83 ± 0.94	77.42 ± 27.69	0.88 / 0.88
	SVD 4	93.00 ± 0.00	96.64 ± 0.50	100.00 ± 0.00	100.00 ± 0.00	96.75 ± 0.87	88.83 ± 1.53	90.67 ± 1.23	72.17 ± 0.58	98.67 ± 0.49	16.67 ± 1.78	85.24 ± 24.39	0.75 / 0.75
	SVD 8	98.89 ± 0.60	98.55 ± 0.52	100.00 ± 0.00	100.00 ± 0.00	99.42 ± 0.51	93.44 ± 0.73	97.22 ± 0.44	82.78 ± 1.64	99.00 ± 0.00	60.00 ± 0.87	93.70 ± 11.59	0.50 / 0.50
	SVD 16	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	99.67 ± 0.50	99.50 ± 0.55	99.67 ± 0.50	100.00 ± 0.00	98.11 ± 0.78	99.69 ± 0.68	0.00 / 0.00
10 diagonal (D)	16 D	83.33 ± 1.53	88.33 ± 0.58	100.00 ± 0.00	97.00 ± 2.00	88.33 ± 1.15	57.00 ± 1.00	48.67 ± 3.21	50.67 ± 4.93	97.67 ± 1.53	5.33 ± 1.15	71.63 ± 29.53	1.00 / 0.90
	32 D	93.00 ± 1.00	95.33 ± 0.58	100.00 ± 0.00	98.00 ± 1.00	93.67 ± 1.53	80.67 ± 2.31	78.67 ± 1.15	68.00 ± 1.73	98.33 ± 0.58	14.67 ± 2.52	82.03 ± 24.99	1.00 / 0.80
	64 D	99.00 ± 0.00	97.00 ± 1.00	100.00 ± 0.00	100.00 ± 0.00	98.00 ± 0.00	90.67 ± 1.53	95.33 ± 1.15	79.67 ± 1.53	99.00 ± 0.00	55.00 ± 4.36	91.37 ± 13.78	1.00 / 0.60
	128 D	100.00 ± 0.00	99.33 ± 0.58	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	96.67 ± 1.53	98.33 ± 1.15	95.67 ± 2.31	100.00 ± 0.00	91.67 ± 3.51	98.17 ± 2.94	1.00 / 0.20
	256 D	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	99.33 ± 1.15	100.00 ± 0.00	95.00 ± 1.00	99.43 ± 1.57	1.00 / -0.60
10 full (F)	16 F	83.00 ± 2.00	93.00 ± 1.00	100.00 ± 0.00	98.33 ± 0.58	91.67 ± 0.58	64.33 ± 3.21	59.33 ± 1.15	52.67 ± 1.53	98.33 ± 0.58	6.33 ± 1.15	74.70 ± 28.71	1.00/0.90
	32 F	91.33 ± 0.58	96.00 ± 1.00	100.00 ± 0.00	98.33 ± 0.58	94.33 ± 0.58	84.00 ± 2.00	83.00 ± 1.73	70.33 ± 1.53	99.00 ± 1.00	22.00 ± 2.65	83.83 ± 22.82	0.99/0.79
	64 F	99.00 ± 0.00	97.33 ± 0.58	100.00 ± 0.00	100.00 ± 0.00	99.33 ± 1.15	91.33 ± 1.53	96.33 ± 0.58	81.67 ± 2.31	99.00 ± 0.00	58.33 ± 1.53	92.23 ± 12.76	0.97/0.57
	128 F	99.67 ± 0.58	99.33 ± 0.58	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	97.67 ± 1.15	100.00 ± 0.00	95.67 ± 1.15	100.00 ± 0.00	91.00 ± 1.00	98.33 ± 2.89	0.88/0.07
	256 F	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	99.67 ± 0.58	99.67 ± 0.58	99.67 ± 0.58	100.00 ± 0.00	98.00 ± 1.00	99.70 ± 0.70	0.50/-1.10
50 diagonal (D)	16 D	52.67 ± 4.51	86.67 ± 3.06	100.00 ± 0.00	85.00 ± 3.46	65.33 ± 3.79	25.33 ± 5.03	10.00 ± 1.00	10.67 ± 10.26	81.00 ± 6.56	0.00 ± 0.00	51.67 ± 36.18	1.00 / 0.98
	32 D	69.67 ± 3.21	88.67 ± 1.53	100.00 ± 0.00	95.00 ± 2.00	80.00 ± 3.00	36.67 ± 5.13	17.00 ± 2.65	26.33 ± 5.03	95.00 ± 2.00	0.00 ± 0.00	60.83 ± 36.02	1.00 / 0.96
	64 D	79.67 ± 2.52	91.00 ± 1.00	100.00 ± 0.00	97.67 ± 0.58	88.00 ± 1.00	52.00 ± 1.00	36.67 ± 5.69	41.33 ± 1.15	96.00 ± 1.00	0.33 ± 0.58	68.27 ± 32.66	1.00 / 0.92
	128 D	90.00 ± 1.00	91.33 ± 0.58	100.00 ± 0.00	98.33 ± 0.58	90.67 ± 2.08	73.67 ± 2.08	63.67 ± 1.53	56.33 ± 0.58	98.00 ± 0.00	7.33 ± 1.15	76.93 ± 27.79	1.00 / 0.84
	256 D	94.67 ± 0.58	96.33 ± 0.58	100.00 ± 0.00	99.67 ± 0.58	96.33 ± 1.15	87.33 ± 0.58	87.00 ± 2.65	71.67 ± 1.53	99.67 ± 0.58	31.67 ± 1.15	86.43 ± 20.41	1.00 / 0.68
50 full (F)	16 F	61.67 ± 3.06	89.67 ± 1.15	99.67 ± 0.58	90.67 ± 2.52	78.33 ± 3.51	34.00 ± 1.00	7.00 ± 3.46	25.00 ± 6.24	90.00 ± 1.00	0.00 ± 0.00	57.60 ± 36.59	1.00/0.98
	32 F	71.00 ± 1.00	89.00 ± 1.73	100.00 ± 0.00	98.00 ± 0.00	85.00 ± 1.00	47.00 ± 1.73	29.00 ± 3.00	35.00 ± 2.00	98.00 ± 1.00	0.00 ± 0.00	65.20 ± 34.00	0.99/0.95
	64 F	81.67 ± 0.58	93.67 ± 1.15	100.00 ± 0.00	98.33 ± 0.58	90.67 ± 2.08	61.67 ± 1.53	54.33 ± 1.53	51.33 ± 1.15	98.33 ± 0.58	3.33 ± 0.58	73.33 ± 29.89	0.97/0.89
	128 F	91.00 ± 1.00	94.33 ± 0.58	100.00 ± 0.00	99.00 ± 0.00	93.33 ± 1.53	81.67 ± 0.58	75.00 ± 1.73	67.67 ± 2.08	98.67 ± 0.58	16.67 ± 0.58	81.73 ± 24.46	0.88/0.72
	256 F	97.00 ± 0.00	98.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	99.67 ± 0.58	92.00 ± 0.00	94.33 ± 1.15	79.67 ± 1.53	99.00 ± 0.00	57.33 ± 2.52	91.70 ± 13.11	0.50/0.18
100 diagonal (D)	16 D	33.00 ± 8.19	79.33 ± 5.69	89.33 ± 5.51	80.00 ± 3.61	35.33 ± 6.03	4.00 ± 1.73	3.00 ± 1.00	0.00 ± 0.00	71.33 ± 4.51	0.00 ± 0.00	39.53 ± 36.15	1.00 / 0.99
	32 D	51.00 ± 7.81	90.00 ± 1.00	100.00 ± 0.00	88.00 ± 7.00	58.67 ± 11.68	17.67 ± 10.26	7.67 ± 2.89	9.33 ± 12.86	86.33 ± 2.52	0.00 ± 0.00	50.87 ± 38.43	1.00 / 0.98
	64 D	68.00 ± 2.65	87.33 ± 1.53	100.00 ± 0.00	94.33 ± 4.04	80.33 ± 2.08	38.00 ± 3.00	19.67 ± 4.51	28.33 ± 1.53	92.67 ± 1.15	0.33 ± 0.58	60.90 ± 34.91	1.00 / 0.96
	128 D	82.00 ± 2.00	90.00 ± 2.00	100.00 ± 0.00	97.33 ± 0.58	85.33 ± 0.58	55.33 ± 2.08	34.33 ± 3.79	36.67 ± 2.52	94.67 ± 0.58	0.00 ± 0.00	67.57 ± 32.95	1.00 / 0.92
	256 D	90.00 ± 1.00	93.00 ± 2.00	100.00 ± 0.00	97.67 ± 0.58	91.67 ± 0.58	71.67 ± 5.13	59.67 ± 1.53	58.00 ± 0.00	97.67 ± 0.58	4.00 ± 1.00	76.33 ± 28.88	1.00 / 0.84
100 full (F)	16 F	49.00 ± 2.00	89.67 ± 3.21	97.00 ± 3.00	84.33 ± 3.06	65.33 ± 2.52	20.67 ± 8.33	6.33 ± 2.08	8.33 ± 4.73	81.33 ± 2.08	0.00 ± 0.00	50.20 ± 37.06	1.00/0.99
	32 F	65.00 ± 3.46	90.33 ± 1.53	100.00 ± 0.00	96.33 ± 1.53	80.00 ± 2.65	41.33 ± 3.21	16.00 ± 0.00	29.33 ± 2.08	92.00 ± 2.65	0.00 ± 0.00	61.03 ± 35.43	0.99/0.97
	64 F	72.33 ± 0.58	89.67 ± 1.53	100.00 ± 0.00	97.67 ± 0.58	86.00 ± 1.00	53.00 ± 1.00	35.33 ± 1.53	38.00 ± 1.73	94.67 ± 0.58	0.00 ± 0.00	66.67 ± 32.54	0.97/0.93
	128 F	84.33 ± 1.53	92.33 ± 1.53	100.00 ± 0.00	98.00 ± 0.00	91.33 ± 0.58	68.67 ± 0.58	56.00 ± 1.00	57.67 ± 1.15	99.00 ± 0.00	5.33 ± 0.58	75.27 ± 28.67	0.88/0.80
	256 F	91.67 ± 1.15	96.67 ± 0.58	100.00 ± 0.00	100.00 ± 0.00	94.33 ± 0.58	84.67 ± 0.58	78.00 ± 0.00	69.67 ± 0.58	99.00 ± 0.00	22.00 ± 1.00	83.60 ± 23.08	0.50/0.34
100 w/clusters (C)	16 C 5	74.67 ± 0.94	91.00 ± 0.82	100.00 ± 0.00	96.67 ± 1.25	87.67 ± 1.70	53.67 ± 2.05	40.67 ± 2.87	41.00 ± 4.55	97.67 ± 1.25	0.67 ± 0.94	68.37 ± 31.46	1.00/0.95
	16 C 7	77.67 ± 0.47	90.33 ± 1.25	100.00 ± 0.00	97.33 ± 0.94	90.33 ± 2.05	58.67 ± 0.94	49.00 ± 2.16	49.00 ± 0.82	97.67 ± 0.47	3.67 ± 1.25	71.37 ± 29.48	1.00/0.93
500 diagonal (D)	16 D	8.00 ± 3.61	51.50 ± 3.54	79.67 ± 4.93	28.00 ± 42.44	56.67 ± 2.89	0.67 ± 1.15	0.33 ± 0.58	0.00 ± 0.00	71.67 ± 6.03	0.00 ± 0.00	28.90 ± 33.54	1.00 / 1.00
	32 D	14.33 ± 11.02	52.50 ± 9.19	80.67 ± 2.08	43.00 ± 31.48	60.67 ± 0.58	0.67 ± 1.15	1.67 ± 1.15	0.00 ± 0.00	74.33 ± 4.04	0.00 ± 0.00	32.10 ± 33.43	1.00 / 1.00
	64 D	25.67 ± 1.15	62.50 ± 12.02	87.33 ± 4.04	78.33 ± 1.15	65.33 ± 3.06	5.33 ± 3.51	3.67 ± 1.15	1.00 ± 1.73	76.67 ± 2.31	0.00 ± 0.00	39.83 ± 35.80	1.00 / 0.99
	128 D	38.33 ± 3.21	85.50 ± 2.12	96.00 ± 3.00	81.33 ± 2.31	65.67 ± 1.15	11.67 ± 4.73	5.33 ± 2.08	2.00 ± 1.00	80.00 ± 5.00	0.00 ± 0.00	45.24 ± 37.78	1.00 / 0.98
	256 D	53.33 ± 0.58	91.00 ± 2.83	100.00 ± 0.00	89.00 ± 2.65	76.00 ± 2.00	20.67 ± 6.81	6.00 ± 1.73	12.00 ± 8.00	86.33 ± 2.31	0.00 ± 0.00	52.14 ± 38.64	1.00 / 0.97
500 full (F)	16 F	8.33 ± 2.08	41.00 ± 5.66	76.67 ± 0.58	78.00 ± 0.00	72.67 ± 0.58	6.00 ± 0.00	5.67 ± 0.58	0.00 ± 0.00	78.00 ± 1.00	0.00 ± 0.00	36.48 ± 35.46	1.00/1.00
	32 F	33.67 ± 4.16	51.00 ± 1.41	92.67 ± 1.53	77.00 ± 1.73	75.00 ± 2.00	14.33 ± 1.53	8.00 ± 0.00	0.00 ± 0.00	80.67 ± 1.53	0.00 ± 0.00	42.97 ± 35.80	0.99/0.99
	64 F	56.00 ± 2.65	85.50 ± 0.71	94.33 ± 0.58	89.33 ± 2.89	74.33 ± 1.15	36.33 ± 1.15	9.00 ± 1.00	2.67 ± 1.15	84.00 ± 1.00	0.00 ± 0.00	52.03 ± 36.92	0.97/0.96
	128 F	69.33 ± 0.58	88.50 ± 0.71	99.00 ± 0.00	96.33 ± 1.53	80.33 ± 1.15	45.00 ± 2.00	16.33 ± 0.58	31.00 ± 1.73	92.00 ± 0.00	0.00 ± 0.00	60.86 ± 35.07	0.88/0.86
	256 F	79.67 ± 0.58	89.50 ± 0.71	100.00 ± 0.00	97.67 ± 0.58	87.33 ± 0.58	57.00 ± 1.00	35.00 ± 1.00	42.00 ± 1.00	95.00 ± 1.00	0.00 ± 0.00	67.59 ± 32.67	0.50/0.47
500 w/clusters (C)	16 C 7	63.00	90.00	99.00	96.00	78.00	31.00	9.00	15.00	89.00	1.00	57.10	1.00/0.98
	16 C 10	69.00	93.00	100.00	98.00	81.00	34.00	8.00	33.00	95.00	1.00	61.20	1.00/0.98
	16 C 25	79.00	90.00	100.00	97.00	88.00	53.00	38.00	48.00	98.00	0.00	69.10	1.00/0.95
	64 C 5	77.00	88.00	100.00	98.00	89.00	56.00	39.00	42.00	99.00	0.00	68.80	0.97/0.93
	64 C 7	76.00	90.00	100.00	97.00	89.00	60.00	48.00	49.00	99.00	3.00	71.10	0.97/0.91
1000 w/clusters (C)	16 C 25	73.00	90.00	100.00	98.00	77.00	39.00	8.00	34.00	96.00	1.00	61.60	1.00/0.97

Table 13: **Absolute** In-Distribution agreement for various tasks and methods

Model Type	Method Type	Tasks										Average
		task039	task190	task280	task290	task391	task442	task620	task1342	task1391	task1598	
SVD	SVD 2	0.29 ± 0.00	0.43 ± 0.00	0.31 ± 0.00	0.40 ± 0.00	0.38 ± 0.00	0.31 ± 0.00	0.37 ± 0.00	0.31 ± 0.00	0.42 ± 0.00	0.30 ± 0.00	0.35 ± 0.05
	SVD 4	0.16 ± 0.00	0.24 ± 0.00	0.16 ± 0.00	0.25 ± 0.00	0.23 ± 0.00	0.17 ± 0.00	0.22 ± 0.00	0.16 ± 0.00	0.25 ± 0.00	0.16 ± 0.00	0.20 ± 0.04
	SVD 8	0.06 ± 0.00	0.09 ± 0.00	0.06 ± 0.00	0.11 ± 0.00	0.10 ± 0.00	0.07 ± 0.00	0.09 ± 0.00	0.06 ± 0.00	0.11 ± 0.00	0.06 ± 0.00	0.08 ± 0.02
10 diagonal (D)	16 D	0.37 ± 0.02	0.51 ± 0.02	0.36 ± 0.01	0.57 ± 0.02	0.55 ± 0.00	0.39 ± 0.02	0.49 ± 0.01	0.36 ± 0.02	0.53 ± 0.03	0.39 ± 0.01	0.45 ± 0.08
	32 D	0.21 ± 0.01	0.28 ± 0.00	0.20 ± 0.01	0.35 ± 0.00	0.33 ± 0.01	0.22 ± 0.01	0.31 ± 0.01	0.20 ± 0.01	0.32 ± 0.01	0.22 ± 0.00	0.26 ± 0.06
	64 D	0.10 ± 0.00	0.11 ± 0.01	0.09 ± 0.00	0.18 ± 0.01	0.18 ± 0.00	0.10 ± 0.00	0.15 ± 0.01	0.09 ± 0.00	0.14 ± 0.00	0.09 ± 0.00	0.12 ± 0.04
	128 D	0.02 ± 0.00	0.01 ± 0.00	0.02 ± 0.00	0.03 ± 0.00	0.04 ± 0.00	0.02 ± 0.00	0.03 ± 0.00	0.02 ± 0.00	0.02 ± 0.00	0.02 ± 0.00	0.03 ± 0.01
	256 D	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00
10 full (F)	16 F	0.35 ± 0.00	0.46 ± 0.00	0.34 ± 0.00	0.51 ± 0.00	0.47 ± 0.01	0.36 ± 0.01	0.45 ± 0.01	0.35 ± 0.01	0.49 ± 0.00	0.35 ± 0.01	0.41 ± 0.06
	32 F	0.20 ± 0.00	0.24 ± 0.00	0.20 ± 0.00	0.30 ± 0.00	0.29 ± 0.00	0.22 ± 0.00	0.27 ± 0.00	0.20 ± 0.00	0.27 ± 0.00	0.21 ± 0.00	0.24 ± 0.04
	64 F	0.10 ± 0.00	0.10 ± 0.00	0.09 ± 0.00	0.13 ± 0.00	0.13 ± 0.00	0.10 ± 0.00	0.12 ± 0.00	0.09 ± 0.00	0.12 ± 0.00	0.10 ± 0.00	0.11 ± 0.02
	128 F	0.02 ± 0.00	0.02 ± 0.00	0.02 ± 0.00	0.01 ± 0.00	0.02 ± 0.00	0.02 ± 0.00	0.02 ± 0.00	0.02 ± 0.00	0.01 ± 0.00	0.02 ± 0.00	0.02 ± 0.00
	256 F	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00
50 diagonal (D)	16 D	0.66 ± 0.01	0.69 ± 0.01	0.88 ± 0.01	0.76 ± 0.03	0.95 ± 0.02	0.91 ± 0.01	0.83 ± 0.02	0.88 ± 0.03	0.72 ± 0.02	0.88 ± 0.02	0.82 ± 0.10
	32 D	0.50 ± 0.01	0.52 ± 0.02	0.73 ± 0.01	0.58 ± 0.03	0.88 ± 0.03	0.79 ± 0.03	0.72 ± 0.01	0.75 ± 0.01	0.57 ± 0.02	0.75 ± 0.01	0.68 ± 0.12
	64 D	0.34 ± 0.01	0.37 ± 0.01	0.52 ± 0.00	0.38 ± 0.01	0.71 ± 0.02	0.58 ± 0.01	0.54 ± 0.00	0.56 ± 0.00	0.44 ± 0.01	0.58 ± 0.01	0.50 ± 0.11
	128 D	0.21 ± 0.01	0.22 ± 0.01	0.31 ± 0.00	0.22 ± 0.00	0.51 ± 0.01	0.42 ± 0.01	0.38 ± 0.00	0.39 ± 0.00	0.27 ± 0.00	0.40 ± 0.00	0.33 ± 0.10
	256 D	0.10 ± 0.00	0.12 ± 0.00	0.16 ± 0.00	0.10 ± 0.00	0.29 ± 0.01	0.21 ± 0.00	0.19 ± 0.00	0.23 ± 0.01	0.15 ± 0.00	0.20 ± 0.00	0.18 ± 0.06
50 full (F)	16 F	0.57 ± 0.01	0.60 ± 0.01	0.86 ± 0.01	0.71 ± 0.02	0.95 ± 0.01	0.88 ± 0.01	0.81 ± 0.00	0.83 ± 0.01	0.67 ± 0.01	0.86 ± 0.01	0.78 ± 0.12
	32 F	0.47 ± 0.01	0.48 ± 0.01	0.71 ± 0.00	0.55 ± 0.01	0.78 ± 0.01	0.69 ± 0.01	0.69 ± 0.00	0.65 ± 0.01	0.53 ± 0.01	0.71 ± 0.00	0.63 ± 0.11
	64 F	0.33 ± 0.00	0.35 ± 0.00	0.45 ± 0.00	0.36 ± 0.00	0.56 ± 0.00	0.50 ± 0.01	0.47 ± 0.00	0.49 ± 0.00	0.39 ± 0.01	0.49 ± 0.00	0.44 ± 0.08
	128 F	0.19 ± 0.00	0.21 ± 0.00	0.25 ± 0.00	0.19 ± 0.00	0.35 ± 0.00	0.30 ± 0.00	0.28 ± 0.00	0.31 ± 0.00	0.24 ± 0.00	0.30 ± 0.00	0.26 ± 0.05
	256 F	0.09 ± 0.00	0.10 ± 0.00	0.10 ± 0.00	0.08 ± 0.00	0.16 ± 0.00	0.13 ± 0.00	0.12 ± 0.00	0.15 ± 0.00	0.11 ± 0.00	0.13 ± 0.00	0.12 ± 0.02
100 diagonal (D)	16 D	0.90 ± 0.01	0.85 ± 0.01	0.87 ± 0.03	0.88 ± 0.02	0.68 ± 0.02	0.91 ± 0.01	0.97 ± 0.01	0.98 ± 0.01	0.96 ± 0.01	1.00 ± 0.00	0.90 ± 0.09
	32 D	0.83 ± 0.02	0.77 ± 0.00	0.77 ± 0.01	0.78 ± 0.00	0.55 ± 0.02	0.79 ± 0.01	0.94 ± 0.02	0.94 ± 0.03	0.87 ± 0.00	0.98 ± 0.01	0.82 ± 0.12
	64 D	0.67 ± 0.01	0.63 ± 0.00	0.59 ± 0.02	0.63 ± 0.01	0.40 ± 0.00	0.62 ± 0.01	0.86 ± 0.02	0.82 ± 0.02	0.71 ± 0.03	0.93 ± 0.00	0.68 ± 0.15
	128 D	0.49 ± 0.01	0.47 ± 0.00	0.42 ± 0.01	0.45 ± 0.00	0.27 ± 0.02	0.44 ± 0.01	0.73 ± 0.01	0.69 ± 0.02	0.59 ± 0.02	0.80 ± 0.02	0.53 ± 0.16
	256 D	0.32 ± 0.00	0.31 ± 0.00	0.26 ± 0.01	0.30 ± 0.00	0.15 ± 0.01	0.28 ± 0.00	0.51 ± 0.02	0.51 ± 0.02	0.40 ± 0.01	0.61 ± 0.01	0.36 ± 0.14
100 full (F)	16 F	0.88 ± 0.00	0.82 ± 0.00	0.84 ± 0.01	0.86 ± 0.00	0.67 ± 0.01	0.88 ± 0.01	0.99 ± 0.00	0.96 ± 0.01	0.91 ± 0.01	1.00 ± 0.00	0.88 ± 0.09
	32 F	0.78 ± 0.00	0.72 ± 0.00	0.73 ± 0.00	0.74 ± 0.00	0.52 ± 0.00	0.74 ± 0.01	0.94 ± 0.01	0.89 ± 0.00	0.77 ± 0.02	0.99 ± 0.00	0.78 ± 0.13
	64 F	0.60 ± 0.00	0.57 ± 0.00	0.57 ± 0.00	0.57 ± 0.00	0.39 ± 0.00	0.56 ± 0.00	0.76 ± 0.00	0.73 ± 0.00	0.60 ± 0.00	0.83 ± 0.01	0.62 ± 0.12
	128 F	0.40 ± 0.00	0.38 ± 0.00	0.35 ± 0.00	0.37 ± 0.00	0.25 ± 0.00	0.37 ± 0.00	0.52 ± 0.00	0.54 ± 0.00	0.45 ± 0.00	0.60 ± 0.00	0.42 ± 0.10
	256 F	0.21 ± 0.00	0.20 ± 0.00	0.18 ± 0.00	0.19 ± 0.00	0.13 ± 0.00	0.19 ± 0.00	0.30 ± 0.00	0.34 ± 0.00	0.26 ± 0.00	0.38 ± 0.00	0.24 ± 0.08
100 w/clusters (C)	16 C 5	0.46 ± 0.00	0.46 ± 0.00	0.45 ± 0.00	0.47 ± 0.01	0.61 ± 0.01	0.65 ± 0.01	0.61 ± 0.00	0.64 ± 0.02	0.45 ± 0.00	0.59 ± 0.01	0.54 ± 0.08
	16 C 7	0.41 ± 0.01	0.42 ± 0.01	0.39 ± 0.01	0.43 ± 0.01	0.51 ± 0.01	0.56 ± 0.01	0.53 ± 0.01	0.55 ± 0.00	0.42 ± 0.01	0.54 ± 0.01	0.48 ± 0.06
500 diagonal (D)	16 D	0.97 ± 0.00	0.73 ± 0.00	0.96 ± 0.00	1.00 ± 0.00	0.99 ± 0.01	0.96 ± 0.01	0.90 ± 0.00	0.92 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	0.94 ± 0.08
	32 D	0.96 ± 0.00	0.70 ± 0.00	0.92 ± 0.01	0.98 ± 0.01	0.96 ± 0.01	0.93 ± 0.01	0.86 ± 0.00	0.89 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	0.92 ± 0.09
	64 D	0.90 ± 0.01	0.65 ± 0.00	0.86 ± 0.01	0.96 ± 0.02	0.90 ± 0.01	0.87 ± 0.01	0.81 ± 0.00	0.83 ± 0.01	0.99 ± 0.01	1.00 ± 0.00	0.88 ± 0.10
	128 D	0.82 ± 0.01	0.60 ± 0.00	0.76 ± 0.00	0.90 ± 0.02	0.83 ± 0.01	0.78 ± 0.02	0.74 ± 0.00	0.74 ± 0.01	0.97 ± 0.01	1.00 ± 0.00	0.81 ± 0.12
	256 D	0.59 ± 0.02	0.51 ± 0.00	0.56 ± 0.01	0.81 ± 0.02	0.70 ± 0.02	0.55 ± 0.01	0.57 ± 0.01	0.54 ± 0.01	0.91 ± 0.01	1.00 ± 0.01	0.67 ± 0.17
500 full (F)	16 F	0.94 ± 0.00	0.67 ± 0.00	0.88 ± 0.00	1.00 ± 0.00	0.98 ± 0.00	0.90 ± 0.00	0.82 ± 0.00	0.83 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	0.90 ± 0.10
	32 F	0.88 ± 0.00	0.61 ± 0.00	0.81 ± 0.00	0.97 ± 0.01	0.94 ± 0.00	0.84 ± 0.00	0.75 ± 0.00	0.77 ± 0.00	0.99 ± 0.00	0.99 ± 0.00	0.86 ± 0.12
	64 F	0.80 ± 0.00	0.55 ± 0.00	0.72 ± 0.00	0.86 ± 0.00	0.82 ± 0.01	0.76 ± 0.00	0.67 ± 0.00	0.70 ± 0.00	0.94 ± 0.00	0.99 ± 0.00	0.78 ± 0.13
	128 F	0.64 ± 0.00	0.46 ± 0.00	0.60 ± 0.00	0.74 ± 0.00	0.65 ± 0.00	0.63 ± 0.00	0.56 ± 0.00	0.58 ± 0.00	0.85 ± 0.00	0.96 ± 0.00	0.67 ± 0.14
	256 F	0.43 ± 0.00	0.35 ± 0.00	0.44 ± 0.00	0.55 ± 0.00	0.49 ± 0.00	0.45 ± 0.00	0.40 ± 0.00	0.42 ± 0.00	0.67 ± 0.00	0.84 ± 0.00	0.50 ± 0.14
500 w/clusters (C)	16 C 7	0.68	0.70	0.64	0.72	0.85	0.90	0.93	0.92	0.71	0.83	0.79
	16 C 10	0.61	0.65	0.61	0.66	0.84	0.86	0.88	0.84	0.62	0.76	0.73
	16 C 25	0.42	0.41	0.42	0.44	0.57	0.64	0.63	0.62	0.40	0.58	0.51
	64 C 5	0.49	0.49	0.45	0.51	0.64	0.66	0.62	0.67	0.50	0.65	0.57
	64 C 7	0.45	0.45	0.41	0.45	0.56	0.58	0.55	0.59	0.44	0.57	0.51
1000 w/clusters (C)	16 C 25	0.58	0.64	0.54	0.64	0.81	0.87	0.90	0.84	0.57	0.74	0.71

Table 14: **Reconstruction error** In-Distribution for various tasks and methods

Model Type	Method Type	Tasks										Average
		task039	task190	task280	task290	task391	task442	task620	task1342	task1391	task1598	
10 full (F)	16 F	0.46 ± 0.01	0.63 ± 0.00	0.50 ± 0.00	0.55 ± 0.01	0.50 ± 0.00	0.49 ± 0.01	0.50 ± 0.01	0.50 ± 0.01	0.61 ± 0.01	0.47 ± 0.01	0.52 ± 0.06
	32 F	0.30 ± 0.01	0.37 ± 0.00	0.31 ± 0.00	0.35 ± 0.00	0.34 ± 0.00	0.31 ± 0.00	0.33 ± 0.00	0.31 ± 0.00	0.38 ± 0.00	0.30 ± 0.00	0.33 ± 0.03
	64 F	0.15 ± 0.00	0.15 ± 0.00	0.16 ± 0.00	0.17 ± 0.00	0.17 ± 0.00	0.16 ± 0.00	0.16 ± 0.00	0.16 ± 0.00	0.17 ± 0.00	0.15 ± 0.00	0.16 ± 0.01
50 full (F)	16 F	0.80 ± 0.02	0.82 ± 0.01	0.85 ± 0.01	0.90 ± 0.02	0.78 ± 0.01	0.95 ± 0.01	0.76 ± 0.01	0.75 ± 0.00	0.79 ± 0.01	0.82 ± 0.01	0.82 ± 0.06
	32 F	0.65 ± 0.01	0.67 ± 0.01	0.72 ± 0.01	0.76 ± 0.02	0.65 ± 0.01	0.82 ± 0.02	0.66 ± 0.01	0.65 ± 0.01	0.67 ± 0.02	0.69 ± 0.00	0.69 ± 0.06
	64 F	0.50 ± 0.01	0.52 ± 0.00	0.52 ± 0.00	0.55 ± 0.01	0.52 ± 0.01	0.62 ± 0.00	0.54 ± 0.01	0.51 ± 0.00	0.54 ± 0.01	0.57 ± 0.00	0.54 ± 0.03
100 full (F)	16 F	0.93 ± 0.02	0.90 ± 0.02	0.93 ± 0.01	0.91 ± 0.02	0.88 ± 0.03	0.98 ± 0.01	0.96 ± 0.01	0.78 ± 0.00	0.82 ± 0.00	0.93 ± 0.02	0.90 ± 0.06
	32 F	0.87 ± 0.01	0.81 ± 0.01	0.85 ± 0.02	0.80 ± 0.01	0.79 ± 0.02	0.91 ± 0.00	0.90 ± 0.01	0.74 ± 0.01	0.70 ± 0.02	0.85 ± 0.02	0.82 ± 0.07
	64 F	0.65 ± 0.04	0.69 ± 0.01	0.71 ± 0.01	0.67 ± 0.01	0.64 ± 0.01	0.76 ± 0.01	0.77 ± 0.01	0.67 ± 0.00	0.61 ± 0.00	0.75 ± 0.06	0.69 ± 0.06
500 full (F)	16 F	0.98 ± 0.04	0.98 ± 0.01	0.99 ± 0.01	1.00 ± 0.00	0.99 ± 0.00	0.96 ± 0.05	0.93 ± 0.10	0.94 ± 0.09	1.00 ± 0.00	0.99 ± 0.00	0.98 ± 0.05
	32 F	0.92 ± 0.07	0.84 ± 0.20	0.92 ± 0.10	0.98 ± 0.02	0.97 ± 0.02	0.89 ± 0.08	0.82 ± 0.13	0.84 ± 0.11	0.99 ± 0.00	0.99 ± 0.02	0.92 ± 0.10
	64 F	0.80 ± 0.00	0.67 ± 0.21	0.78 ± 0.11	0.90 ± 0.07	0.86 ± 0.08	0.76 ± 0.00	0.67 ± 0.00	0.70 ± 0.00	0.96 ± 0.03	0.99 ± 0.00	0.81 ± 0.13

Table 15: **Reconstruction error on random LoRAs** The error is larger in comparison to reconstructing trained (i.e., non-random) LoRAs in Table 14 for the corresponding compression methods.

Model Type	Method Type	Tasks										Average
		task039	task190	task280	task290	task391	task442	task620	task1342	task1391	task1598	
	base	24.44	1.60	19.13	39.22	10.27	35.46	7.85	6.22	17.82	38.87	20.09
	lora	95.00	86.00	99.00	93.67	94.33	74.88	74.40	26.68	95.00	50.32	78.93
10 full (F)	32 F	97.00	90.00	99.00	93.33	94.67	74.09	72.13	27.83	94.00	50.71	79.28
	64 F	95.00	89.00	99.00	93.67	94.67	74.29	74.80	26.63	96.00	51.04	79.41
50 full (F)	32 F	96.00	88.00	99.00	93.67	92.33	72.30	75.97	29.89	94.00	45.68	78.68
	64 F	98.00	89.00	99.00	93.67	93.33	72.74	76.50	29.33	96.00	45.71	79.33
100 full (F)	32 F	92.10	83.00	99.00	93.67	92.00	71.09	63.29	27.87	88.00	42.36	75.24
	64 F	97.00	87.00	99.00	93.67	92.33	72.23	74.69	29.98	95.00	44.71	78.56
500 full (F)	32 F	68.92	43.00	87.00	91.67	90.67	70.08	51.16	14.40	83.00	41.97	64.19
	64 F	93.50	78.00	91.00	92.33	90.33	72.55	57.49	15.44	85.00	42.31	71.80

Table 16: **Performance with convergence** In-Distribution **Rouge-L**

Table 17: Agreement Comparison. 100 LoRAs

Configuration		**Agreement (%)**
Base Model		83.015
Uncompressed LoRAs		100.000
Joint Compression		
Diagonal	Rank 8	87.032
	Rank 16	88.908
	Rank 32	91.545
	Rank 64	94.659
Full	Rank 8	87.686
	Rank 16	90.163
	Rank 32	94.018
	Rank 64	96.918

Table 18: Performance Comparison. 100 LoRAs

Configuration		**Average Performance**
Base Model		32.28
Uncompressed LoRAs		48.32
Join Compression		
Diagonal	Rank 8	41.90
	Rank 16	45.44
	Rank 32	46.89
	Rank 64	47.43
Full	Rank 8	43.88
	Rank 16	45.79
	Rank 32	46.83
	Rank 64	47.66

Table 19: Task-Based Performance Evaluation Across Different Models and Ranks

Task	**Base Model**	**LoRA**	**Diagonal R8**	**Diagonal R16**	**Diagonal R32**	**Diagonal R64**
Causal Judgement	57.47	64.37	55.17	58.62	58.62	58.62
Date Understanding	15.33	23.33	20.67	22.00	21.33	22.67
Formal Fallacies	51.33	56.00	52.67	52.67	53.33	54.67
Hyperbaton	6.67	68.00	57.33	63.33	67.33	68.00
Logical Deduction (5 Objects)	21.33	37.33	32.00	36.67	37.33	37.33
Logical Deduction (7 Objects)	12.67	44.00	31.33	42.67	44.67	45.33
Movie Recommendation	62.67	67.33	62.00	64.67	66.67	67.33
Object Counting	34.67	38.00	35.33	36.67	36.67	38.00
Snarks	50.00	61.54	53.85	56.41	58.97	57.69
Temporal Sequences	16.67	23.33	18.67	20.67	24.00	24.67
Average	32.88	48.32	41.90	45.44	46.89	47.43

Table 20: Task-Based Performance Evaluation Across Different Models and Ranks

Task	**Base Model**	**LoRA**	**Full R8**	**Full R16**	**Full R32**	**Full R64**
Causal Judgement	57.47	64.37	56.32	57.47	58.62	60.92
Date Understanding	15.33	23.33	19.33	22.00	22.67	22.67
Formal Fallacies	51.33	56.00	51.33	52.67	53.33	56.00
Hyperbaton	6.67	68.00	63.33	66.00	69.33	68.00
Logical Deduction (5 Objects)	21.33	37.33	35.33	36.00	35.33	37.33
Logical Deduction (7 Objects)	12.67	44.00	40.00	44.67	44.67	44.67
Movie Recommendation	62.67	67.33	63.33	65.33	67.33	67.33
Object Counting	34.67	38.00	35.33	36.67	37.33	37.33
Snarks	50.00	61.54	53.85	55.13	57.69	58.97
Temporal Sequences	16.67	23.33	20.67	22.00	22.00	23.33
Average	32.88	48.32	43.88	45.79	46.83	47.66